Dell PowerFlex Appliance with PowerFlex 5.x

Administration Guide

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DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

This guide provides procedures for administering PowerFlex appliance.

This guide is applicable for PowerFlex software 5.0 and later managed by PowerFlex Manager 5.0 and later. For PowerFlex Manager 5.1.x managing PowerFlex software 4.5.5 and later, see the PowerFlex rack or PowerFlex appliance with PowerFlex 4.x documentation.

It provides the following information:

- Managing components
- Performing maintenance activities in a PowerFlex cluster
- Monitoring
- Network administration
- Storage administration
- Administering the CloudLink Center
- Backup and restore
- Powering on and off
- Usernames and password management
- System logs
- Migrating to NVMe over TCP on VMware ESXi

The target audience for this document includes system administrators responsible for managing PowerFlex appliance, and Dell personnel responsible for remote management.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex appliance documentation, go to PowerFlex appliance technical documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
device gateway target	DGWT
management virtual machine	MVM
metadata manager	MDM
PowerFlex data server	PDS
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

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Revision history

Table 2. Revisions

Date	Document revision	Description of changes
August 2026	2.1	Editorial updates
June 2026	2.0	Added support for: • Dynamic licensing • SONiC networking • PowerFlex R670/R770/R6725/ R7725 compute-only nodes • CloudLink 8.3
October 2025	1.0	Initial release

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Accessing PowerFlex appliance components

Table 3. Components and method of access

Component	Method of access
PowerFlex Manager	https://pfxm_server_ip_address_or_fqdn
Secure connect gateway	https://<hostname(FQDN) or IP address:5700>
Policy manager for secure connect gateway	https://<hostname(FQDN) or IP address:8443>
VMware vCenter	https://vcenter_server_ip_address_or_fqdn
VMware ESXi	https://esxi-ip-address_or_fqdn
iDRAC	https://idrac-ip-address
CloudLink	https://cloudlinkcenterIP
VMware vCenter server appliance management	https://vcenter_server_ip_address_or_fqdn:5480
Jump server	RDP for embedded operating system based on SLES and VNC for embedded operating system based on CentOS
Network switches	SSH
KVM VM	SSH
PowerFlex nodes	SSH
PowerFlex SCLI	SSH to the primary MDM, and log in to the SCLI

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PowerFlex appliance deployment types

PowerFlex appliance has several options for deployment.

PowerFlex runs on PowerFlex nodes to operate the management and customer storage and tie in workloads. PowerFlex has the following components:

- Storage data client (SDC): Consumes storage from the PowerFlex appliance.
- Storage node: Contributes storage to PowerFlex appliance. The storage node consists of the PowerFlex data server (PDS) and the device gateway target (DGWT).
- PowerFlex metadata manager (MDM): Manages the storage blocks and tracks data location across the system.
- Storage data target (SDT): Manages host connections and controllers, processes NVMe commands, both IO and admin and it sends the IO commands forward to PowerFlex data server and device gateway target through an embedded LibSDC. PowerFlex data server is unaware to the source of the IO commands: SDC or NVMe hosts.

PowerFlex has the following storage components:

- Protection domain: A group of nodes that are defined to behave as an independent system. The maximum number of protection domains is four.
- Device gateway target: A group of complete set of devices and a scope of RAID rebuild.
- PowerFlex data server:
  - There can only be one device group per resource group.
- Storage pool: A group of volumes, allocating capacity from the device group. The size of the storage pool is defined by the user. The maximum size of the storage pool is derived from the device group, and the minimum size is 3 TB. The maximum storage pool per protection domain is four.
  - The protection scheme is set at the storage pool level.
    - 8+2 for 11 or more nodes
    - 4+2 for seven to ten nodes
    - 2+2 for five or six nodes
- Volumes: All volumes are thin volumes.

PowerFlex Manager allows you to specify a non-root user instead of the root user when you configure a template for a compute-only and storage-only.

Table 4. Deployment types and descriptions

Deployment type	Description
Two-layer	Separates compute resources from storage resources, allowing the independent expansion of compute or storage resources. Consists of PowerFlex compute-only nodes (supporting the SDC) and PowerFlex storage-only nodes (connected to and managed by the storage node). PowerFlex compute-only nodes host end-user applications. PowerFlex storage-only nodes contribute storage to the system pool.
Storage-only	Consists of nodes that contribute storage resources to the compute environment. The back-end traffic shares the same PowerFlex data networks. The storage-only PowerFlex appliance provide volumes to an external customer compute limited to Dell PowerEdge servers. No SDC components are installed on these nodes.
Dual network	Consists of a solution integrated into your existing software-defined network (SDN), such as, Cisco application centric infrastructure (ACI). Only PowerFlex compute-only nodes are affected and have two NICs cabled to a pair of customer access/leaf switches. Aggregation switches by default are tied into the customer SDN border for Layer-2 or Layer-3 access.

Table 4. Deployment types and descriptions (continued)

Deployment type	Description
	The solution uses two new customer access switches, two additional network ports on the host, and a new distributed virtual switch to carry traffic seamlessly into the software-defined network.

Two-layer deployments allow rebooting of VMware cluster nodes without PowerFlex ramifications.

When designing initial deployment or specifying later growth, use PowerFlex compute-only nodes and PowerFlex storage-only nodes. You can add PowerFlex compute-only nodes or PowerFlex storage-only nodes as needed.

To control the number of processors or cores, consider separating the compute for the application from the PowerFlex nodes that support storage. This deployment is a pure two-layer deployment. Extra workloads are supported or added on:

- PowerFlex compute-only nodes
- PowerFlex storage-only nodes
- Hybrid deployment: A blend of two-layer and storage-only
- Linux KVM: The management cluster operates with a dedicated instance of PowerFlex Manager to manage it. Both the management cluster's PowerFlex and PowerFlex Manager versions remain on 4.x, while a second instance of PowerFlex Manager 5.x manages the production PowerFlex clusters.
- Linux KVM (customer-provided): Only PowerFlex Manager 5.x is deployed to manage the production PowerFlex clusters.
- VMware vSphere environment: A single VMware vCenter, with separate datacenters designated for management controller virtual machines and the production compute-only cluster. In this setup, only PowerFlex Manager (version 5.x) will run to manage the production PowerFlex clusters.

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Configuring license management

PowerFlex Manager comes by default with a trial or evaluation license for first 90 days. There is no requirement to upload an evaluation license. After 90 days, the license is enforced or it displays an error message and no operation can be performed.

The following licensing are supported:

- Dynamic licensing: Provides an ability to manage license allocations more easily across product registrations, enabling a flexible and keyless experience. It automates and simplifies license management activities such as, registrations, capacity expansions, and renewals. Dynamic licensing is applicable for all systems with PowerFlex Manager 5.1.x and later.

  The following methods are used to activate and use your products with dynamic licensing:
  - Connected method
  - Offline method

- Key-based format licensing: Using this licensing, you can upload a license in PowerFlex Manager. The license is a single file that includes both PowerFlex Manager and PowerFlex licensing information. Once the license capacity is exceeded, you must upload a new license with more capacity.

For information on registering or migrating licenses, see *Dell PowerFlex Family Licensing Guide*.

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Configuring VMware vCenter high availability

VMware vCenter high availability (vCenter HA) protects the VMware vCenter server against host and hardware failures. The active-passive architecture of the solution can also help reduce downtime significantly when you patch the vCenter server.

After you create a three-node PowerFlex cluster that contains active, passive, and witness nodes. Different configuration paths are available, your selection depends on your existing configuration.

VMware vCenter high availability requirements:

- Validate the flex-vcsa-ha networking and VMware vCenter port groups are configured
- Recommended minimum of three VMware ESXi hosts

See the VMware vSphere Product Documentation and select version 8.x for additional requirements and configuration of VMware vCenter high availability.

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Managing components with PowerFlex Manager

Once PowerFlex Manager is configured, you can use it to perform the ongoing tasks necessary to manage your PowerFlex appliance.

The following table describes typical tasks for managing PowerFlex appliance components and what steps to take in PowerFlex Manager to initiate each:

Table 5. Managing components using PowerFlex Manager

If you want to:	Do this action in PowerFlex Manager:
View network topology	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, click the **Port View** tab.
Run inventory (PowerFlex nodes, switches, and VMware vCenter cluster)	1. Click **Lifecycle** > **Resources** and click the **All Resources** tab. 2. Click the check box for the resource that you want to update and click **Run Inventory**. 3. After running the inventory, click **Resource Group Details** under **More Actions** on the **Resource Groups** page for any resource group that requires the updated resource data.
Add an existing resource group	1. Click **Lifecycle** > **Resource Groups**. 2. Click **Add Existing Resource Group**.
Perform PowerFlex node expansion	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, under **Add Resources** and click **Add Nodes**. The procedure is the same for new resource groups and existing resource groups.
Reconfigure MDM roles	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Group** page, select a resource group. 3. On the **Resource Group Details** tab, click **Reconfigure MDM Roles** under **More Actions**. 4. Select the node for the MDM role against the MDM role from the drop-down menu. Click **Next**. 5. On the **Summary** page, verify the changes that are made. Type **`CHANGE MDM ROLES`** to confirm and click **Finish**. You can also reconfigure MDM roles from the **Lifecycle** > **Resources** page. Select a PowerFlex gateway and click **View Details**. Click **Reconfigure MDM Roles**.
Replace a drive	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Group** page, select a resource group. 3. On the **Resource Group Details** page, under **Component Replacement**, click **Drive Replacement**. 4. In the node list, select the node to replace the drive, and click **Next**. 5. In the **Select drive to replace** page, select the drive, click **Blink LED** (if required), and click **Next**. 6. On the **Confirm Drive Replacement** page, review the details and type **`REMOVE DRIVE`**, and click **Confirm**.
Modify an IP address prior to logging in to PowerFlex Manager	1. Log in to the host using SSH. 2. Create new interface network configuration files with the new IP address.

Table 5. Managing components using PowerFlex Manager (continued)

If you want to:	Do this action in PowerFlex Manager:
	3. Bring that interface or IP address up on the host. 4. Repeat for any other IP addresses or hosts where new IP addresses are needed.
Complete the addition of the IP address	1. From the **Lifecycle** dropdown, select **Storage Nodes**. 2. Select the box next to the **Storage Node** where the IP address is to be added. 3. From the **More Actions** dropdown, select **Edit IP Storage Node**. 4. Input the IP address in the empty field. 5. From the **Role** dropdown, select the appropriate role. 6. Click **Add** > **Apply**. A success message should appear.
Remove an IP address from a node	1. From the **Lifecycle** dropdown, select **Storage Nodes**. 2. Select the box next to the **Storage Node** where the IP address is to be removed. 3. From the **More Actions** dropdown, select **Edit IP Storage Node**. 4. Click the **X** next to the IP address you want to remove. 5. Click **Apply**. A success message should appear. Follow any maintenance steps to remove the IP address configuration from the host.

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Enable or disable SSH on the VMware ESXi host

Use this procedure to enable or disable SSH on the VMware ESXi host.

About this task

- PowerFlex Manager disables the SSH service from the backend after the compute-only resource group deployment is complete.
- Manually enable the SSH service whenever it is required.
- Disable SSH after activity is completed on the PowerFlex node.

Steps

1. Log in to the iDRAC virtual console.
2. Provide the VMware ESXi credentials.
3. Click **Troubleshooting Options** and complete the operation for enabling or disabling SSH.

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Enable secure boot enforcement for PowerFlex compute-only nodes

To enable secure boot enforcement for PowerFlex compute-only nodes, you must meet the following prerequisites:

- Boot mode must be set to Unified Extensible Firmware Interface (UEFI) in **System BIOS Settings** > **Boot Settings**.

  NOTE: If the host is not in this mode, you may not be able to change it without reinstalling the operating system.

- The server must have the Trusted Platform Module (TPM) 2.0 installed.
- BIOS firmware must be at the required version for the specific PowerEdge model to support enabling secure boot. For more information, see the Dell Technologies support site.
- Linux distro must be deployed with a Secure-Boot-compatible Linux OS provided by the customer:
  - OS uses a shim-based boot flow
  - Kernel is signed by the OS vendor
- ESXi-based compute-only node is Secure-Boot-ready by default. Readiness is validated by using **secureBoot.py -c** before enabling secure boot.
- Secure boot must be enabled after the PowerFlex Manager deployment of the compute-only node.

Configure Dell PowerEdge iDRAC for secure boot

Use this procedure to configure iDRAC for secure boot.

Steps

1. Log in to the iDRAC web interface and click **Configuration** > **BIOS Settings** > **System Security**.
2. Set **TPM Security** to **On**.
3. Expand **TPM Advanced Settings** and set **TPM2 Algorithm Selection** to **SHA256**.
4. Set **Secure Boot** to **Enabled**.
5. From the **System Security Settings** page, click **Apply**.
6. Click **Apply** > **Reboot**.

Enable secure boot on ESXi-based compute-only nodes

Use the following section to enable secure boot on ESXi-based compute-only nodes for either partial support or full support.

Enable partial support in VMware ESXi

Secure boot must be enabled after PowerFlex Manager deploys the nodes. If it is enabled beforehand, deployments using PowerFlex Manager fail.

About this task

If TPM 2.0, secure boot, and SHA256 are enabled before VMware ESXi is installed (manually, not with PowerFlex Manager), this is automatically configured on the first reboot.

Partial Support: Trusted Boot with Attestation and passes with some partner VIBs.

- UEFI Secure Boot: Verifies bootloader and kernel modules at boot
- TPM Measurements: Stores boot hash measurements in TPM PCRs (used for attestation)
- TPM-Backed Encryption: VM, vSAN, and Core Dump
- VMware vCenter Attestation: Detects if the host booted in a tampered or untrusted state
- vTPM Support on VMs: VMs can be given a virtual TPM for in-guest security functions (also requires vCenter KMS server)

Steps

1. Run the validation script:

   ```
   /usr/lib/vmware/secureboot/bin/secureBoot.py -c
   ```

   If it passes, the following message appears: "Secure Boot CAN be enabled."

   If it fails, it lists unsigned VIBs. You must remove these before proceeding or the host experiences a purple screen on the next boot.
2. Enable SSH on the ESXi host and use any SSH client to connect to the ESXi host using the root user.
3. Verify security level:

   ```
   esxcli system settings encryption get
   ```

   The output should display:
   - Mode: **None**
   - Installed VIBs: **False**
   - Require Secure Boot: **False**
4. Enable TPM mode:

   ```
   esxcli system settings encryption set --mode=TPM --require-secure-boot=true
   ```
5. Reboot the host.
6. When the host is back online, verify the security level:

   ```
   esxcli system settings encryption get
   ```

   The output should display:
   - Mode: **TPM**
   - Installed VIBs: **False**
   - Secure Boot: **True**
7. Sync the configuration to the Bootbank:

   ```
   /bin/backup.sh 0
   ```

Enable full support in VMware ESXi

Use this procedure to enable full support in VMware ESXi.

Prerequisites

All vSphere Installation Bundles (VIBs) must be VMware-signed, or else Execution Control Lockdown fails. Signed VIB Enforcement ensures the following:

- VIBs are not altered
- Only VMware-signed VIBs can be installed
- Only signed VIBs can be loaded during ESXi boot

About this task

Full secure boot support in VMware ESXi includes all features of partial support.

Steps

1. Run the validation script:

   ```
   /usr/lib/vmware/secureboot/bin/secureBoot.py -c
   ```

   If it passes, the following message appears: "Secure Boot CAN be enabled."

   If it fails, it lists unsigned VIBs. You must remove these before proceeding or the host experiences a purple screen on the next boot.

2. Enable SSH on the ESXi host and use any SSH client to connect to the ESXi host using the root user.
3. Verify security level:

   ```
   esxcli system settings encryption get
   ```

   The output should display:
   - Mode: **TPM**
   - Installed VIBs: **False**
   - Require Secure Boot: **False**
4. If the output does not match the above, enable partial support following the instructions above before proceeding.
   a. To get the current settings, type:

      ```
      esxcli system settings encryption get
      ```

   b. To allow the kernel to accept VIB enforcement, type:

      ```
      esxcli system settings kernel set -s execInstalledOnly -v TRUE
      ```

   c. Power off the host and then power the host on (do not use reboot).
   d. To enable the VIB enforcement, type:

      ```
      esxcli system settings encryption set --require-exec-installed-only=T
      ```

   e. Reboot the node to enforce signed VIBs.
   f. Once the node is back online, verify the security level:

      ```
      esxcli system settings encryption get
      ```

   g. To sync the running configuration to the Bootbank, type:

      ```
      /bin/backup.sh 0
      ```

Backup keys and configurations

You must always backup recovery boot key and system configuration. Do not skip this step.

Steps

1. SSH to the VMware ESXi host as root.
2. Display the backup key and copy to a secure location off of the node:

   ```
   esxcli system settings encryption recovery list
   ```

   Copy the Recovery Key (second column) and paste into a text file to save for future recovery. The Recovery ID can be omitted.
3. Generate a host-level backup bundle:

   ```
   vim-cmd hostsvc/firmware/backup_config
   ```
4. Copy the web URL provided to download the backup bundle. Store this bundle in the same location as the recovery key backup text file.

   NOTE: The backup can take up to 30 seconds to complete and it is no longer accessible after 5 minutes.

Enable secure boot on Linux-based compute-only nodes

Use the following section to enable secure boot on Linux-based compute-only nodes.

Enable Linux

Steps

1. Verify secure boot is currently disabled:

   ```
   mokutil --sb-state
   ```

Expected output: `SecureBoot disabled`

2. Verify EFI boot path uses SHIM:

   `efibootmgr -v`

   Expected output: `BootXXXX* <OS> FILE (\EFI\<distro>\shim*.efi)`

   NOTE: The boot entry must already point to shim*.efi. Secure boot enablement does not modify EFI boot order, it only enforces signature validation.

Enable secure boot for Linux

Steps

1. SSH to your Linux host as root and validate secure boot is enabled on your machine:

   `mokutil --sb-stat`

   The output should have secure boot enabled.

2. If SDC is already installed, continue to step 4.
3. If SDC is not installed, install the SDC RPM. The installation should be successful, but the scini driver should fail to load. You should get the error message "scini service failed because the control process exited with error code".

   To check for details on the error:

   `systemctl status scini.service`

   `journalctl -xe`

   If you check dmesg, you should see: Loading of module with unavailable key is rejected
4. Change directory to `/bin/emc/scaleio/scini_sync/certs/`. In this directory are the SDC certificates.
5. Run the following command to confirm that they are valid and not expired:

   `openssl x509 -in <.pem file from directory> -noout -enddate | cut -d= -f2`

   The output is the expiration date.
6. If the certificate is valid, use the mokutil took to import the .der file. You are required to generate a password:

   `mokutil --import <.der file from directory> (Example: emc_scaleio2026.der)`

   There are two prompts to enter the password.
7. If the certificate supplied with the SDC package is expired, you may be supplied a certificate in .pem format which you must convert to .der format using the following command:

   `openssl x509 -in /usr/src/<file.pem> -outform DER -out /usr/src/<file.der>`

   If necessary, contact Dell Technologies Support to get the new signed SDC package and related certificates.
8. Reboot the host.
9. On boot, before the Linux OS boots, you must enter the **Perform MOK management** menu. Enter the MOK management and choose **Enroll MOK**.
   a. On the next screen, choose **View Key 0**.
   b. The next screen displays the key information. Press any key to go to the **Enroll key** page.
   c. On the next screen, enter the password you generated previously in step 6 and choose **Reboot**.
10. Log in to the host after reboot and run this command to validate the SDC is up and running:

    `systemctl status scini.service`
11. For kernel lockdown confirmation, type

    `cat /sys/kernel/security/lockdown`

    Expected output: `[integrity]`

    This indicates kernel lockdown is active due to secure boot.

Enable secure boot for Windows-based compute-only nodes

Use the following procedure to enable secure boot for Windows-based compute-only nodes.

Prerequisites

Ensure that you have configured Dell PowerEdge iDRAC for secure boot.

Steps

1. Log in to the Windows server.
2. Run this command to validate that secure boot is enabled on the server.
   a. Press **Win + R**.
   b. Type **`msinfo32`** and press **Enter**.
      BIOS mode must be **UEFI**.
      Secure Boot State must be **On**.
3. To validate from PowerShell, type **`Confirm-SecureBootUEFI`**
   The output should display **True**.

10

Performing maintenance activities in a PowerFlex cluster

You place a PowerFlex node in maintenance mode to repair, replace, or upgrade hardware components for the customer and management clusters.

Service mode: PowerFlex Manager allows you to place a node into service mode to carry out maintenance tasks. When a node enters service mode, PowerFlex Manager automatically detects if the node is in VMware ESXi or PowerFlex maintenance mode. It then places the node into service mode accordingly and ensures that the associated resource group also transitions into service mode.

Maintenance mode: PowerFlex maintenance mode enables safe execution of hardware or software maintenance tasks on nodes within a PowerFlex cluster, ensuring data availability is preserved and avoiding unnecessary data rebuilds. This mode can be activated from the **Block** tab in PowerFlex Manager when the resource group is in lifecycle mode, allowing the storage node to be placed into maintenance without disrupting cluster operations.

PowerFlex Manager restricts two nodes from being in maintenance mode at the same time.

When the node is in maintenance mode:

- The node is immediately and temporarily removed without building a new copy of the data
- During the maintenance duration, the new writes are written to the rest of the storage node devices
- Degraded reads are used

Once maintenance mode is complete, the system applies only the new writes to the node that was under maintenance. An alert, `STORAGE_NODE_EXCEEDED_IMM_TIMEOUT`, is emitted if the node is in maintenance mode for more than 30 minutes.

Enter maintenance mode

Use this procedure to enter maintenance mode.

Steps

1. Log in to PowerFlex Manager.
2. Click **Lifecycle** > **Storage nodes**.
3. Select the checkbox of the storage node.
4. Click **More actions** > **Enter maintenance mode**.
5. On the **Enter storage node into maintenance mode** page, click **Enter maintenance mode**.
   A message appears that the storage node entered maintenance mode successfully. Click **Dismiss**.

Exit maintenance mode

Use this procedure to exit maintenance mode.

Steps

1. Log in to PowerFlex Manager.
2. Click **Lifecycle** > **Storage nodes**.
3. Select the checkbox of the storage node that is in maintenance mode.
4. Click **More actions** > **Exit maintenance mode**.
5. On the **Exit storage node into maintenance mode** page, click **Exit maintenance mode**.
   A message appears that the storage node exited maintenance mode successfully. Click **Dismiss**.

Enter service mode

Use this procedure to enter and exit service mode.

Steps

1. Log in to PowerFlex Manager.
2. From the menu, click **Lifecycle** > **Resource Groups**.
3. From the **Resource Groups** page, select a resource group.
4. Under **More Actions**, click **Enter Service Mode**.
5. On the **Service Mode** page, select a node to put into maintenance mode on the **Node List** and click **Next**.

   NOTE: Multiple nodes cannot be placed into maintenance mode at the same time.

6. On the **Service Mode** options page, by default, **Maintenance Mode** is selected. Verify that the correct node is selected and click **Finish**.

   PowerFlex Manager displays a yellow warning banner at the top of the **Resource Groups** page. The **Service Mode** page icon displays for the **Deployment State** and **Overall Resource Group Health**, and **Resource Health** for the selected nodes.

   The system is degraded after a rebuild.
7. To exit service mode, click **Actions** > **Exit Service Mode**.

Exit service mode

Use this procedure to exit service mode.

Steps

1. Log in to PowerFlex Manager.
2. From the menu, click **Lifecycle** > **Resource Groups**.
3. From the **Resource Groups** page, select a resource group.
4. Under **More Actions**, click **Exit Service Mode**.

Reboot a PowerFlex node

PowerFlex Manager prevents two nodes from being in service mode simultaneously to protect data.

Prerequisites

Place the PowerFlex node in service mode.

Steps

1. After PowerFlex Manager shows that the node has entered service mode, turn off the PowerFlex node by using the iDRAC interface to run a graceful shutdown.
2. Use the iDRAC interface to power on the PowerFlex node.
3. Exit the node from service mode.

Reboot the management virtual machines

Use this procedure to reboot the management virtual machines.

Steps

1. Open a terminal to the MVM3, and type `kubectl get nodes`

2. To label MVM3 for maintenance, type **`kubectl label node pfmp-mvm03 cmo.maintenance.mode=true`**
3. Drain the node MVM03 where running pods are gracefully evicted from the node. The pods schedule and run on a different node. When the drain process completes, the node reboots. Wait for the node to come back up.

   NOTE: In Linux, if you run two commands joined by && (AND operator) and the first command fails (exits with a non-zero exit code), then the second command is not performed. This behavior is due to short-circuit evaluation in the shell.

4. To drain the node, type **`kubectl drain pfmp-mvm03 --ignore-daemonsets --delete-emptydir-data`**
5. Once the node is drained, to reboot the node, type **`sudo reboot`**
6. SSH to one of the other management virtual machines and type **`watch kubectl get nodes`** to monitor the node that you rebooted to reach a status of **READY**.
7. Once MVM03 reports a **READY** status, type **`kubectl uncordon pfmp-mvm03 ; kubectl label node pfmp-mvm03 cmo.maintenance.mode-`** to uncordon and remove the maintenance label.

   Do not forget to include the DASH symbol. This is required to remove the label from the node.
8. Monitor health from **Monitoring** > **Management Diagnostics**. Wait for all diagnostics to turn green before proceeding to the next management virtual machine.
9. Repeat these steps for MVM02, and then MVM01 if required.

Pausing an inventory job

An inventory job runs daily at midnight. When a PowerFlex management platform or a resource group/devices upgrade is in progress, both jobs compete for resources. The scheduled (nightly) inventory job is automatically disabled at the start of any upgrade and resumes immediately once the upgrade completes minimizing the load.

About this task

In addition to this automatic action on upgrade, you can manually pause scheduled inventory jobs for up to 24 hours using the PowerFlex Manager user interface. This manual block does not effect on-demand inventory actions—you can still run inventory for individual devices through **Lifecycle** > **Resources**.

The status of the inventory job displays on the upper right hand of the screen of the **Resources** page.

Steps

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Inventory Job**.
3. Choose one of the following options:
   - If you want the inventory job to run at the next scheduled time, set the switch to **On**.
   - If you want to pause the inventory job, set the switch to **Off**.

   The name of the user who set the switch, the time it was set, and the time it expires is displayed under the switch.

11

Storage management for PowerFlex management cluster (KVM)

When a PowerFlex management cluster (KVM) is expanded with new management nodes, the procedures in this section can be used to validate the additional nodes, adding new volumes, as well as increasing the size and the corresponding file systems of existing volumes.

The PowerFlex management cluster, hosted in a KVM environment with a minimum of three nodes, contains all management virtual machines. It runs a dedicated instance of PowerFlex Manager for its administration.

Verify the newly added storage data server is a member of the PowerFlex management cluster

Use PowerFlex Manager to verify the status of the newly added management nodes.

Steps

1. Log in to PowerFlex Manager.
2. Go to **Block** > **Storage Systems**.
3. Select the newly added SDS from the list and verify that it is a member of the correct system and protection domain, and provide the correct amount of storage in the **Total Capacity** column.

Add new volumes to a PowerFlex management cluster (KVM)

PowerFlex management cluster (KVM) uses a clustering multi-writer file system on top of mapped PowerFlex devices.

About this task

To add new volumes to a PowerFlex management cluster (KVM), only create new volumes from within the corresponding Resource Group. Adding new volumes using the **Block** tab will not have the same effect and require additional manual steps to complete the process.

Steps

1. Log in to PowerFlex Manager.
2. Click **Lifecycle** > **Resource Groups**.
3. Click the name of the PowerFlex management cluster resource group.
4. Click **Add Resources** > **Add Volumes**.
5. Select **Create New Volumes**, and click **Next**.
6. Click **Add New Volume**.
7. For **Volume Name**, select **Create New Volume** and enter the new volume name and file system name.

   NOTE: The file system name specified here is created as a new path under `/mnt` on the management nodes.

8. Select the storage pool to use.
9. Specify the volume size.
10. Select the volume type: thick or thin.

11. Click **Next**.
12. Review the summary and click **Finish**.

Expand an existing volume in a PowerFlex management cluster (KVM)

To add new volumes to a PowerFlex management cluster (KVM), only grow volumes from within the corresponding resource group.

About this task

Resizing volumes using the **Block** tab will only affect the PowerFlex storage device, and not the corresponding file system. Completing a growth operation from the **Block** tab requires an additional manual step.

Steps

1. Log in to PowerFlex Manager.
2. Click **Lifecycle** > **Resource Groups**.
3. Click the name of the PowerFlex management cluster resource group.
4. In the diagram, click the **Volumes** object.
5. In the resulting **Volume Actions** menu, select **Resize**.
6. Click the **Select Volume** button.
7. Click the name of the volume to expand, and click **Select**.
8. Specify the **New Volume Size**, which must be greater than the current value.
9. Click **Save**.

12

Monitoring system health

Use PowerFlex Manager to monitor system health.

PowerFlex Manager includes the following:

- A dashboard that enables you to see a summary of performance and inventory data of the infrastructure elements and resource groups.
- A **Block** tab that provides a user interface for managing PowerFlex block storage.
- A **Protection** tab that provides a user interface for performing PowerFlex storage protection operations.
- A **Lifecycle** tab that allows you to create and clone templates, and deploy them as resource groups (previously known as services).
- A **Resources** tab that allows you to manage all resource types within the system.
- A **Monitoring** tab that allows you to track events, alerts, and jobs running.
- A **Settings** tab that enables you to configure system settings for PowerFlex Manager.

Monitor system resources

Use PowerFlex Manager to monitor system health.

The following table describes common tasks for monitoring system health and what steps to take in PowerFlex Manager to initiate each:

Table 6. Monitoring system health

If you want to...	Do this in PowerFlex Manager...
Monitor system resources and health	On the **Dashboard**, look at the **Overall Performance**, **Usable Capacity**, and **Data Savings** sections. For information about which resource groups are healthy and in compliance and which are not, look at the **Resource Groups** section.
View alerts	On the menu bar, click **Monitoring** > **Alerts**.
View events	On the menu bar, click **Monitoring** > **Events**.
View jobs	On the menu bar, click **Monitoring** > **Jobs**.
View PowerFlex management pods, nodes, and diagnostics	On the menu bar, click **Monitoring** > **Management diagnostics**.

Management diagnostics

Management diagnostics provide detailed information about the health of PowerFlex management platform pods, nodes, and diagnostics. Only superusers and support users have access to troubleshoot, monitor, and review management diagnostics.

The PowerFlex management platform **Diagnostics** page provides an overview of the latest scheduled run details and manually run diagnostic results.

The **Nodes** page displays the overall health of all the PowerFlex management platform nodes.

The **Pods** page displays the overall health of all the PowerFlex management platform pods. You can restart specified pods from the user interface, copy and download logs from the current pod life cycle or download historical logs from all pod life cycles.

Diagnostics

The **Diagnostics** page displays an overview of the latest run details and diagnostic results.

The following table lists the scheduled frequency of the diagnostics.

Table 7. Scheduled diagnostics

Diagnostic	Description	Scheduled frequency
Pod Status	Monitors the health of all the PowerFlex management platform pods. If there are three consecutive pod failures for a specific pod a failure event is raised in the events log, which usually occurs during upgrades.	1 minute
Node Status	Monitors the health of all the PowerFlex management platform nodes.	2 minutes
Database Health	Monitors the database instance and data replication.	5 minutes
etcd Performance	Evaluates the etcd cluster performance and health under various loads. It provides metrics such as average latency, throughput, and variance. The `etcdctl check perf` command provides a performance measurement and benchmarking tool for etcd clusters. This etcd performance might fail if the cluster components are busy or not fast enough.	N/A
Internal DNS Lookup	Monitors network connectivity through internal DNS queries.	2 minutes
Internal Connectivity	Monitors the internal connectivity between nodes.	5 minutes
Disk Space	Monitors the capacity of the management and operations data under `/opt/platform-provisioner`.	5 minutes
Message Bus Health	Monitors the number of messages in each stream. Warning threshold: 100,000 messages Error threshold: 1 million messages	2 minutes
etcd Certificate	Monitors when one of the nodes secret was created or changed after the node started. The warning threshold is 40 days before the secret was created or changed on one of the nodes. The error threshold is 10 days before the secret was created or changed on one of the nodes.	24 hours

The diagnostic cards display the following details:

- Diagnostic name
- Result icon
- Last run date and time
- Run state

If a diagnostic run receives an error a failure message is displayed.

View the run overview and result details:

- Overview of the run state, start time, and end time.
- Diagnostic results display the result of the latest run.
- If there are errors a failure reason displays.
- Recommendations to remediate failures.
- For more detailed diagnostic result information, click **View Raw Output**.

Optionally, you can edit diagnostic warning and error thresholds and scheduled frequency intervals in the configuration map with *kubectl* or *K9s* command-line tools. After editing the configuration map, restart the Diagnostic Engine pod and the PowerFlex Management Maintenance Manager pod. If the core messaging system is down, only restart the PowerFlex Management Maintenance Manager pod.

NOTE:

- A diagnostic failure event is generated in the events log after a diagnostic fails three consecutive times.
- If the core messaging system for PowerFlex is unavailable, then events and alerts are not available and scheduled diagnostics will not run. You can manually run all diagnostics, except for the etcd performance diagnostic. Historic data will not be available for any of the diagnostics.

Running diagnostics manually

Superusers and support users can manually run the PowerFlex management platform diagnostics to troubleshoot and review their overall status.

Steps

1. Go to **Monitoring** > **Management Diagnostics** > **Diagnostics**.
   The **Diagnostic** page opens which displays PowerFlex management platform diagnostic cards.
2. The result icon, last run date, run time, and run state are listed for each diagnostic card. Click **Run Now** to manually start a diagnostic.

   A notification message confirms that the specified diagnostic has started.

   After a diagnostic run begins, you can cancel it.

   If there are errors, a failure message displays.
3. The overview displays the run state, start time, and end time. The diagnostic result displays failure reasons, recommendations to remediate failures, and raw output only if errors occur. Click **View Raw Output** for more information.

PowerFlex management platform node health

The **Nodes** page displays details of the health of the PowerFlex management platform nodes.

The following table describes the PowerFlex management platform node health details.

Table 8. Node health

Field	Description
Name	Displays the name of the node.
Status	Displays an icon indicating one of the following statuses: • Ready: Traffic can flow through the node • Not ready: Traffic cannot flow through the node because it is failing or unavailable.
IP Address	Displays the IP address of the node.
Disk Pressure	Displays the disk space available for the root file system of the node and image file system. If the disk space is low, then it affects Kubernetes operations such as Kubernetes stops scheduling new pods to the node. The system runs the

Table 8. Node health (continued)

Field	Description
	garbage collector which deletes images which are based on the last time they were used, starting with the oldest. Warnings occur when the disk space for the node or image file systems is less than 10%. Displays an icon indicating that disk pressure is one of the following: ● Greater than 10% ● Less than 10% Critical: Node file system < 5% or Image file system < 5%
Memory Pressure	The node memory pressure uses the default configuration. The pods are evicted when less than 100 MiB are available. Displays an icon indicating the memory pressure is one of the following: ● Greater than 100 MiB is available ● Less than 100 MiB is available
Network Unavailability	Displays an icon indicating the node on the network is one of the following: ● Available ● Unavailable Container Network Interfaces (CNI) are used for cluster networking on the nodes.
PID Pressure	The process ID pressure for nodes measures the ability to start more containers. If the node reaches the specified PID limits, there might be workload failures when trying to obtain a new process ID. Warnings occur when there are less than 10% of available process IDs. Displays an icon indicating that the PID pressure is one of the following: ● Greater than 10% ● Less than 10% Critical available process IDs < 5%

Viewing nodes

Superusers and monitor users can view node details on the **Nodes** page in PowerFlex Manager.

Steps

1. Go to **Monitoring** > **Management Diagnostics** > **Nodes**.
   The **Nodes** page opens which displays a list of PowerFlex management platform nodes.
2. Click **Add filter** by entering case-sensitive free text to control the list of nodes displayed.
3. Select the node that you want to view.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

PowerFlex management platform pod health

The **Pods** page displays the health of the PowerFlex management platform pods.

The following table describes the health details of the PowerFlex management platform pods.

Table 9. Pod health

Name	Description
Pod Name	Displays the name of the pod.
Status	Displays the status of the pod and its containers. The status values are in the pod health status table.
Containers Ready	Displays the number of pod containers that are ready to accept traffic.
Namespace	Isolates groups within a single cluster. The primary namespace for pods is PowerFlex.
Node	Displays the name of the node.
Container Restarts	Displays the number of pod container restarts.
Age	Displays the age of the pod in days, hours, minutes, and seconds.

The following table describes the pod health status values that might be displayed.

Table 10. Pods health status

Icon	Name	Definition
	Running	All the pod containers were created and at least one pod container is running, starting, or restarting.
	Succeeded	All containers within the pod finished their tasks successfully.
	Completed	All pod containers completed their tasks.
	Container creating	The pod is scheduled on a node, and the kubelet on that node is creating the pod containers.
	Pending	The Kubernetes system accepted the pod, but one or more of its containers is pending.
	Terminating	The pod is terminating.
	OOMKilled	The Out-Of-Memory Killed terminated one or more of the pod containers.
	Error	One or more of the pod containers has an issue and failed to start or run correctly.
	Container cannot run	The container is unable to start or run properly.
	Image pull back off	Kubernetes is encountering problems because of incorrect credentials or incorrect container names.
	Create container error	Kubernetes is encountering problems creating containers.

Table 10. Pods health status (continued)

Icon	Name	Definition
	Create container config error	Kubernetes is encountering problems creating containers.
	Failed	A minimum of one pod container terminated with a non-zero exit code.
	Err image pull	An error happened when Kubernetes tried to pull the specified container image from the registry.
	Crash loop back off	One or more pod containers are repeatedly failing to start and are stuck in a restart loop.
	Evicted	One or more pods terminate because the node is running out of resources.
	Deadline exceeded	The pod operation took longer than the specified deadline and was terminated.
	Preempted	The pod was preempted to make room for other high-priority pods.
	Unknown	The state of the pod is unknown.

Restarting and collecting pod logs

Superusers and monitor users can restart pods, collect logs, and view pod details on the Pods page in PowerFlex Manager.

Steps

1. Go to **Monitoring** > **Management Diagnostics** > **Pods**.
   The **Pods** page opens which displays a list of PowerFlex management platform pods.
2. Click **Add filter** to control the list of pods displayed.
3. Enter case-sensitive free text and add additional filters by selecting **Status**, **Namespace**, or **Node Name**.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

4. Select the pod.
5. Restart one of the following pods from the user interface:
   - alcm
   - alert-service
   - asm-manager
   - asmui
   - block-legacy-gateway
   - block-presentation-server
   - cloud-iq-collector
   - historical-metrics-calculator
   - mdm-events-extractor
   - metrics-gateway
   - notifier
   - snmp-listener
   - sso
   - thin-deployer

   Click **Restart**, then click **Approve** to confirm.

   If a pod fails to restart, review the pod status.

   The remaining pods can be restarted using the REST API.

6. Click **Log Collection**.
7. Select one of the following log collection types:
   - **Current pod lifecycle**

     Click **Run**, then select **Copy** or **Download** to review the logs. The default maximum log size is 10 MiB, which can be edited in kubectl.
   - **Full historical logs**

     Click **Download** to review the ZIP file. The historical logs include all pod life cycles.

     NOTE: After downloading the historical logs, there might be empty log directories, which usually occur after system upgrades or when pod logs are unavailable in the system.

Manage compliance

Use PowerFlex Manager to manage software and firmware IC compliance.

The following table describes common tasks and what steps to take in PowerFlex Manager to initiate each:

Table 11. Common PowerFlex Manager tasks

If you want to...	Do this in PowerFlex Manager...
Monitor software and firmware compliance	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, click **View Compliance Report**.
Perform software and firmware remediation	1. From the compliance report, view the firmware or software components. 2. Click **Update Resources** to update non-compliant resources.
Generate a troubleshooting bundle	1. Click **Settings** > **Serviceability**. 2. Click **Generate Troubleshooting Bundle**. **NOTE:** You can also generate the troubleshooting bundle from the **Resource Group** page. Click **Lifecycle** and select **Resource Group**. Click **Generate Troubleshooting Bundle**.
Download a report that lists compliance details for all resources	1. Click **Lifecycle** > **Resources**. 2. Click **Export Report** and click **Export Compliance PDF Report** or **Export Compliance CSV Report**.
Download a configuration report	1. Click **Lifecycle** > **Resources**. 2. Click **Export Report** > **Export Configuration PDF Report**.

Deploying and configuring Dell Technologies connectivity services for the PowerFlex management cluster (KVM)

There are two options to configure events and alerts:

- Connect directly
- Connect using secure connect gateway
- If you connect directly, only the call home option is available
- If you connect through secure connect gateway, all options through secure connect gateway are enabled

Download and launch the PowerFlex management platform deployment wizard

Use this procedure to download and launch the PowerFlex management platform deployment wizard.

Steps

1. Download the PowerFlex management platform deployment wizard package, go to *PowerFlex appliance Intelligent Catalog Software* on the Dell Technologies support site.
2. Copy the package ZIP file to a local jump host that can access the cluster that is being installed.

   Ensure a reliable network connection for image transfers.
3. Unzip the package:

Table 12. Operating systems available

Windows	Linux
Double-click `powerflex-pfmp-deploy-WIN.bat`.	a. Open terminal and navigate to the application folder. b. Run `./powerflex-pfmp-deploy-LINUX.sh`.

Deploy and configure secure connect gateway

Secure connect gateway is an enterprise monitoring technology that monitors your devices and proactively detects hardware issues that may occur.

Prerequisites

Ensure your system complies to the minimum requirements, for more information, see the *Secure Connect Gateway 5.x - Virtual Edition Deployment Guide*.

Steps

Use the PowerFlex management platform deployment wizard. See the help document in the deployment wizard for more information.

Configuring the initial setup and generating the access key and pin

Use the section to generate the access key and pin to register with secure connect gateway and Dell Technologies support site.

Use the following KB article to generate the Dell Technologies Support account and access key and pin: Generate Access Key and PIN for Secure Connect Gateway 5.x.

Customers should work with Dell Technologies Support to get the site ID that is required while generating the access key and pin.

Related information

Configure Dell Technologies connectivity services using the connect directly mode

Register and log in to the secure connect gateway user interface

Use this procedure to log in to the secure connect gateway user interface.

Steps

To log in to the secure connect gateway, see *Registering and signing in to Secure Connect Gateway* in the Secure Connect Gateway 5.x — Virtual Edition Deployment Guide.

Ensure you complete the following procedures:

- *Sign in to Secure Connect Gateway*

- *Register Secure Connect Gateway*
- *Sign in using administrator account or network credentials*

Deploying and configuring policy manager for secure connect gateway

Policy manager for secure connect gateway is a device access management technology that is delivered as a virtual appliance. Policy manager for secure connect gateway can be deployed on a hypervisor.

You can configure policy manager for secure connect gateway to perform the following tasks:
- Control remote access to your devices.
- Maintain an audit log of remote connections and file transfers.
- Access administration actions performed on policy manager for secure connect gateway.

Minimum requirements to deploy and use policy manager for secure connect gateway

The following sections provide information about:

- Minimum system and network requirements for the local system to deploy policy manager.
- Browsers that can be used to access the policy manager user interface.
- Hypervisors that can be used to deploy policy manager.

System requirements

The system requirements to deploy and use policy manager are:

- Number of processor cores—4
- Installed memory (RAM)—8 GB
- Hard drive space—120 GB—Thin provisioning

Deploy policy manager for secure connect gateway

Use this procedure to deploy policy manager for secure connect gateway.

About this task

NOTE: Policy manager for secure connect gateway is optional.

PowerFlex Manager does not manage policies or policy manager for secure connect gateway.

This chapter does not cover procedures to create and manage policies. For information on creating and managing policies, see *Policy Manager for Secure Connect Gateway 5.00.00.xy User's Guide*.

Prerequisites

Before deploying policy manager for secure connect gateway, ensure the following:
- Upgrade secure connect gateway
- Download the latest version of policy manager for secure connect gateway from Dell Technologies support site.

Steps

1. Use the PowerFlex management platform deployment wizard to deploy policy manager. See the help document in the deployment wizard for more information.
2. Configure policy manager using the following steps:
   a. Go to **`https://<hostname(FQDN) or IP address>:8443`**
   b. Enter admin as the user ID.
   c. If you are logging in for the first time, enter the default password.

d. After logging in for the first time, enter the new administrator account password when prompted and click **Save**.
e. Enter the new password and click **Log In**.

Deploy and configure secure connect gateway and policy manager using QCOW2 images

Use this procedure to deploy and configure secure connect gateway and policy manager using QCOW2 images.

Steps

1. Download the policy manager and secure connect gateway Hyper-V VHD files from the Dell Technologies support site.
2. Log in to the KVM host as **delladmin**.
3. Download the VHD files to the KVM host (to `/tmp`).
4. Unzip the secure connect gateway VHD image, type: **unzip <filename>**
5. Convert the VHD image to a QCOW2 image, type: **qemu-img convert -f vpc -O qcow2 <filename> scgvm.qcow2**
6. Unzip the POLICYMGR VHD image, type: **unzip <filename>**
7. Convert the VHD image to a QCOW2 image, type: **qemu-img convert -f vpc -O qcow2 <filename> policymgrvm.qcow2**
8. Move the resulting QCOW2 images to `/mnt/general`, type: **mv *.qcow2 /mnt/general**
9. Create VMs, type: **virt-install**:

```
sudo virt-install --name scgvm --ram 16384 --disk /mnt/general/scgvm.qcow2 --import
--vcpus=4 --network bridge=bridge-<vlan>,model=virtio --osinfo sle15 --noautoconsole
--print-xml --graphics vnc,listen=<PMC host mgmt IP>,port=5901 > /tmp/scgvm.xml

sudo virt-install --name policymgr --ram 16384 --disk /mnt/general/policymgrvm.qcow2
--import --vcpus=4 --network bridge=bridge-<vlan>,model=virtio --osinfo sle15 --
noautoconsole --print-xml --graphics vnc,listen=<PMC host mgmt IP>,port=5902 > /tmp/
scgvm.xml
```

10. Add crm configuration to instantiate both VMs as cluster objects, type:

```
sudo crm configure primitive scgvm VirtualDomain params config="/mnt/general/
scgvm.xml" remoteuri="qemu+ssh://hacluster@%n/system" start_resource=true meta allow-
migrate=true target-role=Stopped op monitor interval=10s timeout=30s

sudo crm configure colocation co-general-scgvm info: scgvm cl-sharedfs-general

sudo crm configure order or-general-scgvm Mandatory: cl-sharedfs-general scgvm

sudo crm configure primitive policymgrvm VirtualDomain params config="/mnt/general/
policymgrvm.xml" remoteuri="qemu+ssh://hacluster@%n/system" start_resource=true meta
allow-migrate=true target-role=Stopped op monitor interval=10s timeout=30s

sudo crm configure colocation co-general-policymgrvm info: policymgrvm cl-sharedfs-
general

sudo crm configure order or-general-policymgrvm Mandatory: cl-sharedfs-general
policymgrvm
```

11. Start the resources, type: **sudo crm resource start scgvm policymgrvm**
12. Temporarily add the firewall rule to host to allow the VNC communication, type: **sudo firewall-cmd --add-port=5901/tcp --add-port=5902/tcp**
13. Use the VNCviewer from the jump server (or laptop connected to service port) to connect to the console of scgvm, see Secure Connect Gateway 5.x — Virtual Edition Deployment Guide for the secure connect gateway and policy manager deployment procedures.
14. Clean up the firewall, type: **sudo firewall-cmd --remove-port=5901/tcp --remove-port=5902/tcp**
15. Clean up the downloaded files (`zip` and `VHD` files), type: **rm ~/POLICYMGR_* ; rm ~/SCG_***

Configure the secure connect gateway and policy manager on KVM VMs

You must set up the initial configuration on the secure connect gateway and policy manager VMs in order to be able to log in to the PowerFlex Manager user interface to configure additional settings.

Steps

1. Connect to the jump server and launch a console.
2. To display the list of VMs running on the host in the terminal, enter: **`virt-viewer -a -c qemu+ssh://delladmin@<KVM Host>/system`**.
3. Select the **SCG / PM VM** to launch the console.
4. To agree the terms, press **ALT** + **A**. Press **ALT** + **N** for next.
5. To edit the network settings, press **ALT** + **I** and complete the following steps for the secure connect gateway VM and the policy manager VM:
   a. Enter the **IP Address** and **Subnet Mask**, press **ALT** + **N** for next.
   b. Press **ALT** + **S** and enter the **Hostname**, **DNS**, and **Domain Search**. Press **ALT** + **N** for next.
   c. Press **ALT** + **U** for routing and press **ALT**+**D** to add the route.
   d. Enter the **Gateway** and select **Device** as **eth0**, press **ALT** + **O**, and then **ALT** + **N** for next.
   e. Select the time zone and press **ALT** + **N** for next.
   f. Enter the root password and press **ALT** + **N** for next.
   g. Retain the SSL as default and press **ALT** + **N** for next.
   h. **NOTE:** This step is only required for policy manager.

      Retain the default user admin settings and press **ALT** + **N** for next.
6. Wait for the configuration to complete.
7. To configure policy manager, repeat steps 3 to 6.

Results

Once the configuration is complete, access the PowerFlex Manager for further configuration.

Configure policy manager settings on secure connect gateway

Policy manager for secure connect gateway is a device access management technology that is delivered as a virtual appliance. Policy manager for secure connect gateway can be deployed on a hypervisor.

About this task

You can configure policy manager for secure connect gateway to perform the following tasks:

- Control remote access to your devices.
- Maintain an audit log of remote connections and file transfers.
- Access administration actions performed on policy manager for secure connect gateway.

NOTE: Policy manager is not managed by PowerFlex Manager.

Steps

1. Sign in to secure connect gateway.
2. Go to **Settings** > **Environment configuration** > **Connectivity details** > **Policy Manager**.
3. Click **Enable remote Policy manager**.
4. Enter the hostname or IP address, port number, username, and password of the policy manager server.

   NOTE: If the port is SSL secured, the port number must be **`8443`**. If the port is not SSL secured, the port number must be **`8888`**.
5. Click **Enable SSL** if policy manager for secure connect gateway is installed on a server that is secured by SSL.
6. If the policy manager server connects to the Internet through a proxy server, perform the following steps in the **Customer proxy server** section:

a. Click **Enable proxy server for Policy manager only**.
b. Enter the hostname or IP address and port number.
c. Click **Proxy requires authentication** and enter the username and password to access the proxy server.

7. Test the connection and click **Apply**.

Configuring Dell Technologies connectivity services on PowerFlex Manager

Depending on the customers requirement, use the following procedures to configure the connect directly or connect using the secure connect gateway. In order to allow remote support, use secure connect gateway. Remote support is not be enabled for direct connect.

Configure Dell Technologies connectivity services using the connect directly mode

Use this procedure to enable Dell Technologies connectivity services using the connect directly mode.

Steps

1. Log in to PowerFlex Manager.
2. Click **Settings** > **Events and alerts**.
3. Click **Notification Policies**.
4. On the **Destinations** pane, click **Dell Technologies connectivity services**.
5. Accept the license and telemetry agreement on the connect support assist page and click **Next**.
6. Choose the connection type **Connect Directly**.

   NOTE: This helps us directly connect to Dell Technologies connectivity services direct. Call to home feature works on connect direct. The proxy setting is not supported.

7. Click **Connect to Dell AIOps**.

   It enables PowerFlex Manager to transport telemetry data, alerts and analytics to assist Dell Technologies in providing support.
8. On the **Authentication details** page, provide the following details.
9. Access key and PIN, see Generate Access Key and PIN for Secure Connect Gateway 5.x.

   You need the software ID for generating the access key and PIN.
10. Choose the **Device type** to be registered like rack, appliance or software.
11. In the **Enterprise License Management Systems** file, enter the software ID used in step 2 while generating the access key and pin.
12. The **Solution serial number** must be provided by customer.
13. In the **Site ID** field, provide the site ID location. If you do not have one, contact Dell Technologies Support to generate one.
14. Click **Next**, provide the contact details for customer, and click **Finish**.

    A window appears on the bottom of the screen configuring Dell Technologies connectivity services and another window appears once it is successfully configured.
15. To activate the policy, click **Configure Now** and enable the policy by making it active.

    Once the policy is active, the policy will move from grayed out mode to available and active mode.

Related information

Configuring the initial setup and generating the access key and pin

Connect Dell Technologies connectivity services using the secure connect gateway

Use this procedure to enable Dell Technologies connectivity services using the secure connect gateway.

Prerequisites

Configure the secure connect gateway.

Steps

1. Log in to PowerFlex Manager.
2. Click **Settings** > **Events and alerts** > **Notification Policies**.
3. Under **Destinations**, click **Dell Technologies connectivity services**.
4. On the **Overview** screen, click **Next**.
5. On the **Telemetry Agreement** screen, select the checkbox to accept the telemetry agreement, and click **Next**.
6. Under **Connection Type**, click **Connect via Gateway Server** and provide the primary gateway IP address or **Primary Gateway IP/Hostname** and **Gateway Port**. Click **Next**.

   NOTE: Connect using the gateway helps register PowerFlex Manager on secure connect gateway and Dell Technologies connectivity services. From here we can enable the proxy setting.

7. Provide the **SCG IP address and Port** number.

   NOTE: Maintain the gateway port as 9443. Port 5700 port is only used for the secure connect gateway UI, API communication with secure connect gateway from the PowerFlex management platform uses port 9443.

8. Click **Connect to Dell AIOPs**.

   It enables PowerFlex Manager to transport telemetry data, alerts, and analytics to assist Dell Technologies in providing support.
9. Enable the **Remote Support** button and click **Next**.
10. On the **Authentication Details** page, provide the following details.
11. Enter the access key and pin gathered during the initial setup. See Generate Access Key and Pin.
12. Choose the **Device** type to register like rack.
13. In the **Enterprise License Management Systems** file, enter the software ID used in while generating the access key and PIN.
14. The **Solution serial number** must be provided by the customer.
15. In the **Site ID** field, provide the site ID location. If you do not have one, contact Dell Technologies Support to generate one.
16. Click **Next**, provide the contact details for the customer, and click **Finish**.

    A popup appears on the bottom of the screen configuring Dell Technologies connectivity services and another pop up appears once it is successfully configured.
17. To activate the policy now, click **Configure Now** and enable the policy by making it active.

    Once the policy is active, it will remove from grayed out mode to available and active mode.
18. To verify the registration, check the PowerFlex management platform log to monitor registration status, type: **`kubectl get supportassist -n powerflex -o yaml`**

    Example of a successful registration:

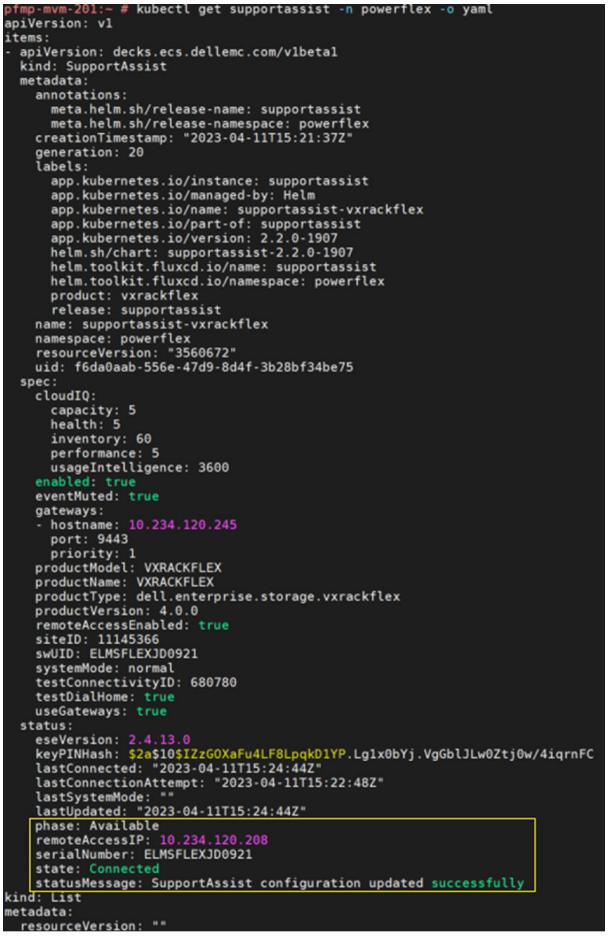


Figure 1. Successful registration of the PowerFlex management platform

Events and alerts

A source is used to configure the receiving of external events and syslog content.

A destination is used to configure the ability to send events and alerts information out. A destination is always external.

Dell Technologies connectivity services, email, SNMP, webhooks, remote syslog are considered destinations.

Notification policies define what information is sent to each destination. Events and alerts exist irrespective of whether notification policies are created.

SNMP sources are not automatically discovered and must be configured to receive events about these sources. PowerFlex Manager is preconfigured and events, and alerts are automatically available. Resources in the PowerFlex appliance are automatically discovered. Any future resources, for example, switch replacements or additional nodes, are considered external and must be added manually as sources.

PowerFlex Manager enables you to register a notification receiver for pushing alert and event messages through webhooks to a target system,for example, BigPanda.

Configure an external source

You must define a source to enable PowerFlex Manager to receive an external event.

About this task

A syslog source can only go to a syslog destination and does not display in events. An SNMP source, either V2 or V3, displays in events even without a defined notification policy.

For a webhook, the resources are discovered in PowerFlex Manager, and the load balancer IP address needs to be configured on the resources individually. The alerts from individual resources are sent to Webhooks from the source resource type.

NOTE: To receive SNMPv3 traps on an external NMS, utilize the SNMP engine ID `0x80001f8804706f776572666c6578`.

SNMPv2 and SNMPv3 are supported for all resource types like switches, CloudLink, and iDRAC.

Steps

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Sources** pane, click **Add**.
3. Enter a source name and description.
4. Configure either SNMP or syslog forwarding and click **Submit** > **Dismiss**:
   - For SNMPv2c:
     a. Enter the community string by which the source forwards traps to destinations.
     b. Enter the same community string for the configured resource. During discovery, if you selected PowerFlex Manager to automatically configure iDRAC nodes to send alerts to PowerFlex Manager, enter the community string that is used in that credential here.
   - For SNMPv3:
     a. Enter the username, which identifies the ID where traps are forwarded on the network management system.
     b. Select a security level from the following:

Table 13. Security levels and passwords

Security level	Description	authPassword	privPassword
Minimum	noAuthNoPriv	Not required	Not required
Moderate	authNoPriv	Required	Not required
Maximum	authPriv	Required	Required

   - If you select Syslog, click **Enable Syslog**.

Configure a destination

Define a location where event and alert data that has been processed by PowerFlex Manager should be sent.

Steps

1. Click **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Destinations** pane, click **Add**.
3. From the **Destinations** page:
   a. Enter the destination name and description.
   b. From **Destination Type** menu, select to configure either SNMP, Syslog, or email (SMTP) forwarding.
   c. Click **Next**.
   d. Depending on the destination type, enter the following information:

Table 14. Destination types and protocol settings

Destination type	Protocol settings
SNMP V2c	• Network name/IP address • Port • Community string
SNMP V3	• Network name/IP address • Port • Username • Security level: ○ Minimal - no more information required

Table 14. Destination types and protocol settings (continued)

Destination type	Protocol settings
	○ Moderate - MD5 authentication password required ○ Maximum - MD5 authentication and DES privacy passwords required
Syslog	● Network name/IP address ● Port ● Protocol: ○ UDP ○ TCP ● Facility: ○ All ○ Authentication ○ Security and authentication
Email (SMTP)	● Destination name ● Description ● Destination type: ○ Server type: ■ SMTP ■ SMTP over SSL ■ SMTPS SMARTTLS ○ Server IP or FQDN ○ Port ○ Sender address and up to five recipient addresses ○ If you choose credentials, enter: ■ Username, password, sender address, and up to five recipient addresses ● Send test email ● Test email server connection
Webhook	● Description ○ Destination name ● Destination Type ● Select **Webhook** (You can have up to three webhook destinations.) ● Webhook destination API URL ○ When using an https Webhook destination, an SSL certificate is being used. Use a DNS or FQDN that matches what the SSL certificate is looking for in the Webhook Destination API URL. ○ For example, if you want to configure a webhook destination for BigPanda, you need to specify the BigPanda endpoint here. ○ You can get this URL from the BigPanda site. ● Enable **Credentials**. ● Credentials ○ For example, for BigPanda, the credential you use must have an app key and a token string. You can generate these values on the BigPanda site and then copy them to the app key and token fields. ● Test **Webhook**.

4. Click **Finish**.

Managing events and alerts

Use this section to manage events and alerts.

Modifying an external source

You can edit the information about how PowerFlex Manager receives an event.

Steps

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Sources** pane, click the source that you want to modify.
   The **Edit Source** window opens.
3. Edit the information and click **Submit**.

Modifying a destination

You can edit the information about where event and alert data that is processed by PowerFlex Manager should be sent.

Steps

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Destinations** pane, click the destination whose information you want to modify.
   The **Edit Source** window opens.
3. Edit the information and click **Submit**.

Add a notification policy

When you add a notification policy, you define the rules for processing events or alerts from sources, and to which destinations that information should be sent.

Steps

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Click **Create New Policy**.
3. Enter a name and a description for the notification policy.
4. From the **Source Type** menu, select how you want events and alerts to be received. The source type options are:
   - **Snmpv2c**
   - **Snmpv3**
   - **Syslog**
   - **Powerflex**
   - **Auditlog**

   PowerFlex related alerts are generated and forwarded to a webhook destination when the notification policy is configured with the **Source Type** set to **Powerflex**.
5. From the **Resource Domain** menu, select the resource domain that you want to add a notification policy to. The resource domain options are:
   - **All**
   - **Management**
   - **Block (Storage)**
   - **Compute (Servers, Operating Systems, virtualization)**
   - **Network (Switches, connectivity etc.)**
   - **Security (RBAC, certificates, CloudLink etc.)**
6. Select the checkbox beside the severity levels that you want to associate with this policy. The severity indicates the risk (if any) to the system, in relation to the changes that generated the event message.

7. Select the required destination. You can choose one or more destinations. For a webhook, you can have up to three destinations defined. For audit logs, you can configure up to four destinations.

   NOTE: The audit log policy is disabled by default. Click **Configure Now** to configure and activate the audit log policy.

8. Click **Submit**.

Next steps

You can send a test event to PowerFlex Manager for a specific or multiple resources by completing the following steps:

1. Select the policy for which you want to send an event.
2. Go to **More Actions** > **Configure to send events in PowerFlex Manager**.
3. Search by text or filter for the resource that you want to send a test event for. The types of filter options available are: **Element Manager**, **Node (Hardware/Software Management)**, and **Switches**.
4. Select one or more checkboxes beside one or more resources that you want to send a test event(s) for from the **Available Resources** table and move them to the **Selected Resources** table.
5. Click **Send Test Event**.

After adding a notification policy, you might need to perform additional configuration steps. For example, If you are setting up a notification policy for a webhook destination that uses BigPanda, you can optionally configure BigPanda to show the severity levels from PowerFlex Manager. To do this, you must configure the status mapping on the BigPanda site. Map the `major` and `minor` severity values from PowerFlex Manager to the `Warning` status for BigPanda.

Configuring SNMP on the resources for webhook

For the SNMP alerts to be forwarded to such a site like BigPanda, the load balancer IP address of the management virtual machine is configured on each of the resources individually.

The resources like CloudLink, iDRAC, and the switches should be discovered and managed in PowerFlex Manager.

SNMPv2 and SNMPv3 are supported for all resource types like CloudLink, iDRAC, and switches.

Identify the SNMP load balancer IP address

Use this procedure to identify the SNMP load balancer IP address on the management virtual machines.

Steps

To identify the load balancer IP address on the management virtual machine, type: **`kubectl get svc -A | grep -i snmp`**.

Example output:

```
delladmin@tb-prdmvm3:~> kubectl get svc -A | grep -i snmp
powerflex       snmp-listener                                LoadBalancer
10.43.1.254   10.118.146.149    162:32306/UDP,514:31860/UDP,514:31809/TCP      48d
powerflex       snmp-listener-net160                         LoadBalancer
10.43.0.115   192.168.161.214   162:30164/UDP,514:31007/UDP,514:31007/TCP      33d
powerflex       snmp-listener-net162                         LoadBalancer
10.43.0.39    192.168.163.214   162:30546/UDP,514:31747/UDP,514:31747/TCP      33d
```

NOTE: If an OOB SNMP load balancer IP address is not listed, see KB article: PowerFlex 4.0.1 SNMP or syslog alerts not working due to missing OOB load balancer IP.

Configure SNMP on CloudLink

Use this procedure to configure SNMP on CloudLink.

Prerequisites

Ensure the following:

- CloudLink is discovered in PowerFlex Manager.

- An external source is configured.

Steps

1. Discover CloudLink on **Resources** page.
2. Log in to CloudLink.
3. Click **Server** > **SNMP** and click **Add**.
4. On the **Add NEW SNMP** configuration window:
   a. Select the **Target Version** as **`SNMPv2`** or **`SNMPv3`**.
   b. On the **Host** field, add the **Load Balancer IP** to forward SNMP traps from CloudLink to PowerFlex appliance.
   c. The port number to be 162.
   d. Add the details as description.
   e. Add the community string, enter the same community string used when creating SNMPv2 source on the PowerFlex management platform appliance, for example: **`public`**.
   f. Click **Add**.
5. Click **Send Test Trap**.
   A test critical/warning alert is sent to PowerFlex Manager.

Related information

Configure SNMP on iDRAC

Configure SNMP on iDRAC

Use this procedure to configure SNMP on iDRAC.

Prerequisites

Ensure the following:

- iDRAC is discovered in PowerFlex Manager.
- An external source is configured.

Steps

1. Log in to iDRAC.
2. Click **Configuration**.
3. Select **System Settings** > **Alert Configuration** > **Alerts** and enable **Alerts**.
4. Expand **Quick alerts Configuration**, select the options as it required to be forwarded to PowerFlex Manager.
   a. Select **Alert Categories**.
   b. Select **Issue Severities**.
   c. Select **SNMP Trap for notification type**.
   d. Click **Apply**.
5. Under **SNMP Traps Configuration**, complete the following:
   a. Select the **Alert Destination** and in the Destination Address field, update the **Load Balancer IP**.
   b. For SNMPv3 traps, choose the user for **SNMP V3 User** under dropdown menu and click **Apply**.

   NOTE: SNMPv3 User must be configured on iDRAC before performing SNMPv3 alert forwarding test.

6. Under **SNMP Settings**, complete the following:
   a. On the **Community String** field, enter the same community string used when creating SNMPv2 source on PowerFlex management platform appliance.
   b. The port number to be 162.
   c. Change the **SNMP Trap format** to **SNMP V2** or **SNMP V3**.
   d. Click **Apply**.

Related information

Configure SNMP on CloudLink

Configure SNMPv2 on a switch

Use this procedure to configure SNMPv2 on a Cisco Nexus or a Dell PowerSwitch switch.

Prerequisites

Ensure the following:

- iDRAC is discovered in PowerFlex Manager.
- An external source is configured.

Steps

Configure SNMPv2:

- On a Cisco Nexus switch:
  a. SSH to the Cisco Nexus switch.
  b. Enable and configure SNMPv2c traps, type the following:

     ```
     # configure terminal
     ```

     ```
     # snmp-server host <Load balancer IP of PFXM> traps version 2c <community string>
     ```

     NOTE: Enter the same community string used when creating SNMPv2 source on PowerFlex management platform, for example: **public**, for example: **# snmp-server host 1.1.1.1 traps version 2c public**

  c. Save the configuration, type the following:

     ```
     # copy running-config startup-config
     ```

  d. Repeat the steps for all of the Cisco Nexus switches.
  e. Run the following command to confirm: **show snmp host**.
- On a Dell PowerSwitch:
  a. SSH to the Dell OS10 Dell PowerSwitch.
  b. Enable and configure SNMPv2c traps, type the following:

     ```
     # configure terminal
     ```

     ```
     # snmp-server host <Load balancer IP of PFXM> traps version 2c <community string>
     ```

     NOTE: Enter the same community string used when creating SNMPv2 source on PowerFlex management platform, for example: **public**, for example: **# snmp-server host 1.1.1.1 traps version 2c public**

  c. Save the configuration, type the following:

     ```
     # copy running-config startup-config
     ```

  d. Repeat the steps for all of the Dell PowerSwitch switches.

Configure SNMPv3 on a switch

Use this procedure to configure SNMPv3 on a Cisco Nexus or a Dell PowerSwitch switch.

Prerequisites

Ensure the following:

- iDRAC is discovered in PowerFlex Manager.
- An external source is configured.

Steps

Configure SNMPv3:

- On a Cisco Nexus switch:
  a. SSH to the Cisco Nexus switch.
  b. Enable and configure SNMPv3 traps, type the following:

     ```
     # configure terminal
     ```

```
# snmp-server user <username> network-admin auth sha <auth-password> priv aes-128
<priv-password>

# snmp-server host <Load balancer IP of PFXM> traps version 3 priv <username>
```

NOTE: Enter the same SNMPv3 username, authentication protocol, authentication password, privacy protocol, and privacy password used when creating the SNMPv3 source on the PowerFlex management platform, for example: **public**, for example:

```
# snmp-server user pfxm network-admin auth sha AuthPass123 priv aes-128 PrivPass123

# snmp-server host 1.1.1.1 traps version 3 priv pfxm
```

c. Save the configuration, type the following:

```
# copy running-config startup-config
```

d. Repeat the steps for all of the Cisco Nexus switches.

e. Run the following command to confirm:

```
show snmp user

show snmp host
```

- On a Dell PowerSwitch:

a. SSH to the Dell OS10 Dell PowerSwitch.

b. Enable and configure SNMPv3 traps, type the following:

```
# configure terminal

# snmp-server group <group-name> v3 priv

# snmp-server user <username> group <group-name> auth sha <auth-password> priv AES-128
<priv-password>

# snmp-server enable traps

# snmp-server host <Load balancer IP of PFXM> community <username>
```

NOTE: Enter the same SNMPv3 username, authentication protocol, authentication password, privacy protocol, and privacy password used when creating the SNMPv3 source on PowerFlex management platform, for example: **public**, for example:

```
# snmp-server group PFXM-GROUP v3 priv

# snmp-server user pfxm group PFXM-GROUP auth sha AuthPass123 priv AES-128
PrivPass123

# snmp-server enable traps

# snmp-server host 1.1.1.1 community pfxm
```

c. Repeat the steps for all of the Dell PowerSwitch switches.

d. Confirm the configuration, type the following:

```
# show snmp user
```

NOTE: Dell recommends SNMPv3 for secure monitoring because it provides authentication and encryption. The SNMPv3 parameters configured on the switch must match those configured on the PowerFlex management platform.

Modify a notification policy

You can modify certain settings that are associated with a notification policy.

About this task

You cannot modify the source type or destination once it is assigned to a notification policy.

Steps

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Select the notification policy that you want to modify.
3. You can choose to modify the notification policy in the following ways:
   - To activate or deactivate the policy, click **Active**.
   - To modify the policy, click **Modify**. The **Edit Notification Policy** window opens.
4. Click **Submit**.

Delete a notification policy

Use this procedure to delete a notification policy.

About this task

NOTE: Once a notification policy is deleted, it cannot be recovered.

Steps

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Select the notification policy that you want to delete.
3. Click **Delete**.
   You receive a message to confirm if you want to delete the policy.
4. Click **Submit** and click **Dismiss**.

CloudLink Center events and alerts

View CloudLink Center details in PowerFlex Manager

View CloudLink Center resource details.

Steps

1. From the menu bar, click **Lifecycle** > **Resources**.
2. From the **Resources** tab, click the row containing CloudLink Center and click **View Details**.
   The **Details** page displays detailed information about the CloudLink Center resource on the following tabs:
   - Summary: Displays cluster details, system performance, alarms, pending machines, and security events. If CloudLink has clustered CloudLink Centers, PowerFlex Manager shows all CloudLink Centers.
   - Machines: Displays machine groups and machines.
   - Approved networks: Displays a list of the names and IP addresses for all approved networks going in and out of the CloudLink Center.

   The **Details** page displays information that is similar to information provided in the CloudLink application. All the data is based on the last inventory, as indicated by the Last Inventory timestamp shown on the left side of the **Details** page under **Resource Information**. Whenever the inventory is updated, either automatically or manually, any changes that are made in the CloudLink Center are reflected in the **Details** page.
3. To view additional details, click **Launch CloudLink Center Console** to launch the CloudLink application.

View CloudLink Center actions

View the actions initiated by users, such as uploading or assigning licenses, accepting a pending machine, or setting the CloudLink Vault mode.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Actions**.
   You can view the actions that happened in the last 10 minutes, 30 minutes, 1 hour or more.

View CloudLink Center events

View the internal activity in the system, as well as activity related to actions and alarms.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Events**.
   You can view the all the events or events that happened in the last 10 minutes, 30 minutes, or 1 hour.

View CloudLink Center security events

View the CloudLink Center security events that happened in the last 10, 30 minutes, one hour or view all the security events.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Security Events**.

View CloudLink Center alarms

An alarm represents a state or condition. When one or more alarm conditions exist, CloudLink Center displays a badge on the Alarms icon in the home page. The badge indicates the number of alarms.

Steps

1. Log in to CloudLink Center.
2. From the home page, click the bell icon or click **Monitoring** > **Alarms**.
   You can view the all the events or events that happened in the last 10 minutes, 30 minutes, or 1 hour.

13

Network management

This section describes common procedures to administer the network in a PowerFlex appliance environment using either CLI or PowerFlex Manager.

For information relating to SONiC PNA, see *Configuring Dell PowerSwitch switches using SONiC networking* under *Optional deployment tasks*.

Networking pre-requisites

Configure the customer network for routing and layer-2 access for the various networks before PowerFlex Manager deploys the PowerFlex appliance cluster.

The pre-deployment customer network requirements are as follows:

- Redundant connections to access switches using virtual link trunking (VLT) or virtual port channel (VPC).
- MTU=9216 on all ports or link aggregation interfaces carrying PowerFlex data VLANs.
- MTU=9216 as default on VMware vSphere vMotion and PowerFlex data interfaces.

The following table lists customer network pre-deployment VLAN configuration options:

- **Example VLAN**: Lists the VLANs that are used in the PowerFlex appliance deployment.
- **Example multi-VLAN**: Lists the multi-VLANs that are used in the PowerFlex appliance deployment.
- **Network type**: Type of networks used in the PowerFlex management cluster deployment.
- **Descripton**: Describes each network or VLAN.

NOTE: VLAN numbers in the table are an example, they may change depending on customer requirements.

In a default PowerFlex setup, two data networks are standard. Four data networks are only required for specific customer requirements, for example, high performance or use of trunk ports. For more information, contact your Dell Technologies sales engineer.

CAUTION: All defined data networks must be accessible from all storage data clients (SDC). If you have implemented a solution with four data networks, all four must be assigned and accessible from each storage data client. Using less than the configured number of networks result in an error in PowerFlex and can lead to path failures and other challenges if not properly configured.

VLAN network requirements:

- VLAN flex-node-mgmt (105) and flex-stor-mgmt (150) must be routable to each other
- VLAN flex-node-mgmt (105) and pfmc-store-mgmt (140) and flex-stor-mgmt (150) must be routable to each other

Network requirements for PowerFlex management cluster (KVM)

NOTE: If a customer does not require VLAN separation between the hypervisor, PowerFlex, and virtual machine management, a single VLAN such as 105, 140, or 150 may be utilized to serve all three functions.

Table 15. Network requirements for PowerFlex management cluster (KVM)

Example VLAN	Traffic type	Network type	Components configured	Description
101	Hardware management	General purpose	• iDRAC • MVMs • Jump VM (OOB management)	For connection to PowerFlex management cluster (KVM) node iDRAC interface and PowerFlex Manager

Table 15. Network requirements for PowerFlex management cluster (KVM) (continued)

Example VLAN	Traffic type	Network type	Components configured	Description
			• Ingress controller	(PowerFlex management platform VMs and ingress controller)
103	VMware vCenter HA	General purpose	• vCenter	**Optional:** For VMware vCenter high availability (vCenter HA) network interface **NOTE:** Only for extended compute only nodes if customer is using as part of management cluster and this is not part of standard offering.
105	Hypervisor management	Hypervisor management	• VMware ESXi • VMware vCenter	**Optional:** VMware ESXi node and vCenter management **NOTE:** Only applicable for PowerFlex management cluster (KVM) compute-only extension (ESXi) nodes. If the customer is using this option as part of management cluster. This is not part of standard offering. Hypervisor management VLAN 105, PowerFlex management (PFMC) VLAN 140 and PowerFlex management (Production) VLAN 150 must be routable to each other.
140	PowerFlex management cluster (KVM)	Hypervisor / PowerFlex management	• Hypervisor (KVM) node • PowerFlex management platform MVMs • CloudLink VMs • Jump server VM • Secure connect gateway VMs • Ingress controller • Installer VM	For management interface of the PowerFlex management cluster (KVM)
141	PowerFlex management cluster (KVM) PowerFlex data 1	PowerFlex data	• Hypervisor (KVM) • PowerFlex management platform MVMs • Ingress controller	For storage node-to-storage node and storage node-to-SDC data path for the PowerFlex management cluster (KVM) nodes
142	PowerFlex management cluster (KVM) data PowerFlex data 2	PowerFlex data	• Hypervisor (KVM) • PowerFlex management platform MVMs	

Table 15. Network requirements for PowerFlex management cluster (KVM) (continued)

Example VLAN	Traffic type	Network type	Components configured	Description
			• Ingress controller	
143	Hypervisor migration for PowerFlex management cluster	VMware vSphere vMotion	• VMware ESXi	**Optional** For VMware vSphere vMotion interface on management controller vSphere cluster **NOTE:** Only applicable for PowerFlex management cluster (KVM) compute-only extension (ESXi) nodes. If the customer is using this option as part of management cluster. This is not part of standard offering.
150	PowerFlex management	Hypervisor / PowerFlex management	• Hypervisor (KVM) • Production MVMs • Ingress controller	For PowerFlex production storage-only management node interface and PowerFlex management MVMs (KVM) production MVMs
151	PowerFlex data 1	PowerFlex data		For storage node-to-storage node and storage node-to-SDC data path production workloads
152	PowerFlex data 2	PowerFlex data		
153	PowerFlex data 3 (if required)	PowerFlex data		
154	PowerFlex data 4 (if required)	PowerFlex data		

Network requirements for PowerFlex management cluster (ESXi)

Table 16. Network requirements for PowerFlex management cluster (ESXi)

Example VLAN	Example multi-VLAN	Traffic type	Network type	Description
101	210-224	Hardware management	General purpose	For connection to PowerFlex management controller node iDRAC interface and PowerFlex Manager (PowerFlex management platform VMs and ingress controller)
103	-	VMware vCenter HA	General purpose	For VMware vCenter high availability (vCenter HA) network interface Use VLAN 103 if a dedicated VLAN is required, otherwise use VLANs 141,142, and 143 for a standard deployment.
105	240-254	Hypervisor management	Hypervisor management	For management interface of KVM, VMware ESXi, VMware vCenter

Table 16. Network requirements for PowerFlex management cluster (ESXi) (continued)

Example VLAN	Example multi-VLAN	Traffic type	Network type	Description
130	-	PowerFlex management controller for a multi-VLAN or multi-subnet network	Hypervisor management	For VMware vCenter, CloudLink center, jump server, secure connect gateway and PowerFlex Manager (PowerFlex management platform VMs and ingress controller)
150	360-374	PowerFlex management	Hypervisor management	For SVM and PowerFlex storage-only node Management interface Use VLAN 150 if a dedicated VLAN is required, otherwise use VLAN 105 for a standard deployment.
151	-	PowerFlex data 1	General purpose	For storage node-to-storage node and storage node-to-SDC data path
152	-	PowerFlex data 2	General purpose	For storage node-to-storage node and storage node-to-SDC data path
153	-	PowerFlex data 3 (if required)	General purpose	For storage node-to-storage node and storage node-to-SDC data path
154	-	PowerFlex data 4 (if required)	General purpose	For storage node-to-storage node and storage node-to-SDC data path

Network requirements for the PowerFlex production cluster

Table 17. Network requirements for the PowerFlex production cluster

Example VLAN	Example multi-VLAN	Network name	Description	Properties
101	210-224	Hardware management	For connection to PowerFlex production node iDRAC interface	Layer-2/layer-3 connectivity, MTU=1500/9216
105	240-254	Hypervisor management	For VMware ESXi management interface on production vSphere cluster	Layer-3 connectivity, MTU=1500/9216
106	270-284	Hypervisor migration	For VMware vSphere vMotion interface on production vSphere cluster	Layer-2 connectivity, MTU=1500/9216 **NOTE:** Layer-3 if multi-subnet/multi-VLAN is used.
150	360-374	PowerFlex management	For PowerFlex storage-only node management interface Use VLAN 150 if a dedicated VLAN is required, otherwise use VLAN 105 for a standard deployment.	Layer-3 connectivity, MTU=1500/9216
151	-	PowerFlex data 1	For storage node-to-storage node and storage node-to-SDC data path	Layer-2 connectivity, MTU=9216
152	-	PowerFlex data 2	For storage node-to-storage node and storage node-to-SDC data path	Layer-2 connectivity, MTU=9216
153	-	PowerFlex data 3 (if required)	For storage node-to-storage node and storage node-to-SDC data path	Layer-2 connectivity, MTU=9216

Table 17. Network requirements for the PowerFlex production cluster (continued)

Example VLAN	Example multi-VLAN	Network name	Description	Properties
154	-	PowerFlex data 4 (if required)	For storage node-to-storage node and storage node-to-SDC data path	Layer-2 connectivity, MTU=9216

Network requirements for VMware NSX (optional)

Table 18. Network requirements for VMware NSX

Example VLAN	Example multi-VLAN	Network name	Description	Properties
101	210-224	Hardware management	For connection to NSX node iDRAC interface	Layer-2/layer-3 connectivity, MTU=1500/9216
105	240-254	Hypervisor management	For VMware ESXi management interface on NSX Edge cluster (Shared with production VMware vSphere cluster)	Layer-3 connectivity, MTU=1500/9216
114	-	nsx-vsan (only if required)	For VMware vSAN interface on NSX Edge cluster (This is optional meaning only if customer chooses to deploy vSAN on NSX Edge cluster)	Layer-2 connectivity, MTU=9216
116	450-464	nsx-vmotion	For NSX vMotion on NSX Edge cluster	Layer-2/layer-3 connectivity, MTU=1500/9216
121	-	nsx-transport	For NSX Transport interface on NSX Edge cluster (Used for NSX overlay)	Layer-2 connectivity, MTU=9216
122	-	nsx-edge1	For NSX Edge external VLAN1 used for BGP uplink	Layer-3 connectivity, MTU=1500
123	-	nsx-edge2	For NSX Edge external VLAN2 used for BGP uplink	Layer-3 connectivity, MTU=1500

Verify the network configuration

Use this procedure to verify configuration before adding a VLAN to the network.

Steps

1. Verify that the virtual port channel (vPC) / virtual link tunneling (VLT) is available between the access switches:
   a. For Cisco Nexus switches, type **`show vpc brief`**.
   b. For Dell PowerSwitch switches, type **`show vlt brief`**.
2. To verify the port channel between servers, hosts, and access switches are configured, type **`show port-channel summary`**.
3. To verify the port channel between customer network switches and access switches are configured, type **`show port-channel summary`**.
4. To verify that the VLAN is created and enabled on the port channel between the network switches and the access switches, type **`show vlan`**.
5. To verify that SVI is configured on the network switches, type **`show ip int brief`**.
6. To verify that you have the VLAN IP address, mask, and gateway to verify the VLAN configuration, type **`show vlans`**.

Add a VLAN to the network

To enable VLANs on the network backbone, you must manually configure the switches.

About this task

When we add new VLANs to the system, PowerFlex Manager configures the server facing port configurations. You must do the switches manually.

NOTE: If multi-VLAN or multi-subnet is enabled, each network may have multiple VLANs except for data networks. For a multi-VLAN or a multi-subnet configuration, include all the associated VLANs in the **<vlan-list>** variable.

Steps

1. To configure access switches in an access-aggregation configuration:

   NOTE: This is applicable for Cisco Nexus and Dell PowerSwitch switches.

   a. Type the following to define the new VLAN on the switch:

   Table 19. Define new VLANs on switches

For...	Do this...

Cisco Nexus switches	`vlan <vlan number>`
	`name <vlan name>`
	`no shutdown`

Dell PowerSwitch switches	`interface vlan <vlan number>`
	`name <vlan name>`
	`no shutdown`

   b. Type the following to add the VLAN to the uplink connected to the aggregation switch:

```
interface port-channel 1900
Description "Uplink Port Channel from Access Switch to AGG-SWITCH"
switchport trunk allowed vlan add <vlan-list>
```

   c. Type the following to add the VLAN to the peer-link:

```
interface port-channel 100
description “vpc-peer-link”
switchport trunk allowed vlan add <vlan-list>
```

   d. Type the following to add the new VLAN to the downstream ports connected to the nodes - only for partial network automation:

```
interface port-channel 81
Description "Port Channel for Connection to Node”
switchport trunk allowed vlan add <vlan number> # add multi-vlan (if applicable)
```

2. To configure aggregation switches in an access-aggregation configuration:

   a. Type the following to define the new VLAN on the switch:

   Table 20. Define new VLANs on switches

For...	Do this...

Cisco Nexus switches	`vlan <vlan number>`
	`name <vlan name>`
	`no shutdown`

Table 20. Define new VLANs on switches (continued)

For...	Do this...

Dell PowerSwitch switches	`interface vlan <vlan number>`
	`name <vlan name>`
	`no shutdown`

b. Type the following to add the VLAN to the downlink connected to the access switch:

```
interface port-channel 1900
Description "Downlink from AGG-SWITCH" to Access Switch
switchport trunk allowed vlan add <vlan-list>
```

c. Type the following to add the VLAN to the peer-link:

```
interface port-channel 100
description "vpc-peer-link"
switchport trunk allowed vlan add <vlan-list>
```

d. Type the following to add the new VLAN to the customer connected link (only required if the VLAN is an Layer-2 VLAN):

```
interface port-channel <900>
Description "Connection to Customer Network"
switchport trunk allowed vlan add <vlan-list>
```

e. Type the following if it is a routed VLAN for Dell PowerSwitch switches:

```
interface vlan<vlan-no>
(Repeat this section for each vlan enabled for multi-vlan on each network-type)
description <vlan-name>
no shutdown
ip address <interface ip address>

ip address <ip-address> secondary (Repeat this line for each subnet enabled with in
vlan, Only if multi-subnet is enabled for any network-type)
vrrp-group <vlan no>
(Repeat this section for each subnet belong to the same vlan)
priority 150
virtual-address <virtual ip address>
```

f. Type the following if it is a routed VLAN for Cisco Nexus switches:

```
interface vlan <x>
ip address <ip address>
ip address <ip address> secondary # repeat for each subnet enabled within a vlan.
Only applicable if multi-subnet is enabled for any network-type
hsrp version 2
hsrp <number> # repeat for each subnet belonging to the same vlan
authentication text <password>
preempt
priority 110
ip <virtual ip>
```

3. To configure leaf/border leaf switches in a leaf-spine configuration:

NOTE: For PowerFlex management cluster (ESXi), repeat this section for all multi-VLANs configured on leaf switches where the PowerFlex management nodes are connected.

a. Type the following to define the new VLAN on the switch:

```
vlan <vlan number>
name <vlan name>
no shutdown
```

b. Type the following to define VLAN to VXLAN mapping:

```
vlan <vlan number>
name <vlan name>
vn-segment <vxlan number>

interface nve1
 source-interface hold-down-time 180
member vni <vxlan number # L2VNI>
suppress-arp
ingress-replication protocol bgp
```

c. Type the following if it is a routed VLAN:

```
interface Vlan<vlan number>
vrf member <vrf name>
fabric forwarding mode anycast-gateway
carrier-delay msec 100
mtu 9216
ip address <distributed gw ip address>
ip address <distributed gw ip address> secondary # repeat for each subnet enabled
within a vlan. Only applicable if multi-subnet is enabled for any network-type
no shutdown
```

d. Type the following if the VLAN is an Layer-2 VLAN:

```
evpn
vni <vxlan number> l2
rd auto
route-target import auto
route-target export auto
```

e. Type the following to add the new VLAN to the downstream ports connected to the nodes - only for partial network automation:

```
interface port-channel 81
Description "Port Channel for Connection to Node”
switchport trunk allowed vlan add <vlan number>
```

Add a network to a resource group using PowerFlex Manager

You can add an available network to a resource group or choose to define a new network for a configuration that was initially deployed outside of PowerFlex Manager. You cannot remove an added network using PowerFlex Manager.

About this task

Before you can add a network to a service, define the network.

You can add a static route to allow nodes to communicate across different networks.

Prerequisites

Ensure that a new VLAN is created on any switches that need access to that VLAN and is added to any management cluster server-facing ports. The VLAN is then added it to any northbound trunks to other switches that it must communicate with.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Lifecycle** > **Resource Groups**.
3. From the **Resource Group Details** page, click **Add Resources**, and select **Add Network**.

4. Click **Add Additional Network** to add an additional network and click **Continue** or to add an additional static route, go to the next step.

   For a multi-VLAN or multi-subnet configuration, consider additional networks only for the following network traffic:
   - General purpose VLAN (visible only for PowerFlex compute-only nodes)
5. Click **Add Additional Static Route** and perform the following to add static routes:
   a. Click **Add New Static Route**.
   b. Select a **Source Network (Data)**.

      The source network must be a PowerFlex data network.

   c. Select a **Destination Network (Rep)**.

      The source network must be a PowerFlex data network.

   d. Type the IP address for the **Gateway**.
   e. Repeat steps 5a through 5d for additional static routes required.
   f. Click **Save**.

Configuring networking

Adding the details of an existing network enables PowerFlex Manager to automatically configure nodes that are connected to the network.

Define a network

Use this procedure to define a network.

Prerequisites

When defining the PowerFlex management cluster management network, ensure that the following is met:

- Deploy a compute service in a multi-VLAN or multi-subnet or customer option in a standard single network build.

Steps

1. On the menu bar, click **Settings** > **Networking** and click **Networks**.
2. Click **Define**.
3. In the **Name** field, enter the name of the network. Optionally, in the **Description** field, enter a description for the network.
4. From the **Network Type** drop-down, select one of the following network types:
   - General purpose LAN
   - Hypervisor management (supports multi-VLAN or multi-subnet configurations)
   - Hypervisor migration (supports multi-VLAN or multi-subnet configurations)
   - Hardware management
   - PowerFlex data
   - PowerFlex data (client traffic only)
   - PowerFlex data (server traffic only)
   - PowerFlex management (supports multi-VLAN or multi-subnet configurations)

   NOTE:
   - For a PowerFlex configuration that uses a two-layer architecture with four dedicated data networks, you typically have two PowerFlex (client-traffic only) VLANs and two PowerFlex data (server-traffic only) VLANs. These network types are used with storage-only and compute-only resource groups.
   - For a multi-VLAN or multi-subnet configuration, include adding additional networks only for the following network traffic:
     - Hypervisor management
     - Hypervisor migration
     - PowerFlex management

- For PowerFlex data and PowerFlex management networks, ensure the network is sized appropriately at the time of creation. These network types will use one IP address as a virtual IP (VIP) in addition to one IP address for each node. Consider future expansion plans for these networks and size them appropriately.

5. In the VLAN ID field, enter a VLAN ID between 1 and 4094.

   NOTE: PowerFlex Manager uses the VLAN ID to configure I/O modules to enable network traffic to flow from the node to configured networks during deployment.

6. Optionally, select the **Configure Static IP Address Ranges** check box, and do the following:
   a. In the **Subnet** box, enter the IP address for the subnet. The subnet is used to support static routes for data networks.
   b. In the **Subnet Mask** box, enter the subnet mask.
   c. In the **Gateway** box, enter the default gateway IP address for routing network traffic.
   d. Optionally, in the **Primary DNS** and **Secondary DNS** fields, enter the IP addresses of primary DNS and secondary DNS.
   e. Optionally, in the DNS Suffix field, enter the DNS suffix to append for hostname resolution.
   f. To add an IP address range, click **Add IP Address Range**. In the row, indicate the role in PowerFlex nodes for the IP address range and then specify a starting and ending IP address for the range. For the role, select either:
      - **Server or Client**: Default; range is assigned to the server and client roles.
      - **Client Only**: Range is assigned to the client role on PowerFlex compute-only nodes.
      - **Server Only**: Range is assigned to the server role on PowerFlex storage-only nodes.

      You can add the virtual IP address value to **Server or Client** and **Server Only**.

      NOTE: Do not add the data networks gateway to this network.

      NOTE: The **Configure Static IP Address Ranges** check box is not available for all network types. For example, you cannot configure a static IP address range for the operating system Installation network type. You cannot select or clear this check box to configure static IP address pools after a network is created.

7. Click **Save**.

Add a new interface

Use the **Add New Interface** feature to create a network interface to match to a network card on a node when deploying a template.

PowerFlex Manager supports a network layout for compute-only, and storage-only deployments. This network layout allows you to use trunk ports and port channels for data networks, instead of access ports. In this type of configuration, both data networks are on both NICs, teamed or bonded together in the operating system.

- For a compute-only deployment, the first port on both interfaces is for trunk traffic, whereas the second port is for data 1 and data 2.
- For a storage-only deployment, the first port is for trunk traffic and data 1, whereas the second port is for data 2. PowerFlex Manager still supports the legacy network layout, but the sample templates use the new configuration.

1. On the node component page, under **Network Settings**, click **Add New Interface**.
2. Enter the following information for the new interface:
   - **Port Layout**—Select the NIC type from the list.

     For a VMware ESXi deployment with an NVIDIA card, you must select the two port, 25-gigabit NIC type.
3. Enter the network VLANs for each port.
   a. Click **Choose Networks** for a port.
   b. To add one or more networks to the port, select **Add Networks to this Port**, then click the check box for each network you want to add from the **Available Networks** list. Alternatively, click the check box in the upper left corner next to the **Name** label to select all the available networks.

      If you want to filter the list by network type, select a **Network Type**, then enter a name or VLAN ID to search.

      Click **>>** to move the selected items to the **Selected Networks** list on the right.
   c. To mirror network settings from another port for which you have already chosen the network VLANs, select **Mirror this Port with Another Port**. Select the other interface and port from which you want to mirror this port.
   d. Click **Save**.

4. To view the list of nodes that match the network configuration parameters, click **Validate Settings**.

   The list of nodes is filtered according to the target boot device and NIC type settings specified.

   When you enable PowerFlex settings for the node, the **Validate Settings** page filters the list of nodes according to the supported storage types (NVMe, All flash, and HDD). Within the section for each storage type, the nodes are also sorted by health, with the healthy (green) nodes displayed first and the critical (red) nodes displayed last.

NOTE: If you select the same network on multiple interface ports or partitions, PowerFlex Manager creates a team or bond on systems with the VMware ESXi operating system. This configuration enables redundancy.

Edit a network

If a network is not associated with a template or resource group, you can edit the network name, the VLAN ID, or the IP address range.

Steps

1. On the menu bar, click **Settings** > **Networking** and click **Networks**.
2. Select the network that you want to modify, and click **Modify**.
3. Edit the information in any of the following fields: **Name**, **VLAN ID**, **IP Address Range**.

   For a PowerFlex data, you can specify a subnet IP address for a static route configuration. The subnet is used to support static routes for data.
4. Click **Save**.

Delete a network

You cannot delete a network that is associated with a template or resource group.

Steps

1. On the menu bar, click **Settings** > **Networking** and click **Networks**.
2. Click the network that you want to delete, and click **Delete**.
3. Click **Yes** when the confirmation message is displayed.

View port details

Use this procedure to view the port details.

About this task

Port View is only available for PowerFlex nodes and is not available for management VMs such as, CloudLink Center, PowerFlex gateway, VMware vCSA, and switches.

Steps

1. Log in to PowerFlex Manager.
2. From the menu, click the **Resources** tab.
3. From the **Resources** tab, select a resource and click **View Details**.
4. Click **Port View**.

Add the VMware NSX service using PowerFlex Manager

Use this procedure only if the PowerFlex nodes are added to the VMware NSX environment.

Prerequisites

- Before adding this service in PowerFlex Manager, verify that the VMware NSX data center is configured on the PowerFlex compute-only nodes.
- Ensure that the iDRAC of nodes, VMware vCenter, and switches (applicable for full networking) are discovered in PowerFlex Manager.
- After adding an NSX node, if you are using PowerFlex Manager, run **Update Service Details** to represent the appropriate environment. If you are using VMware NSX in a PowerFlex Manager service, the service goes into lifecycle mode.

Steps

1. Log in to PowerFlex Manager.
2. From **Getting Started**, click **Define Networks**.
   a. Click **Define** and do the following:

   Table 21. VMware NSX information and values

NSX information	Values
Name	Type **NSX Transport**
Description	Type **Used for east-west traffic**
Network Type	Select **General Purpose LAN**
VLAN ID	Type **121**

   b. Click **Save** > **Close**.
3. Remove the compute-only service:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Click **CO Service**.
   c. Click **More Actions** > **Remove Resource Group**.
   d. Select **Remove Resource Group** from the drop-down under **Resource Group removal type**.
   e. On the **Remove Resource Group** page, select **Leave nodes in PowerFlex Manager inventory and set state to Managed**.
   f. Click **Remove** to remove the service.
4. From **Getting Started**, click **Add Existing Service** and do the following:
   a. On the **Welcome** page, click **Next**.
   b. On the **Service Information** page, enter the following details:

   Table 22. Service information and details

Service information	Details
Name	Type **NSX Service**
Description	Type **Transport Nodes**
Type	Type **Compute-only**
Firmware and software compliance	Select the Intelligent Catalog (IC) version
Who should have access to the service deployed from this template?	Leave as default

   c. Click **Next**.
   d. On the **Network Information** page, select **Full Network Automation/Partial Network Automation**, and click **Next**.

NOTE: For partial network automation, you must finish the complete network configuration required for VMware NSX. Consider the configuration given in this document as a reference.

e. On the **Cluster Information** page, enter the following details:

Table 23. Cluster information and details

Cluster information	Details
Target virtual machine manager	Select vCSA name
Data center name	Select data center name
Cluster name	Select cluster name
Target PowerFlex gateway	Select PowerFlex gateway name
Target protection domain	Select **PD-1**
OS image	Select the VMware ESXi image

f. Click **Next**.
g. On the **OS Credentials** page, select the OS credentials for each node, and click **Next**.
h. On the **Inventory Summary** page, review the summary and click **Next**.
i. On the **Networking Mapping** page, verify that the networks are aligned with the correct dvswitch.
j. On the **Summary** page, review the summary and click **Finish**.

5. Verify that PowerFlex Manager recognizes that VMware NSX is configured on the nodes:
   a. Click **Services**.
   b. Select the compute-only service.
   c. Verify that a banner appears under the **Service Details** tab, notifying that `NSX is configured on a node and is preventing some features from being used.` If you do not see this banner, check if you have selected the wrong service or NSX is not configured on the PowerFlex compute-only nodes.

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Modifying the DNS entries

Modifying the DNS entries in the embedded Linux operating system

Modify or update the DNS entries in the embedded Linux operating system.

Steps

1. SSH to the embedded Linux operating system running on the storage node.
2. Go to the file path: `sudo vi /etc/sysconfig/network/config`
3. Add or modify the line:
   - **`NETCONFIG_DNS_STATIC_SEARCHLIST=xxx.com`**, where *xxx.com* is the domain name
   - **`NETCONFIG_DNS_STATIC_SERVERS=x.x.x.x y.y.y.y`**, where *x.x.x.x y.y.y.y* is the DNS server IP address
4. Press **Esc** and type **`:wq`** to save the file.
5. Type **`sudo netconfig update -f`** to refresh and update the configuration.

Modifying DNS entries in the management virtual machines

Modify or update the DNS entries in the management virtual machines.

Steps

1. Establish an SSH connection to each of the management virtual machines nodes.
2. To view the existing DNS entries, run:

   ```
   sudo cat /etc/sysconfig/network/config | grep -i NETCONFIG_DNS_STATIC
   ```

3. To add or update the DNS entries:
   a. Run the following command:

      ```
      sudo vi /etc/sysconfig/network/config
      ```

   b. Add or modify the following lines:
      - **`NETCONFIG_DNS_STATIC_SEARCHLIST=example.com`**, where *example.com* is the domain name
      - **`NETCONFIG_DNS_STATIC_SERVERS=x.x.x.x y.y.y.y`**, where *x.x.x.x y.y.y.y* are the DNS server IP addresses.

   c. Press **Esc** and type **`:wq`** to save the file.
   d. To refresh and update the configuration, run:

      ```
      sudo netconfig update -f
      ```

4. Update the **coredns** pods on all the management virtual machines after making the DNS entry change for the pods in internal resolutions. Run the following command to view the pod names:

   ```
   kubectl get pods -n kube-system | grep rke2-coredns-rke2-coredns
   ```

5. To restart the **coredns** pods, delete them by running the following command:

   ```
   kubectl delete pods <pod-name> -n kube-system
   ```

   The pods are recreated after they are deleted.

Modifying the DNS entries in VMware vCenter

Modify the VMware vCenter DNS using VAMI.

Steps

1. Log in to VMware vCenter VAMI: **`https://vcenter_ip_or_FQFN:5480`**.
2. Go to the **Networking** section and in **Network Settings**, click **Edit**.
3. From **Edit Network Settings**, select **Network Adapter** and click **Next**.
4. In **Edit Settings**, expand the **Hostname and DNS** field, make the changes, and click **Next**.
5. In **SSO credentials**, enter the administrator credentials, for example, **`administrator@vsphere.local`**.
6. In **Ready to Complete**, review the changes, accept the acknowledgment, and click **Finish**.
7. A window appears with the message **Network update in progress**. When complete, the window displays **Network update succeeded**, click **Close**.

Modifying the DNS entries in the VMware ESXi host

Steps

1. Log in to the VMware ESXi host: **`https://ESXi_host_ip`**.
2. Click **Networking** > **TCP/IP Stacks**. Select **Default TCP/IP stack** and click **Edit Settings**.
3. Under **Basic DNS Configuration**, select **Manually configure the settings for this TCP/IP stack**.
4. Add or update the DNS IP addresses in the **Primary DNS server** and **Secondary DNS server** fields. In the **Search domain** field, add the domain name and click **Save**.

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Storage management

Protection domains

A protection domain is a logical entity consisting of a group of storage nodes, including PowerFlex data server and device gateway target components.

It may also contain NVMe target components that are known as storage data targets (SDTs). Each storage node belongs to one protection domain, so each protection domain is a unique set of storage nodes. You can add, modify, activate, inactivate, or remove a protection domain in the PowerFlex systems.

Rename a protection domain

Rename the protection domain from the PowerFlex system.

Steps

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. Select the protection domain to rename, and click **Modify** > **Rename**.
3. In the **Rename Protection Domain** window, change the name in the **Protection Domain Name** field. Click **Apply**.

Activate a protection domain

Activate the protection domain to enable access to data.

Steps

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. From the list of protection domains, select the relevant protection domain, and click **More Actions** > **Activate**.
3. In the **Activate Protection Domain** dialog box, click **Yes** for **Force Activate** and click **Activate** to enable access to the data on the protection domain.
4. Verify the operation has finished successfully and click **Dismiss**.

Inactivate a protection domain

Use this procedure for a graceful system shutdown.

About this task

While a protection domain is inactivated, the following activities take place:

- Determine if there are any current rebuild/rebalance activities taking place. If so, the shutdown will be delayed (unless it is forced) until they are finished.
- Block future rebuild/rebalance activities.
- Temporarily disable application I/O and disable access to volumes.
- Move the DRL mode of all storage nodes to harden, in preparation for restarting the server.
- Reload of all storage nodes before re-enabling data access.

Steps

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.

2. From the list of protection domains, select the relevant protection domain, and click **More Actions** > **Inactivate**.
3. In the **Inactivate Protection Domain** dialog box, and click **Inactivate**.
4. Verify the operation has finished successfully and click **Dismiss**.

Set I/O priority

PowerFlex includes advanced settings that control I/O priorities and bandwidth use, and can be used to fine-tune system performance. This includes bandwidth and concurrent I/Os for rebuild, rebalance, migration, and maintenance mode.

About this task

NOTE: This feature affects the system performance and should only be configured by an advanced user.

Steps

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. In the list of protection domains, select the relevant protection domain and click **More Actions** > **Set I/O Priority**.
3. In the **Set I/O Priority for PD** dialog box, select one of the policies to use for the secondary I/O priority:
   - Unlimited: I/Os are unlimited.
   - Favor Application I/O: Limit the number of secondary concurrent rebuild I/Os to the values entered in the concurrent I/O limit and bandwidth I/O limit, regardless of user I/O.
4. Click **Apply** > **Dismiss**.

Device groups

A device group is a complete set of devices of all the nodes in the protection domain and there is a scope of RAID rebuild.

There are two types of device groups:

- Storage device
- Persistent memory (optional)

NOTE: There can only be one device group per media type in the protection domain.

Set spare capacity of a device group

You must set the spare capacity of a device group in terms of nodes. The default spare capacity is one node, and the maximum is two nodes.

About this task

Utilizing spare devices is a feature that is recommended only for advanced configurations.

Steps

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. From the list of device groups, select the check box of the device group that you want to set the capacity for, and click **More Actions** > **Spare Capacity**.
3. In the **Set Spare Capacity for Device Group** dialog box, select the spare nodes for the device group.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename the device groups

Use this procedure to rename the device groups.

Steps

1. On the menu bar, click **Block** > **Storage System**.
2. In the **Device Group** tab, click **More Actions** > **Rename**.
3. In the **Rename Device Group** window, type a name in the **Device Group Name** field.
4. Click **Apply**.

Storage pools

A storage pool is a set of physical storage devices in a protection domain.

Each time that you add devices to the system, you must map them to storage pools. Create storage pools before you start adding storage nodes and storage devices to the system.

A volume is distributed over all devices residing in the same storage pool. Add, modify, or remove a storage pool in the PowerFlex system.

One storage pool per protection domain is supported.

Configure storage pool settings

Configure storage pool settings, including checksum, zero padding, and compression.

Steps

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of storage pools, select the relevant storage pool, and click **Modify** > **General Settings**.
3. In the **Storage Pools Settings** dialog box, configure the following settings for the storage pool:
   - Select **Enable Compression** for inline compression to gain more effective capacity.
   - Select **Enable Over Provisioning Control** to enable over provisioning, and set the value from 2 to 1000. If enabled, the default limit is unlimited.
4. Click **Apply**.

Modify storage pool capacity settings

Use this procedure to set capacity alert thresholds.

Steps

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of storage pools, select the relevant storage pool, and click **Modify** > **Capacity**.
3. In the **Storage Pool Capacity Settings** dialog box, set the capacity alert thresholds.
4. Click **Apply**.

Expand a storage pool

Use this procedure to expand a storage pool.

Steps

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of volumes, select the volume and click **Modify** > **Expand**.

3. In the **Expand Storage Pool** dialog box, enter the **Custom Capacity** size or select **All Available Capacity** to utilize the full available capacity of the device group.
4. Click **Apply**.
5. Verify the operation completed successfully and click **Dismiss**.

Volumes

Define, configure, and manage volumes in the PowerFlex system.

Add volumes

Use this procedure to add volumes, giving each volume a meaningful name associated with its operational role.

About this task

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. The default name for each volume object is its ID. The ID is displayed in the Volumes list or can be obtained using a CLI query.

Define each volume name according to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

Prerequisites

There must be five storage nodes in the system and there must be sufficient capacity available for the volumes.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. Click **Create Volume**.
3. In the **Create Volume** dialog box, configure the following items:
   a. Select the storage pool.
   b. Enter the number of volumes to be created.
      - If you type `1`, enter a name for the volume.
      - If you type a number greater than one, enter the **Volume Prefix** and the **Starting Number** of the volume. This number will be the first number in the series that is appended to the volume prefix. For example, if the volume prefix is `Vol%i%` and the starting number value is `100`, the name of the first volume that is created is Vol100, the second volume is Vol101, and so on.
   c. Enter the volume name.
   d. Enter the volume size in GB or TB.
4. Click **Create**.
5. Verify the operation completed successfully and click **Dismiss**.

Next steps

To use the created volume, you must map it to at least one host. If the restricted SDC mode is enabled for the system, you must approve SDCs prior to mapping volumes to them.

Delete volumes

Remove volumes from PowerFlex.

About this task

To prevent causing a data unavailability scenario, avoid deleting volumes or snapshots while the MDM cluster is being upgraded.

CAUTION: All data is erased from a deleted volume.

Prerequisites

Ensure that the volume that you are deleting is not mapped to any hosts. If it is, unmap it before deleting it. In addition, ensure that the volume is not the source volume of any snapshot policy. You must remove the volume from the snapshot policy before you can remove the volume.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the required volume, and click **More Actions** > **Delete**.
3. In the **Delete Volume** dialog box, verify the volumes to be removed, and click **Delete**.
4. In the warning dialog box, click **Delete**.
5. Verify that the operation completed successfully, and click **Dismiss**.

Map volumes

Mapping exposes the volume to the specified host, effectively creating a block device on the host. You can map a volume to one or more hosts.

About this task

For Linux-based devices, the `scini` device name may change on reboot. Dell Technologies recommends that you mount a mapped volume to the unique ID, which is a persistent device name, rather than to the `scini` device name.

To identify the unique ID, run the command `ls -l /dev/disk/by-id/`.

NOTE: You cannot map a volume if the volume is an auto snapshot that is not locked.

Prerequisites

Volumes can only be mapped to one type of host: either SDC or NVMe. Ensure that you know which type of hosts are being used for each volume, to avoid mixing host types.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select one or more volumes, and click **Provision** > **Map**.
3. A list of the hosts that can be mapped to the selected volumes is displayed. If a volume is already mapped to a host, only hosts of the same type, NVMe or SDC, are listed. If the volume is not mapped to a host, click **NVMe** or **SDC** to set the type of hosts to be listed.
4. In the **Map Volume** dialog box, select one or more hosts to which you want to map to the volumes.
5. Click **Map**.
6. Verify that the operation completed successfully, and click **Dismiss**.

Unmap volumes

Unmap one or more volumes from hosts.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the relevant volumes, and click **Provision** > **Unmap**.
3. Select the host from which to remove mapping to the volumes.
4. Click **Unmap**.
5. Verify that the operation was completed successfully, and click **Dismiss**.

Set volume bandwidth and IOPS limits

Setting bandwidth and IOPS limits for volumes lets you control the quality of service (QoS). Bandwidth and IOPS limits are set on a per-host basis.

Prerequisites

Ensure that the volumes are mapped before you set these limits.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the relevant volume, and click **More Actions** > **Set Limits**.
3. In the **Set IO limits for volume** dialog box, enter the required values for **Bandwidth Limits** and **IOPS Limits**, or select the corresponding **Unlimited** check box.
   - The number of IOPS must be larger than 10.
   - The volume network bandwidth is in MB/s.
   - The I/O limits are applied to every mapped SDC.
4. Click **Apply**.
5. Verify that the operation has finished successfully, and click **Dismiss**.

Increase volume size

Increase the capacity of one or more volumes at any time, if there is enough capacity for the volume size to increase.

About this task

You cannot decrease the volume size.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the volume, and click **More Actions** > **Resize**.
3. In the **Resize Volume** dialog box, enter the new volume size.
4. Click **Apply**.
5. Verify the operation has completed successfully, and click **Dismiss**.

Snapshots

Snapshots are the local data protection solution within PowerFlex. They provide a method of recovery for data that has been corrupted or accidentally deleted.

Snapshots are read-only objects and cannot be modified. This immutable property allows snapshots to serve as ransomware protection in the event production data is compromised. Snapshots are pointer-based objects that provide point-in-time copies of data that is stored in volumes or thin clones.

The PowerFlex storage system enables you to take snapshots of existing volumes, up to 1022 per volume. The snapshots are thin provisioned and are quick.

Snapshots are read-only copies and cannot be mapped to the hosts. Once a snapshot is generated, it becomes a new, unmapped volume in the system.

Thin clones are volume copies and called as snapshots. Thin clones are read/write copy of the volume. Volume can be refreshed/restored from snapshot (read-only snapshot) and thin clone (read/write snapshot) can be created from it.

Snapshots cannot be resized, but it can be deleted.

NOTE: Snapshots also consume overall system storage capacity to preserve the point-in-time. Ensure that the storage pool has enough capacity to accommodate snapshots. Removing a volume also removes all its direct snapshots.

NOTE: It is recommended NOT to trim volumes that contain snapshots as the logical capacity may not shrink.

Create a snapshot

Use this procedure to create a snapshot.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Block** > **Volumes**.
3. In the list of volumes, select the relevant volumes, and click **Protect** > **Create Snapshot**.
4. In the **Create Snapshot of Volume** window, enter the name of the snapshot in the **Snapshot Name** field.
5. Optionally, select **Use Secure Snapshot**, enter the **Expiration Time**, and select the unit type from the drop-down menu.

   NOTE: The snapshot cannot be deleted or overwritten before the expiration time.

6. Click **Create**.
7. Verify the operation completed successfully and click **Dismiss**.

Rename a snapshot

Use this procedure to rename a snapshot.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the relevant volume and click **View Details**, and go to the **Snapshots** tab.
3. Click **Modify** > **Rename** and from the **Rename Snapshot** window, enter the name of the snapshot.
4. Click **Apply**.

Restore a snapshot

Restoring of snapshots can only be performed from the snapshots of the source volumes.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the relevant volume and click **View Details**, and go to the **Snapshots** tab.
3. Select the snapshot from the list, and click **Restore**.
4. On the **Restore using Snapshot** window, the volume being restored using the snapshot is displayed, and click **Restore**.
5. Alternatively, in the list of volumes, select the relevant volume, and click **Protect** > **Restore Data**. In the Restore Data of Volume window, select the relevant snapshot from the list and click **Restore Volume**.
6. Verify the operation completed successfully and click **Dismiss**.

Delete the snapshot

Use this procedure to delete the snapshot.

About this task

The secured snapshot cannot be deleted or overwritten before the expiration time.

Steps

1. On the menu bar, click **Block** > **Volumes**.

2. In the list of volumes, select the relevant volume and click **View Details**, and go to the **Snapshots** tab.
3. In the list of snapshots, select the relevant snapshot and click **Modify** > **Delete**.
4. On the **Delete Snapshots** window, **Delete** > **Delete**.
5. Verify the operation completed successfully and click **Dismiss**.

Create a thin clone

A thin clone is a read/write copy of a volume or snapshots.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. Click **Repurpose** > **Create Thin Clone**.
3. In the **Create Thin Clone of Volume** window, add the name for the clone volume and click **Create**.
4. Verify the operation completed successfully and click **Dismiss**.

Refresh data

A refresh can be performed from any node in the vTree to another, or can be done from any snapshot-thin clone volume and snapshot of a thin clone (if any) to the source volume.

Steps

1. On the menu bar, click **Block** > **Volumes**.
2. Click **Repurpose** > **Refresh Data**.
3. In the **Refresh Data of Volume** window, the volumes and snapshots of the volume and thin clone volumes are listed. Select the relevant volume or snapshot to refresh from.
4. Click **Refresh Volume**.
5. Verify the operation completed successfully and click **Dismiss**.

Create a snapshot policy

Use this procedure to create a snapshot policy.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. Click **Create Snapshot Policy**.
3. In the **Create Snapshot Policy** dialog box:
   a. In the **Policy Name** field, enter the policy name.
   b. In the **Snapshot Rules**, define the time interval between snapshots and unit type.
4. Create retention tools:
   a. In the **Retention** field, add the number of snapshots to retain. A retention period is created for the specified number of snapshots times the snapshot interval.
   b. Optionally, to create an additional retention rule, click **Add Retention** and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period. Up to six retention rules can be created.
5. Optionally, select the **Secured** checkbox to prevent the snapshots created by the policy from being modified or deleted before the end of the expiration date. Click **Approve** to enable secured snapshots.
6. Click **Create** to create a policy and not activate, or click **Create and Activate** to create and activate the policy.
   a. To activate a paused policy, select the relevant policy and click **Protect** > **Activate**.
   b. On the **Activate Snapshot Policy** window, click **Activate**.
7. To assign volumes to the policy:
   a. Select the policy from the list, and click **Protect** > **Assign Volumes**.
   b. On the **Assign Volumes to Snapshot Policy** window, the volumes and snapshots are listed. Select the relevant volumes and click **Assign Policy**.

NOTE: The snapshot policy can be assigned only to the volumes.

c. Verify the operation completed successfully and click **Dismiss**.

Delete a snapshot policy

Delete a snapshot policy to stop PowerFlex from automatically creating snapshots for the specified volumes. You cannot delete snapshot policies that are secured.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy and click **More Actions** > **Delete**.
3. In the **Remove Snapshot Policy** dialog box, click **Remove**.
4. Verify the operation completed successfully and click **Dismiss**.

Modify a snapshot

Use this procedure to modify a snapshot.

About this task

The secured snapshot policies cannot be modified.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy and click **Modify** > **Modify Policy**.
3. In the **Modify Snapshot Policy** window:
   a. Under **Snapshot Rules**, define the time interval and unit type. The time interval between snapshots.
   b. Under **Retention Rules**, add the number of snapshots to retain.
4. Click **Modify**.
5. Verify the operation completed successfully and click **Dismiss**.

Secure a snapshot policy

Use this procedure to secure a snapshot policy.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy and click **Modify** > **Secure Policy**.
3. In the **Modify snapshot access of Policy <policy name> to secured snapshots** window:
   a. Optionally, select Apply to existing snapshots (excluding locked snapshots).
   b. Click **Approve**.
   c. On the **Secure snapshots for policy** window, click **Approve**.

   NOTE: When **Secured Snapshots** is enabled, the snapshot policy cannot be modified after creation.

Rename a snapshot policy

Use this procedure to rename a snapshot policy.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy and click **Modify** > **Rename**.
3. In the **Rename Snapshot Policy** dialog box, and click **Apply**.
4. Verify the operation completed successfully and click **Dismiss**.

Activate a snapshot policy

Use this procedure to activate a snapshot policy.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy and click **More Actions** > **Activate**.
3. In the **Activate Snapshot Policy** dialog box, click **Activate**.
4. Verify the operation completed successfully and click **Dismiss**.

Pause a snapshot policy

Use this procedure to pause a snapshot policy.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy and click **Protect** > **Pause**.
3. In the **Pause Snapshot Policy** dialog box, click **Pause**.
4. Verify in the **State** column that the snapshot policy is in a paused state, and click **Dismiss**.

Assign volumes to a snapshot policy

Use this procedure to assign volumes to a snapshot policy.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy and click **Protect** > **Assign Volumes**.
3. In the **Assign Volumes to a Snapshot Policy** dialog box, select the relevant volume and click **Assign Policy**.
   The snapshot policy can only be assigned to the volumes, and not the snapshots.
4. Verify the operation completed successfully and click **Dismiss**.

Unassign a volume from a snapshot policy

Use this procedure to unassign a volume from a snapshot policy.

Steps

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy and click **Protect** > **Unassign Volumes**.
   a. In the **Unassign Volumes to a Snapshot Policy** dialog box, the volumes that are assigned to the policy are listed. Select the relevant volumes.

b. Optionally, to delete auto snapshots of the selected volume, select the **Delete related snapshots** check box.
c. Click **Unassign Policy**.
d. In the **Unassign Volumes to the Snapshot Policy <policy name>** window, click **Yes**.

   If the auto snapshots are locked, they are detached from the snapshot policy, but not deleted. To remove the locked auto-snapshots from the system, manually remove the snapshots by going to **Block** > **Volume**, select the snapshot and click **Snapshots**. Click the relevant snapshot and click **Delete**.

Hosts

Hosts are entities that consume PowerFlex storage for application usage. There are two methods of consuming PowerFlex block storage: using the SDC kernel driver, or using NVMe over TCP connectivity. Therefore, a host is either an SDC or an NVMe host.

Once a host is configured, volumes may be mapped to it. In each case, hosts must be mapped to volumes.

Add a host

Add an NVMe over TCP host to PowerFlex.

Prerequisites

- Ensure that you have the host's NVMe Qualified Name (NQN). If you do not know the NQN, see the documentation for the host operating system.
- Ensure that the host is connected to the Ethernet switch.
- Ensure that the host is configured with the correct VLAN ID and routing rules.

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. Click **Add Host**.
3. In the **Add NVMe Host** dialog box, enter a hostname in the **Host Name** field.
4. In the **Host NQN** field, enter the NQN string for the host.
5. In the **Number of Paths Per Volume** field, enter the maximum number of paths to be provided between the host and each volume (default is 4).
6. In the **Number of System Ports per Protection Domain** field, enter the maximum number of ports to be provided between the host and each protection domain (default is 10).
7. Click **Add**.
8. Verify that the operation has finished and was successful, and click **Dismiss**.

Map hosts to volumes

Use this procedure to map the hosts to volumes.

Prerequisites

Volumes can only be mapped to one type of host: either SDC or NVMe. Ensure that you know which type of hosts are being used for each volume, to avoid mixing host types.

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Provision** > **Map**.
3. In the **Map Hosts to Volumes** dialog box, select the volumes to be mapped to the selected hosts, and click **Map**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Unmap hosts from volumes

Use this procedure to remove the mapping between hosts and volumes.

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Provision** > **Unmap**.
3. In the **Unmap** dialog box, ensure that the correct host is selected.
4. Click **Unmap**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Remove hosts

Use this procedure to remove hosts from PowerFlex.

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Remove**.
3. In the **Remove Host** dialog box, ensure that you have selected the correct host for removal.

   NOTE: You can only remove hosts with unmapped volumes. For SDCs, the host must be disconnected from the PowerFlex cluster before it can be removed.

4. Click **Remove**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Configure or modify approved host IP addresses

When the system's restricted host (SDC) mode is set to `approved IP restriction`, configure host IP addresses before mapping volumes to the hosts.

Prerequisites

Ensure that the hosts have been approved by GUID.

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Modify** > **Modify Approved IP Addresses**.
3. In the **Modify Approved IP addresses** dialog box, enter the IP address of the hosts, and click **Add IP Address**. Repeat this step for additional addresses.

   You can add up to a total of four IP addresses.
4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Approve SDCs

When the restricted host (SDC) mode is set to GUID restriction, approve SDCs before mapping them to volumes.

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select one or more hosts and click **Modify** > **Approve**.
3. In the **Approve host** dialog box, verify that the hosts listed are the ones that you want to approve, and click **Approve**.

4. Verify that the operation has finished and was successful, and click **Dismiss**.

Rename hosts

Use this procedure to rename hosts.

About this task

The hostname must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Modify** > **Rename**.
3. In the **Rename Host** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Modify an SDC performance profile

SDC performance profiles are set by default to high and can be changed to compact. The compact setting may impact the system performance.

About this task

The default setting configures a predefined set of parameters for high-performance use cases.

NOTE: Performance tuning is case-specific. To prevent undesirable effects, Dell Technologies highly recommends that you thoroughly test all changes. For further assistance, contact Dell Technologies Support.

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Modify** > **Modify Performance Profile**.
3. In the **Modify Performance Profile Host** dialog box, select the desired performance profile: **High** or **Compact**.
4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Modify maximum number of paths per volume

Use this procedure to increase the maximum numbers of paths per volume on the NVMe host.

Steps

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Modify** > **Modify Maximum Numbers of Paths Per Volume**.
3. Update the **Number of Paths** field and click **Save**.

Modify maximum number of ports per protection domain

Use this procedure to increase the maximum number of ports per protection domain on the NVMe host.

Steps

1. On the menu bar, click **Block** > **Hosts**.

2. In the list of hosts, select the relevant host, and click **Modify** > **Modify Maximum Numbers of Ports per protection domain**.
3. Update the **Number of Ports** field and click **Save**.

NVMe targets

NVMe targets (or SDT components) must be configured on the system side, in order to use NVMe over TCP technology.

The NVMe target (or SDT component) is a frontend component that translates NVMe over TCP protocol into internal protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.

TCP ports, IP addresses, and IP address roles must be configured for each NVMe target (or SDT component). You can assign both storage and host roles to the same target IP addresses. Alternatively, assign the storage role to one target IP address, and add another target IP address for the host role. Both roles must be configured on each NVMe target.

- The host port listens for incoming connections from hosts, over the NVMe protocol.
- The storage port listens for connections from the MDM.

Once the NVMe targets have been configured, add hosts to , and then map volumes to the hosts. Connect hosts to NVMe targets, preferably using the discovery feature.

On the operating system of the compute nodes, NVMe initiators must be configured. Network connectivity is required between the NVMe targets and the NVMe initiators, and between NVMe targets (or SDT components) and storage nodes.

Add an NVMe target

Add an NVMe target (or SDT component).

Steps

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. Click **Add NVMe Target**.
3. From the **Add NVMe Target** dialog box, configure the following settings:
   a. For **Target Name**, enter a name for the NVMe target (or SDT component).
   b. For **Protection Domain**, select a protection domain.
   c. Accept the default TCP ports, or modify them if necessary.
      - **Storage Port** listens for incoming connections from the MDM, and is only open on an IP address configured with the storage role. Default=12200
      - **I/O Port** listens for incoming connections for I/O, and is only open on an IP address configured with the host role. Default=4420 (which conforms to the NVMe over TCP standard)
      - **Discovery Port** listens for incoming discovery connections, and is only open on an IP address configured with the host role. Default=8009 (which conforms to the NVMe over TCP standard. Use zero to disable the discovery service.)
   d. For **Target IPs**, enter an IP address, and then select a role for the IP address from the **Target IPs Roles** menu. Click **Add Target IP**, and repeat this step until all IP addresses are configured.
4. Click **Add**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Modify an NVMe target

Modify the configuration of an NVMe target (or SDT component).

Steps

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Modify**.
3. In the **Modify NVMe Target** dialog box, modify the desired fields.
4. Optionally, add more target IP addresses, by clicking **Add Target IP**.

5. Click **Modify NVMe Target**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove an NVMe target

Remove an NVMe target (or SDT component).

Steps

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Remove**.
3. In the **Remove NVMe Target** dialog box, verify that you are removing the wanted NVMe target, and click **Remove**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Understanding NVMe over TCP load balancing

PowerFlex supports load balancing with NVMe over TCP.

Persistent discovery

The persistent discovery controller ensures that the host remains connected to the discovery service after discovery. If at any point there is a change in the discovery information that is provided to the host, the discovery controller returns an asynchronous event notification (AEN) and the host requests the updated **Discovery log** page.

Here are some examples of changes in discovery information:

- A new volume is mapped to the host from a new protection domain.
- A new storage data target (SDT) is added to the system.
- Load balancing wants to move the host connection from one storage data target to another.

When configuring NVMe hosts, ensure that every host is connected for discovery at most once per subnet (data IP address subnet). To use this functionality, ensure that the host operating system supports the persistent discovery controller, and that the **Persistent Discovery** flag is set in the discovery. (See the respective operating system for the NVMe over TCP host configuration.)

NVMe over TCP hosts network awareness

Hosts are connected to the storage through Layer-2 or Layer-3 network. While the storage does not manage or need to be aware of the network configuration, there are some aspects of the network that impact the storage.

Load balancing takes the data network/subnet into consideration to ensure there is a balance between the host connections on each data network with a specific subnet for improved performance.

In Layer-3 networks, the system must have routing tables configured.

Multiple objects need to be defined for load balancing to work:

Table 24. Defined objects for load balancing

Object	Description
Host subnet	Networks or subnets used to connect hosts to storage, which can be either layer 2 or layer 3 (routed). You might have to define the data networks before starting the deployment itself. Maximum supported data networks are 4 (in Dell PowerFlex appliance and Dell PowerFlex rack) and 8 in the software-only offering.
System data network	System-wide object that applies to all protection domains. Once a resource group is deployed, configure system data networks/subnets to be used to connect hosts initiator to the storage data target. Maximum allowed system data networks

Table 24. Defined objects for load balancing (continued)

Object	Description
	are 8. Configure two or four system data networks for PowerFlex rack and PowerFlex appliance. The number that you need depends on the number of PowerFlex data (storage node-storage node and storage node-SDT communication) networks that are configured for the deployment.

If you have not defined the system data network, by default, a host can reach all system data networks. Ignoring the data networks/subnets may result in unequal load between the host initiator ports and nonoptimized I/O performance with the PowerFlex system. In addition, it may impact the path resiliency if not all the host initiator ports can connect to the system. For Layer-3 networks, the system must have routing tables configured.

Managing system data networks in PowerFlex Manager

The NVMe over TCP load balancer uses the data networks for balancing the host connections. This section summarizes the steps that you must perform in PowerFlex Manager to add, rename, or remove system data networks.

Table 25. Add, rename, or remove data networks in PowerFlex Manager

If you want to...	Do this in PowerFlex Manager
Add system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Enter the required information and click **Save**.
Rename system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Select the network and click **Rename**. 4. Rename the network and click **Apply**.
Remove system data networks	1. On the menu bar, click **Settings** > **Networking**. 2. Click **System Data networks**. 3. Select the network and click **Remove**. 4. Click **Remove** to remove the network.

Managing system data networks using SCLI

This section provides the SCLI commands that you use to add, rename, or remove system data networks.

For more information on the PowerFlex SCLI commands, see *Dell PowerFlex and PowerFlex Manager CLI Reference Guide*.

Table 26. Using SCLI to manage data networks

If you want to...	Use this SCLI command
Add system data networks	`scli --add_system_network --network_ip <IP> --network_mask <IP> [--network_name <NAME>] (--host_group_id <ID> | --host_group_name <NAME>)`
Rename system data networks	`scli --rename_system_network (--network_id <ID> | --network_name <NAME>) --new_name <NAME>`
Remove system data networks	`scli --remove_system_network (--network_id <ID> | --network_name <NAME>)`

Managing host groups using SCLI

This section provides the SCLI commands that you must perform to add, modify, rename, remove, or query host groups.

This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

The host group holds a subgroup of existing networks that are accessible to a group of hosts. Each host group is associated with a set of system networks. Each host may be associated with multiple host groups. By default, a host is part of the default host group until you create a new host group and associate the host to the new host group. By default, a host supports all data networks and you do not need to configure the host groups. The maximum number of host groups that are allowed per system is 1024.

Table 27. Managing host groups using SCLI

If you want to...	Use this SCLI command
Add host group	`scli --add_host_group [--host_group_name <NAME>] [(--network_id <ID> | --network_name <NAME>)] [(--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)]`
Modify host group	`scli --modify_host_group (--host_group_id <ID> | --host_group_name <NAME>) [(--network_id <ID> | --network_name <NAME>)]`
Rename host group	`scli --rename_host_group (--host_group_id <ID> | --host_group_name <NAME>) [--new_name <NAME>]`
Remove host group	`scli --remove_host_group (--host_group_id <ID> | --host_group_name <NAME>)`
Assign host to group	`scli --assign_host_to_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>) (--host_group_id <ID> | --host_group_name <NAME>)`
Remove host from group	`scli --remove_host_from_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)`
Query host group	`scli --query_host_group (--host_group_id <ID> | --host_group_name <NAME>)`
Query all host group	`scli --query_all_host_group`

Managing network sets using SCLI

This section provides the SCLI commands that you need to perform to add, modify, rename, remove, or query network sets.

This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

A network set is a set of networks that are connected to the same switch. This switch is configured for balancing the I/O traffic between access switches by associating system data networks to network set. The load balancer takes care of the distribution of networks over network sets.

For example, if a system is defined with two system data networks (192.168.150.0 and 192.168.151.0), then you could create two network sets and associate 192.168.150.0 with network set 1 and 192.168.151.0 to network set 2.

Each system data network belongs to exactly one network set. If the network was not assigned to a network set, the network will be considered a network set of its own.

Use the default network set and do not manually create the network set. Creating network sets is applicable for a software-only deployment.

Table 28. Managing network sets using SCLI

If you want to...	Use this SCLI command
Add network set	`scli --add_network_set [--network_set_name <NAME>] [(--network_id <ID> | --network_name <NAME>)]`
Modify network set	`scli --modify_network_set (--network_set_id <ID> | --network_set_name <NAME>) [(--network_id <ID> | --network_name <NAME>)]`
Rename network set	`scli --rename_network_set (--network_set_id <ID> | --network_set_name <NAME>) --new_name <NAME>`
Remove network set	`scli --remove_network_set (--network_set_id <ID> | --network_set_name <NAME>)`
Query network set	`scli --query_network_set (--network_set_id <ID> | --network_set_name <NAME>)`
Query all network sets	`scli --query_all_network_set`

Storage nodes

Add a device to storage nodes

Use this procedure to add a storage node.

Steps

1. Log in to PowerFlex Manager.
2. From the menu, click **Lifecycle** > **Storage nodes**.
3. In the list of storage nodes, select the node, and click **More Actions** > **Add devices**.
4. In the **Add Device to Storage Nodes** window, enter the **Path**, **Name**, and **Device Group**, and click **Add Row**.
5. Drop-down the **Advanced Options** and select **Test and Activate Device**, and click **Add Devices**.

Enter maintenance mode

Use this procedure to enter maintenance mode.

Steps

1. Log in to PowerFlex Manager.
2. Click **Lifecycle** > **Storage nodes**.
3. Select the checkbox of the storage node.
4. Click **More actions** > **Enter maintenance mode**.
5. On the **Enter storage node into maintenance mode** page, click **Enter maintenance mode**.
   A message appears that the storage node entered maintenance mode successfully. Click **Dismiss**.

Exit maintenance mode

Use this procedure to exit maintenance mode.

Steps

1. Log in to PowerFlex Manager.
2. Click **Lifecycle** > **Storage nodes**.
3. Select the checkbox of the storage node that is in maintenance mode.
4. Click **More actions** > **Exit maintenance mode**.
5. On the **Exit storage node into maintenance mode** page, click **Exit maintenance mode**.
   A message appears that the storage node exited maintenance mode successfully. Click **Dismiss**.

Remove the storage node

Use this procedure to remove the storage node.

Steps

1. Log in to PowerFlex Manager.
2. From the menu, click **Lifecycle** > **Storage nodes**.
3. Select the checkbox of the storage node to be removed.
4. Type **`REMOVE STORAGE NODE`** to confirm and click **Remove**.
5. Verify the operation has finished successfully and click **Dismiss**.

Reconfiguring MDM roles

PowerFlex Manager enables you to change the MDM role for a node in a PowerFlex cluster. For example, if you add a node to the cluster, you might want to switch the MDM role from an existing node to the new node.

About this task

NOTE: This is not applicable for PowerFlex in a PowerFlex management cluster (KVM), as there are only three nodes.

You can launch the wizard for reconfiguring MDM roles from the **Resource Groups** page or from the **Resources** page. The nodes that are listed and the operations available are the same regardless of where you launch the wizard.

Steps

1. To access the wizard from the **Resource Groups** page:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group that has the PowerFlex system for which you want to reconfigure MDM roles.
   c. In the right pane, click **View Details**.
   The **Resource Group Details** page is displayed.
   d. On the **Resource Group Details** page, click **Reconfigure MDM Roles** under **More Actions**.
   The **MDM Reconfiguration** page is displayed.
2. To access the wizard from the **Resources** page:
   a. On the menu bar, click **Lifecycle** > **Resources**.
   b. Select the PowerFlex system for which you want to reconfigure MDM roles.
   c. In the right pane, click **View Details**.
   The **Details** page is displayed for the selected PowerFlex gateway.
   d. On the **Details** page, click **Reconfigure MDM Roles**.
   The **MDM Reconfiguration** page is displayed.
3. Review the current MDM configuration for the cluster.
4. Find each MDM role that you want to reassign and choose the new hostname or IP address for the role in the **Select New Node for MDM Role** list.

You can reassign multiple roles at one time.

5. Click **Next**.
   The **Summary** page is displayed.
6. Type **CHANGE MDM ROLES** to confirm your changes.
7. Click **Finish**.

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Secure component verification and TPM enablement on the PowerFlex nodes

The secure component verification tool ensures that the server hardware has not been changed (a component moved or replaced) between leaving the Dell factory and arriving at the customer site. If any changes are made to the node, the secure component verification tool will report the difference.

Secure component verification is an optional feature, and TPM is enabled at the factory on the nodes. The following nodes are supported:

- PowerFlex R650 nodes
- PowerFlex R750 nodes
- PowerFlex R6525 nodes
- PowerFlex R7525 nodes
- PowerFlex R660 nodes
- PowerFlex R760 nodes
- PowerFlex R6625 nodes
- PowerFlex R7625 nodes
- PowerFlex R670 nodes
- PowerFlex R770 nodes
- PowerFlex R6725 nodes
- PowerFlex R7725 nodes

The secure component verification license is tagged with the service tag of the node, and is uploaded at the iDRAC from the factory.

Verify no hardware changes using secure component verification tool

Use the following procedure to verify no hardware changes using secure component verification tool.

Steps

1. Download the Windows or Linux version of the secure component verification tool from the Dell Technologies support site.
2. To run the secure component verification tool on Windows, double click the secure component verification software and click **unzip**.
   a. Navigate to the file path in file explorer, default `c:\SecureComponentVerification`, and click on setup to install.
   b. Click **Next**, accept the license, click **Next** > **Install**. Click **Finish**.
   c. In the command prompt, run the secure component verification commands using iDRAC credentials.

      ```
      C:\Users\Administrator>

      scv ValidateSystemInventory -r <idrac IP> -i

      Username: root

      Password:
      ```

   d. To check the logs, navigate to the file path that is displayed in the command prompt.
3. To run secure component verification on Linux, in the terminal, navigate to the directory where iDRAC tools file is downloaded and unzip the file: **`# tar -xvf Dell-SCV-Web-LX-1.91.0-308_A00.tar.gz`**
   a. Navigate to the directory `SCVTools` after the files have been extracted and execute the install_scv.sh script using the command **`sh install_scv.sh`**.

NOTE: To uninstall secure component verification tool you can use the command **`sh uninstall_scv.sh`** to execute the `uninstall_scv.sh` script.

b. Once secure component verification tool is installed, run the command **`scv ValidateSystemInventory`** to start the validation process.

NOTE: Use the command **`scv help`** to get more information on secure component verification tool and how to run it.

c. To check the logs, navigate to the file path that is displayed in the command prompt.

4. Once the system runs the secure component verification tool application successfully, it should give the result **Validating System Inventory: Match and Mismatch** if the component has mismatched.

Enabling secure boot and disabling Trusted Platform Module

This section provides information about enabling secure boot and disabling the Trusted Platform Module.

Enabling secure boot for PowerFlex nodes

The secure boot option is not available by default. Contact your Dell Technologies account representative to evaluate and enable the secure boot option for PowerFlex nodes through a request for product qualification (RPQ) process. For information on enabling secure boot, see PowerFlex: Enable Secure Boot Enforcement for PowerFlex Compute Nodes.

Disabling the Trusted Platform Module attestation alarm

Trusted Platform Module is enabled by default for PowerFlex nodes.

When the Trusted Platform Module is enabled and the secure boot is disabled, VMware ESXi raises a Trusted Platform Module attestation alarm prompting you to activate the secure boot option. You can disable the alarm or disable the Trusted Platform Module if not in use.

Disable the Trusted Platform Module attestation alarm

Use this procedure to disable the Trusted Platform Module attestation alarm in a VMware environment.

About this task

This alarm is raised when the secure boot option is disabled on the VMware ESXi host, irrespective of whether the Trusted Platform Module is used or not used in the environment.

Steps

1. Log in to the VMware vSphere Client.
2. Browse to the VMware vCenter Server object in the inventory.
3. Go to **Configure** > **Alarm Definitions**.
4. In the filter field, next to **Alarm Name**, type **TPM**.
5. Select the radio button next to **Host TPM attestation alarm**.
6. Click **Edit**.
7. In the alarm configuration dialogue box, click **Next** until you reach the **Review** screen.
8. Select the **Disable this alarm** option.
9. Click **Save** to apply the changes.

Disable the Trusted Platform Module

Use this procedure to disable the Trusted Platform Module.

Steps

1. Disable Trusted Platform Module security in iDRAC:
   a. Log in to the iDRAC of the host using a web browser.
   b. Click **Configuration** > **BIOS Settings** > **System Security**.
   c. Select **TPM Security** > **Off**.
   d. Click **Apply And Reboot** or **At Next Reboot**, as required.
2. Disable Trusted Platform Module security using command line:
   a. Open an administrator-level command prompt on a system with RACADM tools installed.
   b. Run the command **`racadm -r xx.xx.xx.xx -u root -p <password> set BIOS.SysSecurity.TpmSecurity Off`**.
   Sample output:

   ```
   RAC1017: Successfully modified the object value and the change is in pending state.
   ```

   c. Create a commit job, run **`racadm -r xx.xx.xx.xx -u root -p <password> jobqueue create BIOS.Setup.1-1`**.
   d. Reboot the host as necessary.

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Administering the CloudLink Center

NOTE: A CloudLink license is not required for CloudLink 8.3 and later.

Add the CloudLink Center license in PowerFlex Manager

Use this procedure to add CloudLink Center in PowerFlex Manager.

About this task

NOTE: Use the following procedure if the system is CloudLink 8.2 and earlier.

Steps

1. Log in to PowerFlex Manager.
2. Click **Settings** > **License Management**.
3. Click **Other Software Licenses**.
4. Click **Add**.
5. Under **Upload License**, click **Choose License**.
6. Browse and select the license to upload and click **Open**.
7. Select the **Type** as **CloudLink** and click **Save**.
8. From **Lifecycle** > **Resource**, select the CloudLink VMs, and click **Run inventory**.

Adding and managing CloudLink Center licenses

Perform the following procedures to add CloudLink Center licenses and manage CloudLink Center licenses through PowerFlex Manager.

NOTE: A CloudLink license is not required for CloudLink 8.3 and later.

License CloudLink Center

Use this procedure to add licenses to CloudLink Center.

About this task

CloudLink license files determine the number of machine instances, CPU sockets, encrypted storage capacity, or physical machines with self-encrypting drives (SEDs) that your organization can manage using CloudLink Center. License files also define the CloudLink Center usage duration.

NOTE: CloudLink center can act as a key management interoperability protocol (KMIP) server if you upload a KMIP license to it.

NOTE: Use the following procedure if the system is CloudLink 8.2 and earlier.

Steps

1. Log in to CloudLink Center.

2. Select **System** > **License**.
3. Click **Upload License**.
4. Browse to select the software or hardware encryption license file and click **Upload**.

   NOTE: If the CloudLink environment is managed by PowerFlex Manager, after you update the license, go to the **Resources** page, select the CloudLink Center VMs, and click **Run Inventory**.

Remove expired or unused CloudLink Center licenses from PowerFlex Manager

Use this procedure to remove expired or unused CloudLink Center licenses from PowerFlex Manager.

About this task

After upgrading to CloudLink 8.3 and later, the license can be removed from PowerFlex Manager.

Steps

1. Log in to PowerFlex Manager.
2. Click **Settings** > **License Management** > **Other Software Licenses**.
3. Select the license you want to remove, and click **Remove**.
4. From **Resource**, select the CloudLink VMs, and click **Run inventory**.

Configure syslog and audit logs in CloudLink

CloudLink supports four multiple syslog configurations from CloudLink 8.1. The audit log and system log are retrieved locally and available for 15 days.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog** > **Add Syslog Server**. The **Add** window appears.
   a. Provide the syslog IP address, port number, and select the facility and protocol. Click **Add**.

      NOTE: For audit logs, choose the facility as audit if the logs are transmitted using PowerFlex Manager.

   b. If the service status of the syslog server is not active, click **Resume**. In the **Confirm Syslog Mode Change** window, click **Resume**.

      The service status is Active.

      NOTE: The **Change Syslog Format** and **Postpone** options can only be done globally.

3. To retrieve the logs locally:
   a. To enable SSH from CloudLink, click **Server** > **Network** > **Change SSH**.
   b. Select the **Enabled** option and click **Change**.
   c. SSH to the CloudLink IP address. The logs are in the `/var/log/CLC_audit.log` file.

Manage a self-encrypting drive (SED) from CloudLink Center

Use this procedure to manage an SED device through CloudLink Center.

About this task

When managing SEDs from CloudLink Center, be aware of the following:

- CloudLink Center can manage encryption keys for self-encrypting drives (SEDs).
- Managing SEDs with CloudLink Center is functional when the CloudLink agent is installed on machines with SEDs.
- When managed by CloudLink Center, SED encryption keys are stored in the current keystore for the machine group they are in.
- The functionality for managing SEDs requires a separate SED license.
- If the SED cannot retrieve the key from CloudLink Center, the SED remains locked.

Steps

From the CloudLink Center, select **Agent** > **Machines**, click **Actions** and select **Manage SED**. Ownership of the encryption key is enabled.

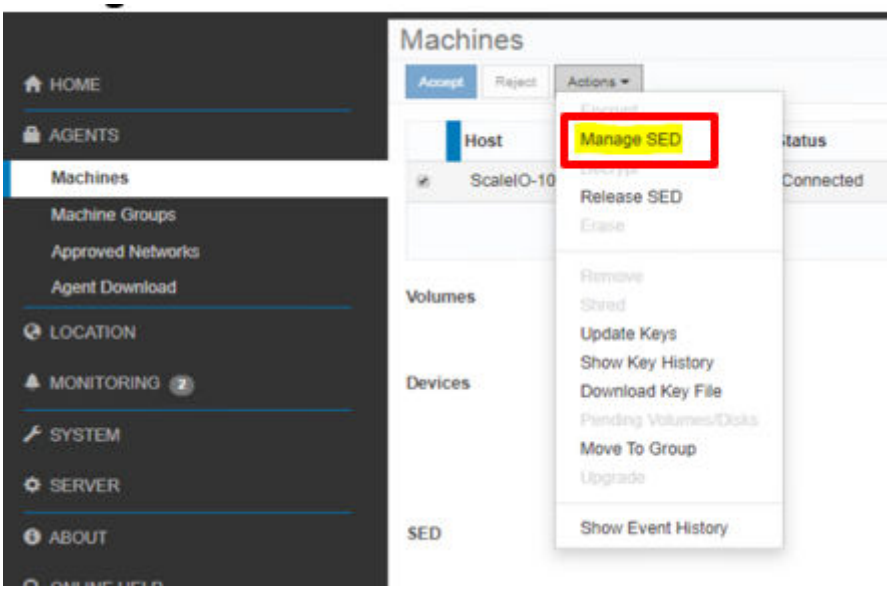


Figure 2. Manage SED selection

NOTE: This option is only available if an SED license is uploaded and an SED is detected in the physical machine managed by CloudLink Center. The **Manage SED** option does not change data on an SED it only takes ownership of the encryption key.

Manage a self-encrypting drive from the command line

As an alternative to CloudLink Center, use the command line to manage an SED.

Steps

1. Log in to the storage node.
2. To manage the SED from the command line, type **`svm manage [device name]`**.

   For example, **`svm manage /dev/sdb`**.

Release a self-encrypting drive

Use this procedure to release an SED that is managed by CloudLink.

About this task

This option allows you to release ownership of an SED that is managed by CloudLink. This option is only available if an SED license is uploaded and an SED is detected in the physical machine managed by CloudLink Center.

When CloudLink releases an SED, the encryption key is released in CloudLink Center.

Steps

1. From CloudLink Center, click **Agents** > **Machines** and select **Storage Nodes Machine**. Click **Release SED**.
2. From **RELEASE SED**, use the drop-down menu to select the SED drive that you want to release and click **Release**.

   The status of the SED drive changes to **Releasing Control**.

   Once CloudLink releases the control, the SED device status shows as **Unmanaged**.

   NOTE: The **Release SED** option does not change any data on the SED.

Release management of a self-encrypting drive from the command line

Use this procedure to release an SED using the command line.

Steps

1. Log in to the storage node.
2. To release the SED from the command line, type **`svm release [device name]`**.

   For example, **`svm release /dev/sdb`**.

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Backup and restore

This section contains instructions for backing up and restore for PowerFlex appliance components. Backups of collected data is critically important in data management which allows you to restore to a previously known state.

This section also provides instructions for backing up and restoring CloudLink Center.

VMware vCenter Server

Back up VMware vCenter Server

Manually backup the VMware vCenter.

Prerequisites

You must have an FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB server up and running with sufficient disk space to store the backup.

NOTE: If you back up a vCenter Server High Availability cluster, the backup operation only backs up the primary vCenter Server instance. Before restoring a vCenter Server High Availability cluster, you must power off the active, passive, and witness nodes. The restore operation restores the vCenter Server in non-vCenter Server High Availability mode. You must reconstruct the cluster after the restore operation completes successfully.

Steps

1. Go to the vCenter Server Management Interface, **`https://appliance-IP-address-or-FQDN:5480`** and log in as **`root`**.
2. In the vCenter Server Management Interface, click **Backup**.
3. Click **Backup Now**.
4. Enter the backup location details.
5. (Optional) Enter an encryption password if you want to encrypt your backup file.
   If you select to encrypt the backup data, you must use the encryption password for the restore procedure.
6. (Optional) Select **Stats**, **Events**, and **Tasks** to back up additional historical data from the database.
7. (Optional) In the Description text box, enter a description of the backup.
8. Click **Start** to begin the backup process.

Restore VMware vCenter Server

Perform this procedure to deploy the VMware vCenter Server Appliance planned for vCenter deployment.

About this task

Deploying VMware vCenter Server Appliance (vCSA) is a two-step process:

- Deploy a new appliance to the target VMware vCenter server or ESXi host
- Copy data from the source appliance to the VMware vCenter Server Appliance

Steps

1. Mount the ISO and open the VMware vCSA 7.x or 8.x installer from `\vcsa-ui-installer\win32\installer.exe`.
2. Select **Restore** from the VMware vCSA 7.x or 8.x installer.

3. Click **Next** in Stage 1: Deploy vCenter Server wizard.
4. Accept the End User License Agreement and click **Next**.
5. Enter the backup details:
   - Location: The following protocols are supported: FTP, FTPS, HTTP, HTTPS, SFTP, NFS, or SMB
   - ex. `ftp://<server_IP_address>/tmp/vCenter/hostname_vcenter.com/<backup_filename>`
   - Username: Enter the user name of a user with read privileges on the backup server
   - Password: Enter the password of the user with read privileges on the backup server
6. Click **Next**.
7. Review **Backup Details** and click **Next**.
8. Type the host FQDN of the PowerFlex management node:
   a. Provide all the log in credentials.
   b. Click **Next** > **Yes**.
9. Accept the certificate warning.
10. Select the deployment size to large and leave storage as default. Click **Next**.
11. In the **Select Datastore** page, select the datastore location.
12. Select **Enable Thin Disk Mode**.
13. Click **Next**.
14. Verify the network settings populated from the backup file of the vCenter Server.
15. Click **Next**.
16. Review the summary and click **Finish**.
17. Wait for the OVA deployment to finish.
18. Copy data from the source appliance to the VMware vCenter Server Appliance:
    a. Click **Continue** and click **Next** on the **Introduction** page.
    b. Review the backup details and click **Next**.
    c. Enter single sign-on username and password and click **Validate and Recover** > **Next**.
    d. On the Ready to complete page, review the details, and click **Finish** > **OK**.
    e. After the restore finishes, click **Close**.

PowerFlex Manager

Performing a backup saves all user-created data to a remote share from which it can be restored. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Perform frequent backups to guard against data loss and corruption. The best practice is to take a snapshot of PowerFlex Manager every time you perform a backup and restore, or before conducting any changes in the environment.

The **Backup** page displays information about the last backup operation that was performed on PowerFlex Manager. The information provided applies to both manual and automatically scheduled backups and includes the following:

- Last backup date
- Last backup status
- Back up directory path to a CIFS share
- Back up directory username

The **Backup** page also displays information about the status of automatically scheduled backups (enabled or disabled).

On this page, you can:

- Manually start an immediate backup
- Edit general backup settings
- Edit automatically scheduled backup settings

After performing a backup operation, you need to run a script outside of PowerFlex Manager to restore from the backup. The user interface does not support the ability to restore from a backup.

Edit the backup settings and details

You can change the location where backup files are saved or the password that is required to access a backup file.

Steps

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Settings**.
4. Indicate the network share location where the backup file is saved, enter a backup directory path in the **Backup Directory Path** box.
   Use the following format: CIFS—`\\host\share`
5. If the username and password are required to access the network share, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes.
6. Click **Test Connection** to confirm that the backup settings you provided are correct.
7. To open the backup file, enter a password in the **Encryption Password** box.
8. To verify the encryption password, enter the password in the **Confirm Encryption Password** box.
9. To schedule automatic backups, next to **Scheduled Backups**, select **Enabled**. To discontinue automatically scheduled backups, deselect **Enabled**.
10. To specify the days on which backup must occur, select the days under **Frequency**.
11. From the **Run Time** drop-down list, select the time.
12. Click **Save**.

Back up PowerFlex Manager

In addition to automatically scheduled backups, you can manually run an immediate backup.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Settings** > **Serviceability**.
3. Click **Backup**.
4. Click **Backup Now**.
5. Select one of the following options:
   - To use the general settings that are applied to all backup files, select **Use Backup Directory Path and Encryption Password from Settings and Details**.
   - To use custom settings:
     a. In the **Backup Directory Path** box, enter a file path where the backup file is saved. Use this format: CIFS—`\\host\share\`
     b. Optionally, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes, if they are required to access the location you provided.
     c. Click **Test Connection** to confirm that the backup settings you provided are correct.
     d. In the **Encryption Password** box, enter a password that is required to open the backup file, and verify the encryption password by entering the password in the **Confirm Encryption Password** box.
6. Click **Backup Now**.

Completing the restore of PowerFlex Manager

Restoring PowerFlex Manager returns user-created data to an earlier configuration that is saved in a backup file. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

About this task

CAUTION: Restoring an earlier configuration restarts PowerFlex Manager and deletes data created after the backup file to which you are restoring. Any running jobs could be terminated as well.

Perform frequent backups to prevent data loss and corruption. Perform a back up of PowerFlex Manager after completing a restore.

NOTE: Before removing the PowerFlex management platform installer VM, take a backup of the **`PFMP_Config.json`** file located at `/opt/dell/pfmp/PFMP_Installer/config`. The **`PFMP_Config.json`** file is required when restoring PowerFlex Manager.

Restore the `PFMP_Config.json` file to `/opt/dell/pfmp/PFMP_Installer/config` that was backed up before removing installer VM. This is typically backed up to the jump server.

Install the same version or release of the installer that deployed the PowerFlex management platform. See the *Dell PowerFlex Appliance with PowerFlex Deployment Guide* for more information.

Steps

1. From VMware vCSA, power on the PowerFlex management platform installer VM.
2. From the KVM host, power on the PowerFlex management platform installer VM, type **`sudo virsh start <VM name>`**.
3. Log in to the PowerFlex management platform installer VM.
4. Run the restore script that is included with the installer bundle:

   ```
   ./restore_backup.sh [BACKUP_LOCATION] [ENCRYPTION_PASSWORD] {CIFS_USERNAME]
   [CIFS_PASSWORD}
   ```

   The *BACKUP_LOCATION* path supports CIFS. The *CIFS_USERNAME* and *CIFS_PASSWORD* parameters are optional.

5. During a back up and restore, if it fails with the error `No such containter`, the required containers are not running on the installer VM:
   a. Type **`docker PS`** to check on the installer VM to see if the containers are running.
   b. To check if the containers are in an exited state, type **`docker start <container ID>`**.
   c. To check if the containers are in a started state, type **`docker ps -a`**.
   d. After all of the containers are running, start the restore process.

Results

The restore process prints out status information until the restore is complete. Perform a back up of PowerFlex Manager after completing the restore.

Restoring compliance file data

Use this procedure to return compliance files to a healthy state.

About this task

After performing a restore operation, the compliance files will be in an ERROR state. The compliance file data is not part of the backup payload, so after a restore, the compliance files appears in an error/unhealthy state. Any changes previously made to compliance files using single component upgrade, for example BIOS, iDRAC, or ESXi version modifications, will not be preserved through the backup and restore process.

Steps

1. Resynchronize the compliance file from the original source to restore it to a good state.

2. If any components were previously modified using single component upgrade (BIOS, iDRAC, ESXi depot), edit the compliance file using single component upgrade to reapply those changes.
3. Confirm that the compliance file is no longer in ERROR state and all components reflect the expected versions.

Integrated Dell Remote Access Controller

Back up iDRAC

Use this procedure to back up iDRAC.

Steps

1. Log in to the iDRAC using root credentials.
2. Click **Configuration** > **Server Configuration Profile**.
3. Click **Export**.
4. Select one of the following to specify the location type:
   - **Local** to save the configuration file on a local drive.
   - **Network Share** to save the configuration file on a CIFS or NFS share.
   - **HTTP or HTTPS** to save the configuration file to a local file using HTTP/HTTPS file transfer.

   Depending on the location type, you must enter the Network Settings or HTTP/HTTPS settings. If proxy is configured for HTTP/HTTPS, proxy settings are also required.
5. Select the components that you need to back up the configuration for.
6. Select the **Export** type:
   - Basic
   - Replacement Export
   - Clone Export
7. Select an **Export file format**.
8. Select **Additional export items**.
9. Repeat the steps on all the PowerFlex nodes.

Restore iDRAC

Steps

1. Click **Configuration** > **Server Configuration Profile**.
2. Select one of the following to specify the location type:
   - **Local** to save the configuration file on a local drive.
   - **Network Share** to save the configuration file on a CIFS or NFS share.
   - **HTTP or HTTPS** to save the configuration file to a local file using HTTP/HTTPS file transfer.

   Depending on the location type, you must enter the Network Settings or HTTP/HTTPS settings. If proxy is configured for HTTP/HTTPS, proxy settings are also required.
3. Select the components listed in the **Imported Components** option.
4. Select the **Shutdown** type.
5. Select the Maximum wait time to specify the wait time before the system shuts down after the import is complete.
6. Click **Import**.

Network switch configuration back up and restore

Back up and restore running configuration of the Cisco and Dell switches.

About this task

If switches are owned by customers, advise customers to take the switch backup immediately after successful PowerFlex appliance deployment.

NOTE: The customer is responsible for providing the backup location and maintaining the backup as well. Dell Technologies recommends storing the backup in a separate shared location, for example, not on the jump server.

Steps

1. Connect to the Cisco Nexus or Dell PowerSwitch switch, either using the console cable, Telnet, or SSH using admin credentials, type **`#copy running-config scheme://server/[url]/filename`**.

   For the scheme argument, you can enter **`tftp:`**,**`ftp:`**, **`scp:`**, or **`sftp:`**.

   The server argument is the address or name of the remote server, and the URL argument is the path to the source file on the remote server. The server, URL, and file name arguments are case-sensitive.

   For example:

   ```
   switch# copy running-config tftp://10.10.10.1/sw1-run-config.bak
   switch# copy running-configuration scp://root:calvin@10.11.10.12/tmp/backup.txt
   ```

2. Restore the network configuration, connect to the Cisco Nexus or Dell PowerSwitch switch, either using the console cable, Telnet, or SSH using admin credentials, type **`#copy scheme://server/[url]/filename running-config`**.

   For the scheme argument, you can enter **`tftp:`**,**`ftp:`**, **`scp:`**, or **`sftp:`**.

   The server argument is the address or name of the remote server, and the URL argument is the path to the source file on the remote server. The server, URL, and file name arguments are case-sensitive.

   For example:

   ```
   switch# copy tftp://10.10.10.1/my-config running-config
   switch# copy scp://root:calvin@10.11.10.12/tmp/backup.txt running-configuration
   ```

Backing up and restoring CloudLink Center

Use this procedure to create a CloudLink Center backup file.

View backup information

You can view the Backup page information.

To view the backup information, log in to the CloudLink Center, and click **System** > **Backup**. The **Backup** page lists the following information:

- **Backup File Prefix**—Prefix used for the backup files.
- **Current Key ID** —The identifier for the current RSA-2048 key pair.
- **Current Backup File** —The name of the current backup file.
- **Current Backup Time**—The date and time that the current backup file was generated.
- **Backup Schedule**—The schedule for generating automatic backups.
- **Next Backup In** —The time remaining before the next automatic backup is generated.

When a backup file is downloaded, the **Backup** page lists the following additional information:

- **Last Downloaded File** —The name of the backup file that was last downloaded. Only shown when a backup file has been downloaded.
- **Last Downloaded Time** —The date and time of the last backup file download. Only shown when a backup file has been downloaded.
- **Backup Store** —The backup store configuration type. If you have not configured a backup store, the value is Local, which is stored on the local desktop.

You can also use the FTP or SFTP servers as backup stores. To change the backup store, click **Actions** > **Change Backup Store**.

If you have configured an FTP or SFTP backup store, the following additional information is available:

- **Host** —The remote FTP, SFTP, or FTPS host where you saved the CloudLink Center backups. You can set this value to the host IP address or hostname (if DNS is configured).
- **Port**—The port used to access the backup store.
- **User**—The user with permission to access the backup store.
- **Directory**—The directory in the backup store where backup files are available.

Change the schedule for automatic backups

CloudLink Center automatically generates a backup file each day at midnight (UTC time).

To change the schedule for generating automatic backups, click **System** > **Backup** > **Actions** > **Change Backup Schedule**.

Back up CloudLink Center

Use this procedure to generate a back up manually, if you want to preserve CloudLink Center before the next automatic back up. Download the back up file when you generate a back up file manually.

Steps

1. Log in to the CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Generate New Backup**. A back up file is generated.

Generate a backup key pair

You can generate a new backup key pair. For example, if the private key for a backup key pair is lost, you can generate a new key pair. You cannot access your backup files without the associated private key. When you generate a new key pair, CloudLink Center automatically generates a new backup file to ensure that the current backup can be opened with the private key of the current key pair.

Dell recommends the following practices when you generate a new backup key pair.

1. Download the private key to the Downloads folder for the current user account. For example, `C:\Users\Admnistrator\Downloads`.

   NOTE: The previously generated backup key will not open the backup file created, after a new key is generated.

2. Generate the new key pair by clicking **System** > **Backup** > **Actions** > **Generate and Download New Key**.
3. Click **Generate and Download**.

Download the current backup file

You can download the current backup file at any time. The current backup file is either:

- The last backup file that CloudLink Center automatically created.
- The last backup file that you manually generated after the last automatic backup.

To download the backup file:

1. Click **System** > **Backup** > **Actions** > **Download Backup**.
2. In the **Download Current Backup** dialog box, click **Download**.

When you download the current backup file, CloudLink Center shows the age of the backup file.

Restore the CloudLink backup

Use this procedure to restore the CloudLink backup.

Steps

1. Log in to the CloudLink Center.
2. Click **System** > **Backup** > **Actions** > **Restore Keystores**.
3. In the **Restore Keystores** dialog box, complete the following steps:
   a. In the **Key** box, browse to the private key file.
   b. In the **Backup** box, browse to the backup file.
   c. In the **Unlock** box, type the passcode that was set during the initial configuration of the CloudLink Center.
   d. Click **Restore**.
   A **Restore Keystores succeeded** message is displayed.

   NOTE: If the CloudLink backup is not associated with a key pair, the file is corrupted or key mismatch error message is displayed. In such a scenario, generate a new key pair and download the backup file.

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Powering on and off

Power on a PowerFlex appliance

To safely power on the system, power on one component at a time, in the order specified in this procedure.

PowerFlex appliance provides the following management controller options. Refer to the appropriate sections for powering on.

- PowerFlex management cluster (KVM)
- Single ESXi PowerFlex management node
- Customer-provided infrastructure for PowerFlex management (KVM/Linux)
- Customer-provided infrastructure for PowerFlex management (ESXi)

Power on the PowerFlex management cluster (KVM)

Use this procedure to power on the PowerFlex management cluster (KVM).

About this task

Power on workflow for the PowerFlex management cluster (KVM):

- Power on the PDUs
- Power on the switches
- Power on the PowerFlex management node as part of PowerFlex management cluster (KVM)
- Verify that the management virtual machines running on BOSS storage (Three management virtual machines are used for managing the management infrastructure) are powered on
- Activate protection domain
- Power on the management virtual machines running on PowerFlex storage (Three management virtual machines are used for managing the production infrastructure)
- Power on all other virtual machines:
  - Jump VMs
  - CloudLink Center (optional)
  - Secure connect gateway (optional)
  - Policy manager (optional)

Powering on must be completed in this order for the components that you have in your environment. Prioritize and power on the PowerFlex storage-only nodes with the PowerFlex metadata manager (MDM) first.

Steps

1. Log in to the iDRAC and power on all the PowerFlex management nodes.
   Monitor the virtual console and wait for the SLES server to appear.
2. Log in to eachPowerFlex management node and verify that all the PowerFlex services are active and running, type:
   `sudo systemctl status lia`
   `sudo systemctl status mdm`
   `sudo systemctl status scini`
   `sudo systemctl status activemq` (MDMs only)
3. Confirm that the management virtual machines are running, type **`sudo crm cluster run "virsh list"`**.
4. Log in to the management virtual machines running the management PowerFlex management platform processes:
   a. To check the status of the rke2-server, type **`> systemctl status rke2-server`**
   b. Do the following depending on the rke2-server status:

Table 29. Status of the rke2-server

Status of the rke-2 server	Do the following
active	Go to the next step.
activating	Repeat the command to check the rke2-server status until active.
failed	Attempt to start the service, type **> systemctl start rke2-server**

5. Once the rke2-server is running on all the PowerFlex management platform nodes, log in to the management virtual machines on the first host to ensure the nodes are in **ready** state. Type **> kubectl get nodes**.

   If you see an error message, wait a few minutes and try again. Once the nodes are in **ready** state, go to the next step.
6. To gather the IP address of the database service, type:

   # export NAMESPACE=powerflex

   # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
7. To start the database pods and service, type:

   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartup"
8. To check the status of the startup, type:

   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartupstatus"
9. Continue to run the status startup command until the status returns verified, see the following example:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"startup_status":"verified"}
```

10. Log in to PowerFlex Manager managing the PowerFlex management cluster KVM:
    a. Go to **Monitoring** > **Management Diagnostics** > **Diagnostics**.

       The diagnostic page opens which displays PowerFlex management platform diagnostic cards.
    b. In the result icon, last run date, run time, and run state are listed for each diagnostic card. Click **Run Now** to manually start a diagnostic.

       A notification message confirms that the specified diagnostic has started. After a diagnostic run begins, you can cancel it. If there are errors, a failure message displays.
    c. The overview displays the run state, start time, and end time. The diagnostic result displays failure reasons, recommendations to remediate failures, and raw output only if errors occur. Click **View Raw Output** for more information.
11. Check for the status of any ongoing rebalance activity prior to power on steps using the **Dashboard** or the **Block** tab.
12. Activate the management protection domain:
    a. Click **Block** > **Protection Domain** and select the box to the left of the protection domain.
    b. Click **Protection Domain**.
    c. Select the **Protection Domain**.
    d. Click **More Actions**.
    e. Leave **Force Activate** set to **No**.
    f. Click **Activate**.

       In the resulting dialog box, click **Activate**, and wait for the status to change to Active.
13. Log in to any management controller using SSH.
14. Begin the CRM startup:
    a. Type **sudo crm status**.

       The output should show three nodes in an online state, and various resources in a Stopped (Disabled) state.
    b. Start the resources, type:

       sudo crm resource start stonith-sbd

       sudo crm resource start cl-dlm

c. Validate that the stonith device and dynamic lock manager have started before proceeding, type **sudo crm status**.

   The output should show **Started** for **stonith-sbd** and **cl-dlm** objects. If they are not, contact Dell Technologies Support.

d. Start the remaining crm resources, type:

   sudo crm resource start cl-shared-lock

   sudo crm resource start cl-shared-fs1 (repeat for all shared FSes)

   sudo crm resource start <resource name> (repeat for all resources)

e. Validate that all resources are in a running state, type **sudo crm status**.

15. Power on the management virtual machines running on PowerFlex storage (Three management virtual machines are used for managing the production infrastructure).
16. Log in to the PowerFlex management cluster node as **delladmin**.
17. Obtain a list management virtual machines and nodes and type **$ sudo crm status**.

    Record the node on which each MVM resource is running.
18. Verify the status of PowerFlex management platform.
19. SSH to any of the management virtual machines:
    a. To check the status of the rke2-server, type **> systemctl status rke2-server**
    b. Do the following depending on the rke2-server status:

Table 30. Status of the rke2-server

Status of the rke-2 server	Do the following
active	Go to the next step.
activating	Repeat the command to check the rke2-server status until active.
failed	Attempt to start the service, type **> systemctl start rke2-server**

20. Once the rke2-server is running on all the PowerFlex management platform nodes, log in to the management virtual machines on the first host to ensure the nodes are in **ready** state. Type **> kubectl get nodes**.

    If you see an error message, wait a few minutes and try again. Once the nodes are in **ready** state, go to the next step.
21. To gather the IP address of the database service, type:

    # export NAMESPACE=powerflex

    # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
22. To start the database pods and service, type:

    # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartup"
23. To check the status of the startup, type:

    # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartupstatus"
24. Continue to run the status startup command until the status returns verified, see the following example:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"startup_status":"verified"}
```

25. To identify the port number of the PowerFlex management platform monitor utility, type:

    kubectl get services monitor-app -n powerflex -o jsonpath="{.spec.ports[0].nodePort}{\"\n\"}"

    Wait for 20-30 minutes and check the overall health status of the PowerFlex management platform.
26. Go to **http://<node IP>:port/**.

    Where the node IP address is a management IP address that is configured on any of the management virtual machines (not the ingress or PowerFlex Manager IP address).
27. Click **PFMP status** and wait for all entries to turn green.

Contact Dell Technologies Support if the PowerFlex management platform status persists as red, or the main user interface is not accessible after 20 to 30 minutes.

Power on the PowerFlex management cluster (KVM) compute-only extension (ESXi-based)

Use this procedure to power on the PowerFlex management cluster (KVM) compute-only extension (ESXi-based) (optional).

Steps

1. Log in to the iDRAC and power on the node.
2. Identify the host where the VMware vCenter server is running and make a note of the IP address or hostname.
3. Log in to the VMware ESXi host client and power on the VMware vCenter server virtual machine:
   a. Log in to the VMware vSphere client.
   b. Click **vSphere Client** > **Shortcuts** > **Hosts and Clusters** and browse to the VMware vCenter virtual machine.
   c. Make note of the host IP address from the **Summary** page.
4. Power on all of the management virtual machines.

Power on the single ESXi PowerFlex management node

Use this procedure to power on the single ESXi PowerFlex management node.

About this task

Power on workflow for the single ESXi PowerFlex management node:

- Power on the PDUs
- Power on the switches
- Power on the single ESXi PowerFlex management node
- Power on the VMware vCenter server (if not automatically started with VMware ESXi)
- Power on the management virtual machines running on the VMware ESXi node (Three management virtual machines are used for managing the production infrastructure)

Steps

1. Log in to the iDRAC and power on all the PowerFlex management nodes.
   Monitor the virtual console and wait for the VMware ESXi server to appear.
2. Power on the VMware vCenter server:
   a. Log in to VMware ESXi using the host client.
   b. Click **Virtual Machines**, select the VMware vCenter server virtual machine, and click **Power on**.
3. Power on all of the PowerFlex management virtual machines that manage the production PowerFlex cluster:
   a. Log in to the VMware vCenter server using the VMware vSphere client.
   b. Click **Virtual Machines**, select the management virtual machine, and click **Power on**.
   c. Repeat to power on each management virtual machine.
4. Power on the virtual machines:
   - Jump server virtual machine
   - CloudLink virtual machines (if applicable)
   - Secure connect gateway virtual machines (if applicable)
   - Policy manager virtual machines (if applicable)
5. Log in to the management virtual machines running the production PowerFlex management platform processes:
   a. To check the status of the rke2-server, type `> systemctl status rke2-server`
   b. Do the following depending on the rke2-server status:

Table 31. Status of the rke2-server

Status of the rke-2 server	Do the following
active	Go to the next step.
activating	Repeat the command to check the rke2-server status until active.
failed	Attempt to start the service, type **> systemctl start rke2-server**

6. Once the rke2-server is running on all the PowerFlex management platform nodes, log in to the management virtual machines on the first host to ensure the nodes are in **ready** state. Type **> kubectl get nodes**.

   If you see an error message, wait a few minutes and try again. Once the nodes are in **ready** state, go to the next step.
7. To gather the IP address of the database service, type:

   # export NAMESPACE=powerflex

   # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
8. To start the database pods and service, type:

   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartup"
9. To check the status of the startup, type:

   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartupstatus"
10. Continue to run the status startup command until the status returns verified, see the following example:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"startup_status":"verified"}
```

11. To identify the port number of the PowerFlex management platform monitor utility, type:

    kubectl get services monitor-app -n powerflex -o jsonpath="{.spec.ports[0].nodePort}{\"\n\"}"

    Wait for 20-30 minutes and check the overall health status of the PowerFlex management platform.
12. Go to **http://<node IP>:port/**.

    Where the node IP address is a management IP address that is configured on any of the management virtual machines (not the ingress or PowerFlex Manager IP address).
13. Click **PFMP status** and wait for all entries to turn green.

    Contact Dell Technologies Support if the PowerFlex management platform status persists as red, or the main user interface is not accessible after 20 to 30 minutes.

Power on the customer-provided infrastructure for PowerFlex management (ESXi)

Use this procedure to power on the customer-provided infrastructure for PowerFlex management (ESXi).

About this task

Power on workflow: Power on the production management virtual machines (MVMs) running on the VMware ESXi cluster.

Steps

1. Power on all of the PowerFlex management virtual machines that manage the production PowerFlex cluster:
   a. Log in to the VMware vCenter server using the VMware vSphere client.
   b. Click **Virtual Machines**, select the management virtual machine, and click **Power on**.
   c. Repeat to power on each management virtual machine.
2. Power on the virtual machines:

- Jump server virtual machine
- CloudLink virtual machines (if applicable)
- Secure connect gateway virtual machines (if applicable)
- Policy manager virtual machines (if applicable)

3. Log in to the management virtual machines running the management PowerFlex management platform processes:
   a. To check the status of the rke2-server, type **> systemctl status rke2-server**
   b. Do the following depending on the rke2-server status:

Table 32. Status of the rke2-server

Status of the rke-2 server	Do the following
active	Go to the next step.
activating	Repeat the command to check the rke2-server status until active.
failed	Attempt to start the service, type **> systemctl start rke2-server**

4. Once the rke2-server is running on all the PowerFlex management platform nodes, log in to the management virtual machines on the first host to ensure the nodes are in **ready** state. Type **> kubectl get nodes**.

   If you see an error message, wait a few minutes and try again. Once the nodes are in **ready** state, go to the next step.

5. To gather the IP address of the database service, type:

   # export NAMESPACE=powerflex

   # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')

6. To start the database pods and service, type:

   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartup"

7. To check the status of the startup, type:

   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartupstatus"

8. Continue to run the status startup command until the status returns verified, see the following example:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"startup_status":"verified"}
```

9. To identify the port number of the PowerFlex management platform monitor utility, type:

   kubectl get services monitor-app -n powerflex -o jsonpath="{.spec.ports[0].nodePort}{\"\n\"}"

   Wait for 20-30 minutes and check the overall health status of the PowerFlex management platform.

10. Go to **http://<node IP>:port/**.

    Where the node IP address is a management IP address that is configured on any of the management virtual machines (not the ingress or PowerFlex Manager IP address).

11. Click **PFMP status** and wait for all entries to turn green.

    Contact Dell Technologies Support if the PowerFlex management platform status persists as red, or the main user interface is not accessible after 20 to 30 minutes.

Power on the customer-provided infrastructure for PowerFlex management (KVM/Linux)

Use this procedure to power on the customer-provided infrastructure for PowerFlex management (KVM/Linux).

About this task

Power on workflow: Power on all of the PowerFlex management virtual machines (MVMs) running on the management virtual machines (MVMs) that manage the production PowerFlex cluster.

Steps

1. Power on all of the PowerFlex management virtual machines, type **sudo virsh start <MVM name>**.
2. Confirm that the management virtual machines are running, type `sudo virsh list-- all`.
3. Log in to the management virtual machines running the management PowerFlex management platform processes:
   a. To check the status of the rke2-server, type **> systemctl status rke2-server**
   b. Do the following depending on the rke2-server status:

Table 33. Status of the rke2-server

Status of the rke-2 server	Do the following
active	Go to the next step.
activating	Repeat the command to check the rke2-server status until active.
failed	Attempt to start the service, type **> systemctl start rke2-server**

4. Once the rke2-server is running on all the PowerFlex management platform nodes, log in to the management virtual machines on the first host to ensure the nodes are in **ready** state. Type **> kubectl get nodes**.

   If you see an error message, wait a few minutes and try again. Once the nodes are in **ready** state, go to the next step.
5. To gather the IP address of the database service, type:

   # export NAMESPACE=powerflex

   # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
6. To start the database pods and service, type:

   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartup"
7. To check the status of the startup, type:

   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclusterstartupstatus"
8. Continue to run the status startup command until the status returns verified, see the following example:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"startup_status":"verified"}
```

9. To identify the port number of the PowerFlex management platform monitor utility, type:

   kubectl get services monitor-app -n powerflex -o jsonpath="{.spec.ports[0].nodePort}{\"\n\"}"

   Wait for 20-30 minutes and check the overall health status of the PowerFlex management platform.
10. Go to **http://<node IP>:port/**.

    Where the node IP address is a management IP address that is configured on any of the management virtual machines (not the ingress or PowerFlex Manager IP address).
11. Click **PFMP status** and wait for all entries to turn green.

Contact Dell Technologies Support if the PowerFlex management platform status persists as red, or the main user interface is not accessible after 20 to 30 minutes.

Power on the VMware NSX Edge nodes

Use this procedure to power on the VMware NSX Edge nodes (if applicable).

Steps

1. Power on the VMware NSX Edge nodes.
2. Verify that VMware ESXi has booted and you can ping the management IP address.
3. Power on the VMware NSX Edge virtual machines.

Power on PowerFlex storage-only nodes

Use this procedure to power on PowerFlex storage-only nodes.

Steps

1. From iDRAC, power on all PowerFlex storage-only nodes and allow them time to boot completely.
2. Log in to PowerFlex Manager:
   a. Click **Lifecycle** > **Storage Nodes**. Verify all the storage nodes are healthy.
   b. Click **Block** > **Storage System** > **Devices**. Verify all the devices are healthy.
   c. Click **Block** > **Storage System** > **Protection Domains**, select the protection domain, and click **More actions** > **Activate**.
3. Log in to all host iDRACs as `root` and confirm the NTP, syslog, and SNMP settings.
4. Use SSH to connect to all network switches.
5. To verify that connected interfaces are not in a **not connected** state, type:

   ```
   show interface status
   ```

6. Do the following:
   a. Log in to the jump server on the controller stack.
   b. At the command prompt or the terminal, type **`nslookup`**, against the correct DNS and verify that the DNS is correct.

      For example, `nslookup eagles-r640-f-158.lab.vce.com (node hostname) 10.234.134.100 (dns server ip address).`
7. Log in to the management PowerFlex Manager and check for the status of any ongoing rebalance activity prior to power on steps using the **Dashboard** or the **Block** tab.

Power on PowerFlex compute-only nodes

Use this procedure to power on PowerFlex compute-only nodes with VMware ESXi (if applicable).

Steps

1. From iDRAC, power on all PowerFlex compute-only nodes and allow them time to boot completely.
2. For PowerFlex compute-only nodes with VMware ESXi:
   a. Log in to the VMware vSphere Client if the PowerFlex appliance includes VMware vSphere.
   b. Take each PowerFlex compute-only node with VMware ESXi out of maintenance mode.
3. Power on the VMware vCenter server.
4. Power on all other virtual machines.

Complete the powering on of PowerFlex appliance

Complete the power on process for PowerFlex compute-only nodes with VMware ESXi (if applicable).

Steps

1. From VMware vCenter, power on the remaining virtual machines of all PowerFlex compute-only nodes with VMware ESXi.
2. From the VMware vSphere Client:
   a. Rescan to rediscover datastores.
   b. Power on the remaining virtual machines.
3. For VMware vSphere, enable HA, DRS, and affinity rules.
4. Resume replication if using:
   a. On the menu bar, click **Protection**.
   b. Click the relevant replication consistency group check box, and click **More Actions** > **Resume**.
   c. In the **Resume RCG** dialog box, click **Resume**.
   d. Verify that the operation has finished and was successful, and click **Dismiss**.

Power off a PowerFlex appliance

To safely power off the system, power off one component at a time, in the order specified in this procedure.

PowerFlex appliance provides the following management controller options. Refer to the appropriate sections for powering off.

- PowerFlex management cluster (KVM)
- Single ESXi PowerFlex management node
- Customer-provided infrastructure for PowerFlex management (KVM/Linux)
- Customer-provided infrastructure for PowerFlex management (ESXi)

Power off workflow:

- Power off the management virtual machines running on PowerFlex storage
  - Check PowerFlex health, and rebuild status
  - Power off all workload virtual machines on VMware vCenter (except the management virtual machine for PowerFlex management cluster (KVM) and VMware vCenter virtual machine)
  - Power off VMware NSX Edge nodes
  - Power off PowerFlex compute-only nodes with VMware ESXi
  - Inactivate the protection domain from production PowerFlex Manager 5.0.x
  - Power off PowerFlex storage-only nodes
  - Power off all infrastructure virtual machines (except the first set of management virtual machines deployed on BOSS for PowerFlex management cluster (KVM))
  - Power off all other virtual machines:
    - Jump server virtual machines
    - CloudLink Center
    - Secure connect gateway
    - Policy manager
    - UCC (optional)
  - Power off all three management virtual machines running on PowerFlex storage
  - Power off the extended PowerFlex compute-only nodes (if applicable)
  - Inactivate the protection domain from production PowerFlex Manager 4.x
  - Power off all three management virtual machines running on BOSS
  - Power off the PowerFlex management node
  - Power off PDUs
  - Power off the switches

Power off PowerFlex compute-only nodes

Use this procedure to power off PowerFlex compute-only nodes with VMware ESXi (if applicable).

Steps

1. Log in to the VMware vCenter, click **Home**, and click **Inventory**.
2. Shut down the virtual machines, except the VMware vCenter server virtual machine.
3. Disable DRS and HA on the customer cluster.
4. Place all PowerFlex compute-only nodes running VMware ESXi into maintenance mode, except for the node hosting the VMware vCenter server.
5. Power off the PowerFlex compute-only nodes with VMware ESXi, except for the node hosting the VMware vCenter server.
6. Log in to the VMware ESXi host client, the node hosting the VMware vCenter server and shut down the VMware vCenter server.
7. Shut down the compute-only node currently hosting the VMware vCenter server.

Power off the VMware NSX Edge nodes

Use this procedure to power off the VMware NSX Edge nodes.

Steps

1. In the management VMware vCenter, right-click the NSX Edge virtual machines and click **Power** > **Shut down Guest OS**.
2. In the management VMware vCenter, right-click the NSX Edge virtual machines and click **Power** > **Shut down**.

Power off the PowerFlex storage-only nodes

Use this procedure to power off the PowerFlex storage-only nodes.

Steps

1. Check PowerFlex health and rebuild status:
   a. Log in to PowerFlex Manager and select **Block** > **Storage System** > **Protection Domain**, and choose the respective protection domain and locate the **Rebuild/Rebalance** wizard on the right-hand side of the screen.
   b. Confirm there is no error and rebuild or rebalance is running.
2. Pause replication if using:
   a. On the menu bar, click **Protection**.
   b. Click the relevant replication consistancy group check box, and click **More Actions** > **Pause**.
   c. In the **Pause RCG** dialog box, click one of the following options:
      - **Stop Data Transfer**: this option saves all the data in the source journal volume until there is no longer any available capacity.
      - **Track Changes**: this option enables manual slim mode where only metadata in the source journal volumes is saved.
   d. Click **Pause**.
   e. Verify that the operation has finished and was successful, and click **Dismiss**.
3. Shut down all of the customer virtual machines running on the PowerFlex production cluster.
4. Inactivate the protection domain using PowerFlex Manager:
   a. Log in to PowerFlex Manager.
   b. Click **Block** > **Storage System** > **Protection Domain**.
   c. For each protection domain, click **More Actions** > **Inactivate**.
   d. Click **OK** and type the administrator password when prompted. Repeat for each protection domain and verify that each is inactivated.
   e. Repeat for each protection domain and verify that each is inactivated.
5. Log in to iDRAC and gracefully shut down the PowerFlex storage-only nodes.

Inactivate the protection domain using PowerFlex Manager

Use this procedure to inactivate the protection domains using PowerFlex Manager.

About this task

This procedure is only applicable to the PowerFlex production cluster.

Steps

1. Log in to PowerFlex Manager.
2. Click **Block** > **Storage System** > **Protection Domain**.
3. For each protection domain, click **More Actions** > **Inactivate**.
4. From the **Inactivate Protection Domain** pop-up window, click **Inactivate**.

Power off the PowerFlex management cluster (KVM) compute-only extension (ESXi-based)

Use this procedure to power off the PowerFlex management cluster (KVM) compute-only extension (ESXi-based) (optional).

Steps

1. Identify the host where the VMware vCenter server is running and make a note of the IP address or hostname.

   If VMware vCenter high availability is enabled, make a note of the IP addresses of the hosts where active, passive, or witness nodes are running.
2. Log in to the VMware ESXi host client and power on the VMware vCenter server virtual machine:
   a. Log in to the VMware vSphere client.
   b. Click **vSphere Client** > **Shortcuts** > **Hosts and Clusters** and browse to the VMware vCenter virtual machine.
   c. Make note of the host IP address from the **Summary** page.
3. Shut down all of the management virtual machines gracefully.
4. Shut down the VMware vCenter server virtual machine gracefully. If you have VMware vCenter high availability that is configured, shut down the active, passive, and witness nodes:
   a. Log in to the VMware ESXi host client.
   b. Click **Virtual Machines** > **Select the vCenter VM** and click to open a browser console.
   c. Press **F12** to shut down the virtual machine.
   d. Enter the root password and click **OK**.
5. Shut down the VMware ESXi servers gracefully:
   a. Log in to the VMware ESXi host client.
   b. Right-click **Host** and click **Shut down**.
   c. To confirm the shut down of the selected host, click **Shut down**.
   d. Repeat for all remaining nodes.

Power off the PowerFlex management cluster (KVM)

Power off the management virtual machines running on PowerFlex storage (three management virtual machines are used for managing the production infrastructure).

Prerequisites

Ensure that your production environment is gracefully shut down before proceeding to power off the management infrastructure. Ensure you have powered off the PowerFlex storage-only nodes, the PowerFlex compute-only nodes, the VMware NSX Edge notes, and the PowerFlex management cluster (KVM) compute-only extension (ESXi-based).

All workloads should be halted before you power off.

Steps

1. Shut down the PowerFlex management cluster (KVM)
   a. Log in to any of the management virtual machines.
   b. To gather the IP address of the database service, type:

      ```
      # export NAMESPACE=powerflex
      # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
      ```

   c. To shut down the database pods and service, type:

      ```
      # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdown"
      ```

   d. To check the status of the shutdown, type:

      ```
      # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdownstatus"
      ```

   e. Continue to run the status shut down command until the status returns verified, see the following example:

      ```
      HTTP/1.1 200 OK
      Date: Mon, 05 Aug 2024 18:36:14 GMT
      Content-Length: 47
      Content-Type: text/plain; charset=utf-8
      {"return_status":0,"shutdown_status":"verified"}
      ```

2. Shut down the CRM cluster:
   a. Log in to any one of the PowerFlex management cluster nodes.
   b. Stop the CRM cluster services and resources, type **`sudo crm cluster stop --all`**.

      This stops all virtual machines running on management powerflex storage, logical volumes, shared file system, and the stonith device (SBD watchdog). This must occur prior to inactivating the management PowerFlex protection domain. This step may take some time, due to shutdowns of the virtual machines.

Inactivate the protection domain on PowerFlex management cluster nodes

Use this procedure to inactivate the protection domains on PowerFlex management cluster nodes.

Steps

1. Log in to the management PowerFlex Manager.
2. Click **Block** > **Protection Domains**.
3. For each protection domain, click **More Actions** > **Inactivate**.
4. From the **Inactivate Protection Domain** pop-up window, click **Inactivate**.
5. Wait for the state to change to **Inactive**.

Power off the management virtual machines running on BOSS storage

Power off the management virtual machines running on BOSS storage (three management vitual machines are used for managing the management infrastructure).

Steps

1. To gather the IP address of the database service, type:

   ```
   # export NAMESPACE=powerflex
   # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
   ```

2. To shut down the database pods and service, type:

   ```
   # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdown"
   ```

3. To check the status of the shutdown, type:

```
# curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdownstatus"
```

4. Continue to run the status shut down command until the status returns verified, see the following example:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"shutdown_status":"verified"}
```

5. Shut down the management virtual machines running on BOSS:
   a. On each PowerFlex management cluster, type

      sudo virsh list

      sudo virsh shutdown <MGMT MVM name on host>

      The graceful shutdown can take one to two minutes.
   b. Type **sudo virsh list --all** and monitor the output until all of the virtual machines display in shutdown state.
   c. Type **sudo poweroff** to power off all hosts.

Power off the single ESXi PowerFlex management node

Use this procedure to power off the single ESXi PowerFlex management node.

Prerequisites

Ensure that your production environment is gracefully shut down before proceeding to power off the management infrastructure. Ensure you have powered off the PowerFlex storage-only nodes, the PowerFlex compute-only nodes, the VMware NSX Edge notes, and the PowerFlex management cluster (KVM) compute-only extension (ESXi-based).

All workloads should be halted before you power off.

Steps

1. Power off all workloads virtual machines including customer virtual machines, jump server virtual machines, CloudLink Center, secure connect gateway, and policy manager, on VMware vCenter except the production management virtual machines.
2. Shut down the management platform which manages the production PowerFlex cluster.
   a. Log in to any of the management virtual machines.
   b. To gather the IP address of the database service, type:

      # export NAMESPACE=powerflex

      # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
   c. To shut down the database pods and service, type:

      # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdown"
   d. To check the status of the startup, type:

      # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdownstatus"
   e. Continue to run the status shut down command until the status returns verified, see the following example:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"shutdown_status":"verified"}
```

3. Gracefully shut down all three management virtual machines.
4. Power off all VMware ESXi hosts.

Power off the customer-provided infrastructure for PowerFlex management (ESXi)

Use this procedure to power off the customer-provided infrastructure for PowerFlex management (ESXi).

Prerequisites

Ensure that your production environment is gracefully shut down before proceeding to power off the management infrastructure. Ensure you have powered off the PowerFlex storage-only nodes, the PowerFlex compute-only nodes, the VMware NSX Edge notes, and the PowerFlex management cluster (KVM) compute-only extension (ESXi-based).

Steps

1. Shut down the management platform which manages the production PowerFlex cluster.
   a. Log in to any of the management virtual machines.
   b. To gather the IP address of the database service, type:

      ```
      # export NAMESPACE=powerflex
      # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
      ```

   c. To shut down the database pods and service, type:

      ```
      # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdown"
      ```

   d. To check the status of the shut down, type:

      ```
      # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdownstatus"
      ```

   e. Continue to run the status shut down command until the status returns verified, see the following example:

      ```
      HTTP/1.1 200 OK
      Date: Mon, 05 Aug 2024 18:36:14 GMT
      Content-Length: 47
      Content-Type: text/plain; charset=utf-8
      {"return_status":0,"shutdown_status":"verified"}
      ```

2. Gracefully shut down all three management virtual machines.

Power off the customer-provided infrastructure for PowerFlex management (KVM/Linux)

Use this procedure to power off the customer-provided infrastructure for PowerFlex management (KVM/Linux).

Prerequisites

Ensure that your production environment is gracefully shut down before proceeding to power off the management infrastructure. Ensure you have powered off the PowerFlex storage-only nodes, the PowerFlex compute-only nodes, the VMware NSX Edge notes, and the PowerFlex management cluster (KVM) compute-only extension (ESXi-based).

Steps

1. Shut down the management platform which manages the production PowerFlex cluster.
   a. Log in to any of the management virtual machines.
   b. To gather the IP address of the database service, type:

      ```
      # export NAMESPACE=powerflex
      # export DBSA_IP=$(kubectl get service/postgres-monitor-leader -n $NAMESPACE -o jsonpath='{.spec.clusterIP}')
      ```

   c. To shut down the database pods and service, type:

      ```
      # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdown"
      ```

   d. To check the status of the shut down, type:

      ```
      # curl -i -H "Accept: application/json" "http://$DBSA_IP:8080/pgclustershutdownstatus"
      ```

e. Continue to run the status shut down command until the status returns verified, see the following example:

```
HTTP/1.1 200 OK
Date: Mon, 05 Aug 2024 18:36:14 GMT
Content-Length: 47
Content-Type: text/plain; charset=utf-8
{"return_status":0,"shutdown_status":"verified"}
```

2. Gracefully shut down all three management virtual machines.

Complete the powering off of PowerFlex appliance

Use this procedure to complete the powering off of your PowerFlex appliance. Ensure that switches and PDUs are not powered off if other nodes remain connected to the environment, in order to prevent disruption to dependent systems.

Prerequisites

Ensure you complete a back up before powering off.

Steps

1. Connect to all of the switches using SSH:
   For Cisco Nexus switches, type **`copy running-config startup-config`**.
   For Dell PowerSwitch switches, type **`copy running-config tftp://hostip/filepath`**.
2. On Zone B (BLUE), turn off all PDU power breakers (OPEN position).
3. On Zone A (RED), turn off all PDU power breakers (OPEN position).
4. To verify there is no power beyond the PDUs, disconnect the AC feeds to all PDUs.

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Credentials management

Proper credential management is an essential part of PowerFlex Manager and the ability to manage, deploy, and upgrade the PowerFlex environment.

Administrators of PowerFlex should routinely verify the credentials within PowerFlex Manager are updated for each component in inventory. Any password changes that occur external to PowerFlex Manager need to be reflected within PowerFlex Manager by updating the associated credential to ensure product functionality. It is recommended to validate credentials before performing any deployment or upgrade procedures.

Using PowerFlex Manager to manage credentials

Update the PowerFlex Manager password

When you first log in to PowerFlex Manager, you need to set your password. You can also change your password at any time after the first login.

Steps

1. Click the user icon in the upper right corner of PowerFlex Manager.
2. Click **Change Password**.
3. Type the password in the **New Password** field.
4. Type the password again in the **Verify Password** field.
5. Click **Apply**.

Create credentials for supported system resources

Use this procedure to create credentials for root and non-root users in PowerFlex Manager.

Prerequisites

To import and create the SSH keys for a PowerFlex node, switch, **OS Admin**, **OS User**, ensure you generate SSH key pairs of RSA type without passphrase.

About this task

You can now use a non-root user instead of the root user for PowerFlex system administration functions. This enhances security by disabling the root user during node discovery, operating system installation, and non-disruptive updates. The default non-root username is **`pflex`**.

NOTE: The credential type **OS Admin** is used for root users, and **OS User** is used for non-root user. **OS Admin** and **OS User** credential types apply for the deployment resource groups.

For non-root user authentication, after the deployment of the resource group, the SSH access to the user root is disabled, password is still available to take console access for troubleshooting.

NOTE: PowerFlex Manager requires iDRAC username and password for all Redfish API communication. SSH keys may be uploaded to the iDRAC credential object, but their use is limited to SSH/racadm access. Keys do not replace the need for a password in Redfish operations.

PowerFlex Manager allows you to specify a non-root user when you configure a template for a compute-only or a storage-only deployment.

PowerFlex Manager allows you to use an LDAP user for PowerFlex system administration functions. When you create or edit an operating system user credential, you can optionally specify the LDAP domain. This allows you to use an active directory (AD) user rather than a local user for administration functions.

NOTE: If you are using **Windows Active Directory (AD)** as an **LDAP server**, the hostname cannot exceed **15** characters. A hostname that exceeds the 15-character limit may cause issues during the deployment and integration.

Steps

1. On the menu bar, click **Settings** > **Security**.
2. Click **Resource Credentials**. The Credentials Management page opens.
3. Click **Create**.
4. In the **Create Credentials** dialog box, from the **Credential Type** drop-down list, select one of the following resource types for which you want to create non-root credentials:
   - **Node**
   - **Switch**
   - **vCenter**
   - **Element Manager**
   - **Power Distribution Unit (PDU)**
   - **PowerFlex Gateway**
   - **OS Admin**
   - **OS User**
   - **PowerFlex Management System**
   - **Webhook**

   The **OS Admin** and **OS User** credential types apply to deployed items, not to PowerFlex Manager. If you are creating an OS user credential set for the management virtual machines on a PowerFlex management cluster resource group, select **OS User**.
5. In the **Credential Name** field, enter the name to identify the credential.
6. Click **Enable Key Pairs** to enable log in with SSH key pairs and perform the following:

Table 34. Key pairs options

To...	Do this...
Enable key pairs for the **Node** or **Switch credential**:	a. Click **Import SSH Key Pair**. **NOTE:** Manually generate the SSH keys pairs. b. Click **Choose File** and browse to the file that contains the private key. c. Type a name for the key pair. d. Click **Import**.
Create keys using PowerFlex Manager for the **OS Admin** or **OS User** credential and enable key pairs:	a. Click **Create a new key**. b. Click **Create & Download Key Pair**. c. On **Key Pair Name**, type the name for the key pair. d. Click **Create**. e. Click **Download Public Key**.
To manually generate and import existing key pair for the **OS Admin** or **OS User** credential:	a. Click **Import SSH Key Pair**. **NOTE:** Manually generate the SSH keys pairs. b. Click **Choose File** and browse to the file that contains the public and private key. c. Type a name for the key pair. d. Click **Import**.

If you enable SSH key pairs for a **Node** or **Switch credential** and use that credential for discovery, PowerFlex Manager uses public or private RSA key pairs to SSH into your node or switch securely, instead of using a username and password.

If you enable SSH key pairs for an **OS User** or **OS Admin** credential and use that credential for a deployment, PowerFlex Manager uses RSA public or private key pairs for the deployment operations.

NOTE: PowerFlex Manager does not consume SSH keys for all component types. For example, if you enable SSH key pairs for an admin credential, the SSH keys are not used for the deployment of a CloudLink Center VM. Instead, the user name and password are used instead for all communication.

7. To enable LDAP for an **OS User** (optional):
   a. On the **Create Credentials** page, in the **Credential type** field, enter **`OS User`**.
   b. In the **Credential Name** field, enter **`LDAP`**.
   c. Enter the domain name and username in the **Domain** and **User Name** fields.
   d. Enter the passwords and click **Save**.

   NOTE: Create username with domain name on active directory server. NTP server and active directory server time must sync. Configure DNS server and prefix on the management network configuration.

8. In the **User Name** field, enter the username for the credential.

   For **Nodes** (iDRAC), **`root`** is the only valid username for root-level credentials. For a non-root username, enter the default non-root username.

   For the **OS Admin** credential type, the **User Name** field is disabled because the user is assumed to be root. You must use the root user for new deployments.

   For the **OS User** credential type, enter the default non-root username.

   For the embedded operating system, this user account must have SSH enabled and have sudo access. For VMware ESXi, the account must be configured with the administrator role on the local server permission setting, which should enable SSH and other tools like esxcli. You can add existing resource groups with a non-root user. The account on the PowerFlex storage-only nodes for the **OS User** credential type must have a `/home` directory and have the correct group permissions.

9. In the **Password** and the **Confirm Password** boxes, enter the password for the credential.

Related information

Generate SSH key pairs

Generate SSH key pairs

To create the SSH keys for a PowerFlex node, switch, **OS Admin**, **OS User**, use this procedure to manually generate the SSH keys that are of RSA type and ensure the key is generated without a passphrase.

Steps

1. Generate a private key using the following command:

   `ssh-keygen -b 2048 -t rsa`

2. Enter a file path to save the public and private keys or proceed with the default file path without a passphrase:

   Sample output:

```
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase: # do not enter a passphrase
Enter same passphrase again: # do not enter a passphrase
```

   Sample output that indicates the public and private keys are saved in the default file path:

```
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:fKIllqp5UXeuEW7xuYeE7PNHpvjAUatNx6I/bisRqRU root@PFMP2-4
The key's randomart image is:
+---[RSA 2048]----+
|                 |
|        E        |
|         o.      |
|       .oB..o    |
|      .+*SO=.o   |
```

```
|     .o.=BO=oo     |
|     ...o=*.*      |
|    o.    *o* o    |
|   o.      B*=     |
+----[SHA256]-----+
```

3. Verify that the public and private keys are saved in the default file path: `/root/.ssh/`.

   Sample output:

```
cd /root/.ssh
ls
```

   - `id_rsa.pub` indicates a public key.
   - `id_rsa` indicates a private key.

Related information

Create credentials for supported system resources
Update a credential in PowerFlex Manager

Update a credential in PowerFlex Manager

If the password for a resource or component is manually changed outside of PowerFlex Manager, you must update the credential in PowerFlex Manager with the new password. Administrators should also consider any external tools that may push new passwords to components routinely. If this is occuring, the credentials that PowerFlex Manager uses to communicate with those components also needs to be updated at that time.

Prerequisites

To modify the SSH keys of a PowerFlex node, switch, OS, admin OS user, generate SSH keys of RSA type and ensure the key is generated without passphrase.

About this task

If you change the password for a resource, you must update all similar resources (for example, element manager, node, switches, or VMware vCenter) to have the same password.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Settings** > **Security**. Under **Security**, click **Resource Credentials**.
3. On the **Resource Credentials** page, select the resource whose password you want to edit, and click **Modify**.
4. In the **Edit Resource Credentials** dialog box, perform the following to import new SSH keys:
   a. Generate the SSH key pairs.
   b. Click **Import existing key**.
   c. Click **Import SSH Key Pair**.
   d. Click **Choose File** and browse to the file that contains your public and private key.
   e. Type a name for the **Key Pair Name field**.
   f. Click **Import**.
   g. Click **Save**.
5. Click **Save**.
6. On the menu bar, click **Lifecycle** > **Resource Groups** and select the deployed service that contains the resource whose password is updated.
7. Under **More Actions**, click **Update Resource Group Details**.

Related information

Generate SSH key pairs

Managing local and external user accounts

The **User Management** page allows you to manage local users, LDAP users, and directory services for PowerFlex Manager. This controls who can log in to PowerFlex Manager and identifies the type of user role to assign to each user or group.

Under **Settings** > **User Management**, you can find three pages:

- Local Users
- LDAP Users
- Directory Services

User roles (erasure coding)

User roles control the activities that different types of users can perform when using PowerFlex Manager.

Ensure that you configure the active directory before assigning roles. The roles that can be assigned to local users and LDAP users are identical. Each user can only be assigned one role. If an LDAP user is assigned directly to a user role and also to a group role, the LDAP user is provided with permissions of both roles.

The following table summarizes the activities that can be performed for each user role:

Table 35. User roles and activities

Role	Description	Activities
SuperUser	A SuperUser can perform all system operations.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Manage users, certificates • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
SystemAdmin	A SystemAdmin can perform all operations, except for user management and security ones.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
StorageAdmin	A StorageAdmin can perform all storage-related front-end operations including block system operations, such as create a volume. **NOTE:** Operations such as create storage pool cannot be performed by the StorageAdmin role, but can be performed by the LifecycleAdmin role.	• Manage storage resources • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
LifecycleAdmin	A LifecycleAdmin can manage the life cycle of hardware and PowerFlex systems.	• Manage life cycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations

Table 35. User roles and activities (continued)

Role	Description	Activities
		• View resource groups and templates • System monitoring (events, alerts)
SnapshotManager	SnapshotManager is a subset of StorageAdmin, working only on existing systems. This role includes all operations that are required to set up and manage snapshots.	• View storage configurations, resource details • System monitoring (events, alerts)
SecurityAdmin	The SecurityAdmin manages PowerFlex role-based access control (RBAC), and LDAP user federation. It includes all security aspects of the system.	• Manage users, certificates • System monitoring (events, alerts)
Technician	The Technician can perform all hardware FRU operations on the system, including entering a node into maintenance mode.	• Replace drives • Hardware operations • System monitoring (events, alerts) • Perform serviceability operations
DriveReplacer	This is a subset of the technician role. The DriveReplacer can perform only the operations that are required for a drive replacement.	• Replace drives • System monitoring (events, alerts)
Monitor	The monitor role has read-only access to the system, including topology, alerts, events, and metrics.	• View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
Support	The support role is a special kind of SystemAdmin (all activities except for user/ security management operations) to be used only by Dell Technologies support staff and developers. This user role has access to undocumented, special operations and options for common operations, required only for support purposes. **NOTE:** This special role should be used only by Dell Technologies support. It opens special, often dangerous, commands for advanced troubleshooting.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Special Dell Technologies Support operations

Mapping 5.0 or later roles to legacy roles

For reference, the tables below map user and group roles from earlier releases to the roles used in the current release.

Role mapping per legacy user interface

Table 36. Role mapping

Legacy role (prior to version 5.0)	New role (from 4.0 and later)
Monitor	Monitor
Back-end configurator	LifecycleAdmin
Frontend configurator	StorageAdmin
Configurator	SystemAdmin
Security	SecurityAdmin
Administrator	StorageAdmin
local Super User	SuperUser
technician commands	Support

Table 36. Role mapping (continued)

Legacy role (prior to version 5.0)	New role (from 4.0 and later)
Administrator	SuperUser
Read-only	Monitor
Standard owner	LifecycleAdmin
Standard member	LifecycleAdmin
Operator	DriveReplacer

Legacy LDAP users and groups

LDAP user role permissions were modified in version 4.0. For reference purposes, the following table shows the mapping of legacy users and groups to the roles now supported by .

Prior to version 5.0, mapped LDAP users and groups to roles. This mapping allowed assignment of multiple roles to a single group. Assignment of multiple roles to a group is no longer supported.

All legacy roles have a proximate hierarchy. The group's single role is the highest ranking role.

Table 37. LDAP hierarchy

Ranking	Legacy LDAP role (earlier than v4.0)	v4.0 or later group role without security officer permissions	v4.0 or later group role with security officer permissions
1	Super User	N/A (local user)	N/A (local user)
2	Administrator	Admin	SuperUser
3	Configurator	Admin	SuperUser
4	Frontend configuration and backend configuration	Admin	SuperUser
5	Frontend configuration or backend configuration	StorageAdmin or LifecycleAdmin	SuperUser
6	Monitor	Monitor	SecurityAdmin
Not ranked	Security Officer (SE)	SecurityAdmin	SecurityAdmin

Local users

You can create and manage local users within PowerFlex Manager.

Creating a user

Perform this task to create a local user and assign a role to that user.

Steps

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, click **Create**.
4. Enter a unique **User Name** to identify the user account.
5. Enter the **First Name** and **Last Name** of the user.
6. Enter the **Email** address.
7. Enter a **New Password** that a user enters to access PowerFlex Manager. Confirm the password in the **Verify Password** field.

   The password must be at least 8 characters long and contain one lowercase letter, one uppercase letter, one number, and one special character. Passwords cannot contain a username or email address.
8. In the **User Role** box, select a user role. Options include:

- SuperUser
- SystemAdmin
- StorageAdmin
- LifecycleAdmin
- SnapshotManager
- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support

9. Select **Enable User** to create the account with an `Enabled` status, or clear this option to create the account with a `Disabled` status.
10. Click **Submit** and click **Dismiss**.

Results

PowerFlex Manager creates the new user with the specified password. The first time you login with the new user and password, PowerFlex Manager asks you to change the password.

Modifying a user

Perform this task to edit a PowerFlex Manager user profile.

Steps

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select the user account that you want to edit.
4. Click **Modify**. For security purpose, confirm your password before editing the user.
5. You can modify the following user account information from this window:
   - **First Name**
   - **Last Name**
   - **Username**
   - **User Role**
   - **Email**
   - **Enable User**

   If you select the **Enable user** checkbox, the user can log in to PowerFlex Manager. If you disable the checkbox, the user cannot log in.
6. Click **Submit** and click **Dismiss**.

Deleting a user

Perform this procedure to remove an existing local user.

Steps

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select one or more user accounts to delete.
4. Click **Delete**.

   Click **Apply** in the warning message to delete the user. Click **Dismiss**.

Resetting the password for a user

You can reset the password for a local user in PowerFlex Manager.

Steps

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select one user account to reset the password.
4. Click **Reset Password**.
5. Enter **New Password** and **Verify Password**.
6. If you wish to set the password as a temporary one, check **Temporary Password**.
7. Click **Apply** and click **Dismiss**.

LDAP users

You can add and modify LDAP users and groups in PowerFlex Manager.

The LDAP realm name must not exceed the 155 character limit defined by Microsoft.

Directory services

You can create a directory service that PowerFlex Manager can access to authenticate users.

An Active Directory or Open LDAP user is authenticated against the specific directory domain to which a user belongs.

The **Directory Services** page displays the following information about PowerFlex Manager active directories:

- LDAP configuration
- User search settings
- Group search settings

From this page, you can:

- Add a directory service (only available when no service is defined in the system)
- Modify a directory service
- Remove a directory service

Add a directory service

Add a directory service for user authentication.

About this task

The hostname character limitation is 15 characters.

Steps

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Add**.
4. For **LDAP Configuration**, configure the following:
   a. In the **Address** box, enter the address of the authentication server.
      The address must be specified in URL-like format:
      - Enter **`ldap://`*`HOSTNAME or IP ADDRESS`*** for a plaintext LDAP connection.
      - Enter **`ldaps://`*`HOSTNAME or IP ADDRESS`*** for a secure LDAP connection.

      For example: **`ldap://100.68.68.1`**
   b. In the **Bind DN** box, enter the bind distinguished name attributes.
      The Bind Distinguished Name (DN) uniquely identifies an entry and its position in the hierarchy of entries contained in a directory server.

The user account entered here is the user that PowerFlex Manager uses to connect to your directory service.

For example, for the administrator account in the **Users** folder of **asm.delllabs.com**, enter the following: **CN=administrator,CN=Users,DC=asm,DC=delllabs,DC=net**.

c. In the **Bind DN Password** box, enter the Bind DN password.

This is the password used to access the LDAP server.

d. In the **Timeout** box, enter a value in milliseconds.
For example: **1000**

5. For **User Search Settings**, configure the following:

a. In the **Username LDAP Attribute** box, enter the name of an LDAP attribute that is mapped as the username. For many LDAP servers, it can be **uid**. For Active Directory, it can be **sAMAccountName** or **cn**. The attribute should be filled in for all LDAP users you want to import from LDAP to PowerFlex.

NOTE: If you are using Microsoft Windows Active Directory (AD) as an LDAP server, the hostname cannot exceed 15 characters. A hostname that exceeds the 15-character limit may cause issues during deployment and integration.

For example: **sAMAccountName**

b. In the **ID Attribute** box, enter the ID attribute for users.
For example: **sAMAccountName**

c. In the **Object Class** box, enter an object class.
For example: **top,person,organizationalPerson,user**

d. In the **Search Path** box, enter the search path.

The search path is used to identify and retrieve entries in the directory information tree that match a set of criteria.

For example: **CN=Users,DC=asm,DC=delllabs,DC=net**

6. For **Group Search Settings**, configure the following:

a. In the **Group Member Attribute** box, enter a group member name.
For example: **member**

b. In the **Group ID Attribute** box, enter the group ID.
For example: **name**

c. In the **Group Object Class** box, enter the group object class.
For example: **group**

d. In the **Group Search Path** box, enter the group search path.

The search path is used to identify and retrieve entries in the directory information tree that match a set of criteria for groups.

For example: **CN=Users,DC=asm,DC=delllabs,DC=net**

7. Click **Test Connection**.

If the test is successful, the **Submit** button will become active. If the test fails, you will not be able to proceed until you fix the connectivity issue.

8. When you have finished making your changes, click **Submit**.

Modify a directory service

The **Modify** option allows you to edit the existing directory service settings.

Steps

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Modify**.
4. In the **LDAP Settings** dialog box, edit the desired fields.

Note that the **Bind DN Password** must be reentered.

5. When you have finished making your changes, click **Test Connection**.

If the test is successful, the **Submit** button becomes active. If the test fails, you will not be able to proceed until you fix the connectivity issue.

6. Click **Submit**.

Remove a directory service

The **Remove** option allows you to remove the directory service configuration from .

Steps

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Remove**.
4. In the warning dialog box, click **Submit**.

Add LDAP users or groups

Add existing LDAP users or groups to PowerFlex, and assign roles to them.

About this task

These roles control access permissions for the corresponding LDAP user or group. These users and groups must also be configured in the directory service. On the PowerFlex side, these users and groups are mapped to PowerFlex roles.

Steps

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory Services**.
3. Click **Add**.
4. In the **Add LDAP User/Group** dialog box, select a **Type** option.
   - `User`—a user definition will be configured for an individual user.
   - `Group`—a group definition will be configured for a specific group.
5. In the **User Name** box, enter the user or group name.
6. In the **User Role** box, select the role to be assigned to the user or group. Options include:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - ReplicationManager
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician
   - Monitor
   - Support
7. Click **Apply**.

Modify an LDAP user or group

Modify the user role of an LDAP user or group. User roles control access permissions to the corresponding user or group.

Steps

1. On the menu bar, click **Settings** > **LDAP Users**.
2. Click **Modify**.
3. In the **Modify LDAP User/Group** dialog box, change the user role by selecting one of the **User Role** options:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - ReplicationManager
   - SnapshotManager

- SecurityAdmin
- DriveReplacer
- Technician
- Monitor
- Support

4. Click **Apply**.

Modify PowerFlex Manager credentials for resources

Updating passwords for nodes

You can update the passwords for one or more nodes from PowerFlex Manager.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Resources**.
3. On the **All Resources** tab, select one or more nodes for which you want to change the passwords.
4. Click **Update Password**.
   PowerFlex Manager displays the **Update Password** wizard.
5. On the **Select Components** page, specify which passwords you want to update for the selected nodes by clicking one or more of the following check boxes.
   - **iDRAC Password**
   - **Node Operating System Password**
   - **MVM Operating System Password**
6. Click **Next**.
7. On the **Select Credentials** page, create a credential with a new password or change to a different credential.
   a. Open the **iDRAC (*n*)** object under the **Type** column to see details about each node you selected on the **Resources** page.
   b. To create a credential that has the new password, click the plus sign (**+**) under the **Credentials** column.

      Specify the **Credential Name** and the **User Name** for which you want to change the password. Enter the new password in the **Password** and **Confirm Password** fields.

   c. To modify the credential, click the pencil icon for the nodes under the **Credentials** column and select a different credential.
   d. Click **Save**.

   You must perform the same steps for the node operating system and MVM operating system password changes. For a node operating system credential, only the OS Admin credential type is updated.
8. Click **Finish**.
9. Click **Yes** to confirm.

Results

PowerFlex Manager starts a new job for the password update operation, and a separate job for the device inventory. The node operating system operating components are updated only if PowerFlex Manager is managing a cluster with the operating system. If PowerFlex Manager is not managing a cluster with these components, these components are not displayed and their credentials are not updated. Credential updates for iDRAC are allowed for managed and reserved nodes only. Unmanaged nodes do not provide the option to update credentials.

Updating passwords for system components with PowerFlex Manager

You can update the passwords for some system components from PowerFlex Manager.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Resources**.
3. On the **All Resources** tab, select one or more resources of the same type for which you want to change passwords.
   For example, you could select one or more PowerFlex gateway components.
4. Click **Update Password**.
   PowerFlex Manager displays the **Update Password** wizard.
5. On the **Select Components** page, select one or more components for which you want to update a password and click **Next**.
   The component choices vary depending on which resource type you initially selected on the **Resources** page.
6. On the **Select Credentials** page, create a credential or change to a different credential having the same username.
7. Click **Finish** and click **Yes** to confirm the changes.

Update credentials on the management virtual machine cluster

Modify the root user to delladmin on the management virtual machine cluster

Use this procedure to change the root user to delladmin on the deployed management virtual machines.

About this task

The default username of the management virtual machines is delladmin.

Prerequisites

- It is recommended to take a snapshot of each of the management virtual machines.
- Log in using SSH to the management virtual machines using the delladmin credentials, and type kubectl get nodes. If this command fails with a permission issue, you must follow this procedure.

Steps

1. Log in to all three management virtual machines and to allow the user delladmin to execute the kubectl commands, type:
   `sudo chown delladmin /etc/pki/trust/anchors`
   `sudo chown delladmin /etc/rancher/rke2`
   `sudo chown delladmin /etc/rancher/rke2/*`
   `sudo chown delladmin /etc/docker/certs.d/*`
   `sudo chown delladmin /etc/helmrepo/certs.d/*`
2. To block the root SSH access, make an edit in the `/etc/shh/sshd_config` file on all three management virtual machines.
   a. Type **`vi /etc/shh/sshd_config`**
   b. Change **PermitRootLogin** to **No**.
   c. Type **ESC+wq** to save.
   d. To restart the sshd service, type **`systemctl restart sshd`**
3. Log in to the management virtual machine using the default credentials to run the kubectl commands.

4. For PowerFlex Manager 4.6 or earlier, after changing the credentials to delladmin, the backup will fail as the system uses the **root** account to collect logs. To update the management virtual machine credentials in PowerFlex Manager, go to *Update the management virtual machine credentials for backup in PowerFlex Manager*.

   For PowerFlex Manager 4.6 or later, the PowerFlex Manager user interface allows the user to enter the credentials.

Update management virtual machine credentials for backup in PowerFlex Manager

Use this procedure to update management virtual machine credentials for backup in PowerFlex Manager.

About this task

To update the OS password after deployment, see the KB article.

Prerequisites

To verify the credentials of the management virtual machine, type **kubectl -n powerflex get secret platform-nodes-credentials -o jsonpath='{.data.platform_nodes_credentials\.json}' | base64 -d | jq**

Steps

1. Log in to the one of the management virtual machines using the default username.
2. To extract the existing credential, type **kubectl -n powerflex get secret platform-nodes-credentials -o jsonpath='{.data.platform_nodes_credentials\.json}' | base64 -d > platform_nodes_credentials.json**

   The `platform_nodes_credentials.json` is the file name.
3. In the extracted file, change the credentials:
   a. Type **vi platform_nodes_credentials.json**.
   b. Change the credentials and type **ESC+wq** to save.
4. Decode the edited file and store them into a variable `new_creds_b64`, type **new_creds_b64=$(cat platform_nodes_credentials.json |base64 | tr -d '\n'; echo)**
5. To create a new secret patch file, type **echo '[{"op":"replace","path":"/data/platform_nodes_credentials.json","value":"'"$new_creds_b64"'"}]' > secret-patch.json**
6. To patch the secret file, type **kubectl -n powerflex patch secret platform-nodes-credentials --type='json' --patch "$(cat secret-patch.json)"**
7. To verify the credentials are updated, type **kubectl -n powerflex get secret platform-nodes-credentials -o jsonpath='{.data.platform_nodes_credentials\.json}' | base64 -d | jq**

Maintaining PowerFlex Manager credential resources

Changing a VMware ESXi host root password

For security reasons it might be necessary to change the password for the root user on a VMware ESXi host after installation.

Use any of the following methods to change the root password for the VMware ESXi host:

- VMware vSphere Client
- VMware ESXi shell command
- VMware ESXi host System Customization menu

Change the password using VMware vSphere Client

Steps

1. Log in to the host UI https://esxi_IP_address using root credentials.
2. In the left pane, under **Host**, click **Manage**.

3. Select the **Security & Users** tab, click **Users** and click the **root** user.
4. Click **Edit User**.
5. In the **Edit User** dialog box, enter and confirm a new password.
6. Click **Save**.

Change the password using the VMware ESXi shell command

Prerequisites

Log in to the VMware ESXi host service console as **root** user.

Steps

1. When prompted, enter the current password.
2. To change the root password, enter: **passwd root**.
3. Enter the new root password and press **Enter**.
4. Verify the password by entering it again.

Change the password using the VMware ESXi host system customization menu

Prerequisites

Log in to the VMware ESXi host service console as **root** user.

You can also acquire root privileges by executing the **su** command.

Steps

1. From the **System Customization** menu of the VMware ESXi host, use the keyboard arrows to select **Configure Password** and press **Enter**.
2. In the **Configure Password** dialog box, fill in the required fields to change the password:
   a. Enter the **Old Password** of the VMware ESXi host.
   b. Enter the new root password in the **New Password** field. Re-type it in the **Confirm Password** field.
   c. Press **Enter**.

Modify the VMware vCenter Server single sign on default administrator password

Use this procedure to change the default password for the VMware vCenter single sign on administrator account.

Prerequisites

Log in to the VMware vSphere Client and connect to vCenter.

Steps

1. In the left pane, select **Administration**.
2. Under Single Sign On, select **Users and Groups**.
   The **admin** user displays in the right pane.
3. On the **Users** tab, select the domain from the drop-down menu, and select the **administrator** user. Click **Edit**.
4. Set and confirm the password for the admin user account. Be sure to use a strong password as the system validates the password before accepting it.
5. Click **Save**.

Change the administrator password for VMware vCenter Server Appliance

Use this procedure to change the VMware vCenter Server Appliance administrator password.

Steps

1. Log in to the VMware vCenter Server Appliance Web console.
2. On the **Admin** tab, enter your current password in the **Current administrator password** box.
3. Enter the new password in the **New administrator password** and **Retype new administrator password** boxes.
4. Click **Change password**.

Resetting the VMware vCenter Server Appliance root password

Use this procedure to reset a forgotten vCenter Server Appliance root password.

Steps

1. Take a snapshot or backup of the vCenter Server Appliance. Do not skip this step.
2. Reboot the vCenter Server Appliance.
3. After the operating system starts, press the **E** key to access the **GNU GRUB Edit Menu**.
4. Locate the line that starts with the word `Linux`. Append the following entries to the end of the line:

   `rw init=/bin/bash`

5. Press **F10**.
6. At the command prompt, type the following command: **`passwd`**
7. Enter a new root password and re-enter the password to confirm it.
8. Unmount the file system by running the following command: **`umount /`**
9. Reboot the vCenter Server Appliance by running the following command: **`reboot -f`**
10. Confirm that you can access the vCenter Server Appliance using the new root password.
11. Remove the snapshot taken in step 1 if applicable.

Change the storage data client (SDC) password

The compute-only nodes within the PowerFlex deployment have the storage data client (SDC) running on them.

About this task

The default username is **`root`**.

This procedure is valid for SDC nodes that are ESXi hosts managed by PowerFlex Manager. Any other SDC would need to be changed directly on the server and then update the credentials used by PowerFlex Manager by following the steps in the section titled *Update a credential in PowerFlex Manager*.

Steps

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Resources**.
3. On the **All Resources** tab, select one or more resources of the same type for which you want to change passwords.
   For example, you could select one or more PowerFlex gateway components.
4. Click **Update Password**.
   PowerFlex Manager displays the **Update Password** wizard.
5. On the **Select Components** page, select the **Node OS** password and click **Next**.
   The component choices vary depending on which resource type you initially selected on the **Resources** page.

6. On the **Select Credentials** page, create a credential or change to a different credential having the same username.
7. Click **Finish** and click **Yes** to confirm the changes.

Change CloudLink passwords

CloudLink software is installed on the CloudLink Center VMs.

Unlock secadmin user password

Use this procedure to unlock the secadmin password.

Steps

1. Log in to the controller vCenter web client, and launch the CloudLink Center VM console.

   NOTE: For a single vCenter environment, log in to single vCenter with a controller and customer datacenter. Go to the controller datacenter to launch the CloudLink center VM console.

2. Log in using the CloudLink user credentials.
3. In the **Update Menu** dialog box, select **Unlock User**, and click **OK**.
   A user unlocked message is displayed.
4. Click **OK**.

Managing passwords external to PowerFlex Manager

If you change a password for a component that PowerFlex Manager has discovered, go back to PowerFlex Manager and change the credentials used by PowerFlex Manager. Refer to Update a credential in PowerFlex Manager.

Change the system and setup passwords

Use this procedure to change the system BIOS password.

Prerequisites

If the password status is set to `locked`, you cannot change or delete the system or setup password.

Steps

1. Enter System Setup by pressing **F2** immediately after turning on or restarting your system.
2. On the System Setup Main Menu, click **System BIOS** > **System Security**.
3. On the **System Security** screen, ensure that **Password Status** is set to **`Unlocked`**.
4. In the **System Password** field, change or delete the existing system password and press **Enter**.
5. In the **Setup Password** field, change or delete the existing setup password and press **Enter**.

   If you change the system or setup password, a message prompts you to re-enter the password. If you delete a password, a message prompts you to confirm the deletion.

6. Press **Esc** to return to the **System BIOS** screen. Press **Esc** again. A message prompts you to save the changes.

Manage EmbeddedOS15.x users credentials for the jump server

Create users

About this task

useradd allows you to add users and specify certain criteria such as: comments, the users home directory, shell type, and many others account properties for SUSE Linux operating system.

Steps

1. SSH to the jump server using the admin user, and type: **Server1:~# useradd -mU -G trusted <newuser>**.

   The following table that explains what each qualifier is used for:

   Table 38. Qualifier and descriptions

Qualifier	Description
-m	This qualifier makes the useradd command create the users home directory.
-G	This qualifier makes the useradd command to add the user to the group directory

2. Set the associated password, type: **server1:~ # passwd <newuser>**.

   Once the password is set, the user can successfully log in to the server.

Set up desktop icons for a new user

Use this procedure to set desktop icons for a new user.

Steps

1. Using SSH, log in as the new user.
2. Copy desktop files from admin user home directory to the new user home directory, type:

   cp /home/admin/Desktop /home/<newuser>/

   and change the ownership the files to the new user:

   cd /home/<newuser>/Desktop

   sudo chown <newuser> *
3. Log in as the new user on the user interface.
4. Right-click on the desktop, click **Open in Terminal**.
5. Type the following commands on the terminal to show the desktop icons:

   gsettings set org.gnome.shell.extensions.desktop-icons show-home false

   gsettings set org.gnome.shell.extensions.desktop-icons show-trash false

   gnome-extensions enable desktop-icons@csoriano
6. Right-click on each .desktop file and click **Allow Launching**.

Delete users

The command to delete users is **userdel** and is specified with the -r qualifier which removes the home directory and mail spool.

Steps

SSH to the server and type: **server1:~ # userdel -r <newuser>**

Once you have issued the **userdel** command, you will notice that the `/home/<newuser>` directory is removed. If you only want to delete the user but leave their home directory intact, you can issue the same command but without the **-r** qualifier.

Enable sudo on a user

Steps

1. SSH to the server and log in as **root**.
2. Type **sudo usermod -a -G wheel <newuser>**.

EmbeddedOS Jump server authentication

This is an option section to authenticate the jump server using SSH key pair or configuring with the existing LDAP.

With SSH key pair enabled authentication, jump server access can be taken only for the command line.

Configure SSH key pair

The SSH key pair can be generated in any of the Linux boxes.

Steps

1. Generate a private key, type **ssh-keygen -b 2048 -t rsa**.
2. Enter a file path to save the public and private keys or proceed with the default file path without a pass phrase:
   Sample output:

```
Enter file in which to save the key (/root/.ssh/id_rsa): /home/admin/.ssh/id_rsa  ##
admin is the username for jumpserver
Enter passphrase: # do not enter a passphrase
Enter same passphrase again: # do not enter a passphrase
```

3. Verify that the public and private keys are saved in the default file path: `/root/.ssh/.`
   Sample output:

```
cd /root/.ssh
id_rsa.pub indicates a public key.
id_rsa indicates is a private key.
```

4. Copy the public key **id_rsa.pub** using the `vi authorized_keys` file, and enter **Esc+wq**.
5. Create a key that is compatible with PuTTY, using the PuTTYgen application, load the existing private key file and click **Save private key**. The key will save in ppk format.
6. Log in using PuTTY, and provide the hostname details.
   a. Click **Connection** > **SSH** > **Auth** > **Credentials** and browse to the private key in ppk format.
   b. Click **Open**.

Configure an LDAP user in a SLES based jump server

Map the existing Active Directory based LDAP account to the jump server.

About this task

When LDAP credentials are used for the jump server, all other applications accessing the jump server will continue to work as expected. The applications use different credentials to log in and are independent from the log in credentials of the jump server.

Prerequisites

Ensure that the DNS entry for the LDAP and the jump server are made, and that NTP is configured for LDAP and the jump server and are in sync.

Steps

1. Use **realm** and **sssd** to activate the Active Directory based LDAP on the jump server. To install the packages, type **sudo zypper install realmd adcli sssd sssd-tools sssd-ad**.
2. Discover and join the LDAP domain using the realm command:
   a. To discover the LDAP, type **sudo realm -v discover <domain name>**.
   b. Join the domain, at the prompt, enter the password of the LDAP administrator account, type **sudo realm join -v <domain name>**.

      To use a different account other than the administrator account, type **sudo realm join -v <domain name> -U <LDAP username>**.
3. Verify that the jump server is added in Active Directory.
4. After the discovery and joining the existing LDAP, the realm will make an entry on `/etc/sssd/sssd.conf`.

   For example:

```
[sssd]
config_file_version = 2
services = nss, pam
domains = flexlab.in

[domain/flexlab.in]
default_shell = /bin/bash
krb5_store_password_if_offline = True
cache_credentials = True
krb5_realm = FLEXLAB.IN
realmd_tags = manages-system joined-with-adcli
id_provider = ad
fallback_homedir = /home/%u@%d
ad_domain = flexlab.in
use_fully_qualified_names = True
ldap_id_mapping = True
access_provider = ad
```

5. Enable and start the sssd service, type

   systemctl enable sssd.service

   systemctl start sssd.service
6. Edit the `/etc/krb5.conf` configuration files to reflect the realm name and domain-to-realm mappings.

   For example:

```
includedir /etc/krb5.conf.d

[libdefaults]
    default_realm = FLEXLAB.IN
    dns_lookup_relam = true
    dns_lookup_kdc = true
    rdns = false


 [domain_realm]
    .flexlab.in = FLEXLAB.IN
```

```
    flexlab.in = FLEXLAB.IN

[logging]

 [realms]
    FLEXLAB.IN = {

       kdc = win-ljji3bn74mk.flexlab.in
    admin_server = win-ljji3bn74mk.flexlab.in

}
```

7. After the configuration is successful, verify the LDAP user, type **id username@<domain name>**.

   For example:

```
jumpserver:~# id Administrator@FLEXLAB.IN

uid=895800500(administrator@flexlab.in) gid=895800513(domain users@flexlab.in)
groups=895800513(domain users@flexlab.in),895800520(group policy creator
owners@flexlab.in),895800519(enterprise admins@flexlab.in),895800512(domain
admins@flexlab.in),895800518(schema admins@flexlab.in),895800572(denied rodc password
replication group@flexlab.in)

jumpserver:~# id xxxx@FLEXLAB.IN

uid=895801105(aish@flexlab.in) gid=895800513(domain
users@flexlab.in) groups=895800513(domain
users@flexlab.in),895800519(enterprise admins@flexlab.in),895800512(domain
admins@flexlab.in),895801104(ldaptest@flexlab.in),895800572(denied rodc password
replication group@flexlab.in)
```

8. Log in to the user interface with the LDAP account using VNC.

Changing the IPI appliance password

Use this procedure to change the password for the Intelligent Physical Infrastructure (IPI) appliance.

Steps

1. Open a web browser and access the following URL: **https://< *IPI_Appliance_IP_address*>**
2. Click the **Setup** tab on the top menu bar.
3. Click **Users** in the left navigation pane.
4. Set the username, password, and/or access level. You can set a unique username for individuals requiring web management access to the appliance unit. There are three user account levels:
   - Administrator: Full control of IPI appliance configuration settings.
   - Controller: Can view configuration settings.
   - Viewer: Can view configuration settings.

   User 1/admin is the primary administrator. Do not remove administrator rights from the admin user as it might result in no one having administrator access. If this occurs, a reset to factory defaults is the only solution.
5. Click **Save** to confirm changes.

Changing a user account password on the IPI appliance

Use this procedure to change the password for a user account on the Intelligent Physical Infrastructure (IPI) appliance.

Steps

1. Open a web browser and access the following URL: **https://*<IPI_Appliance_IP_address>***

2. From the **Setup** menu, click **Users**.
3. Locate the account for which you are changing the password.
4. Enter the new password in the field.

   Do not make any changes to user name or security role.

5. Click **Save**.

Changing the operating system password

If you lose the password on storage VM (SVM), start up in recovery mode to reset it.

Steps

1. Log in to PowerFlex Manager.
2. Select the SVM, and click **Enter Maintenance Mode**.
3. Select **Protected**, and click **Enter Maintenance Mode**.
4. Log in to VMware vCenter. Select the cluster and click **Related Objects** > **Virtual Machines** > **Open Console**.
5. Restart the SVM.
6. Enter **ESC** during restart and click **OK** to enter text mode.

   NOTE: You might have to hold down the **ESC** key.

7. Type `E` to edit the password.
8. Select the second line (starts with `kernel /boot…`) and type `E`.
9. Append `init=/bin/bash` to the end of the line.
10. Select `B` to restart in recovery mode.
11. After the PowerFlex node restarts, type `passwd`, and enter a new password.
12. Type `reboot`, and allow the SVM to restart using the new password.

Minimum VMware vCenter permissions

PowerFlex Manager supports managing VMware vCenter objects without root permissions. There are three PowerFlex Manager VMware vCenter management modes available:

- Monitoring mode
- Lifecycle mode
- Management mode

These procedures provide information for creating VMware vCenter user accounts with access for the various modes listed (monitoring mode etc). VMware vCenter default permissions to meet the specified requirements so no additional permission changes are needed.

Create a user in monitoring mode

Steps

1. Log in to VMware vSphere client, select **Administration** > **Users and Groups**.
2. Click **Add User** to create a user account and enter the username and password.
3. In root view of the VMware vSphere client, click **Administration** and select **Roles**.
   a. Create a new role and select the following permissions:
      - **Profile-driven storage** > **Profile-driven storage view**
      - **VM** > **Read customization specifications (under provisioning)**
   b. Assign a name to the new role.
4. Click **Hosts and Clusters** and right-click the VMware vCenter.
   a. Choose **Add permission** and select the user account previously created.

b. Select the name of the role you created and check **Propagate to children**.

5. Log in to PowerFlex Manager.
6. Click **Settings** and create a new credential of type **vCenter**.

   NOTE: Ensure the username and password coincide with the vSphere credentials created earlier.

7. Create a user credential for the vCenter server that matches the account created in vCenter earlier.
8. Add the vCenter server object to the inventory using those credentials from PowerFlex Manager. For more information on the PowerFlex Manager credential creation see the *Dell PowerFlex Manager User Guide*.

Create a user in lifecycle mode

Steps

1. Log in to VMware vSphere client, select **Administration** > **Users and Groups**.
2. Click **Add User** to create a user account and enter the username and password.
3. In root view of the VMware vSphere client, click **Administration** and select **Roles**.
   a. Create a new role and select the following permissions:
      - **Profile-driven storage** > **Profile-driven storage view**
      - **VM** > **Read customization specifications (under provisioning)**
      - **Host: Connection, firmware, maintenance, power, query patch, system management, system resources**

   b. Assign a name to the new role.
4. Click **Hosts and Clusters** and right-click the VMware vCenter.
   a. Choose **Add permission** and select the user account previously created.
   b. Select the name of the role you created and check **Propagate to children**.
5. Log in to PowerFlex Manager.
6. Click **Settings** and create a new credential of type **vCenter**.

   NOTE: Ensure the username and password coincide with the vSphere credentials created earlier.

7. Create a user credential for the vCenter server that matches the account created in vCenter earlier.
8. Add the vCenter server object to the inventory using those credentials from PowerFlex Manager. For more information on the PowerFlex Manager credential creation see the *Dell PowerFlex Manager User Guide*.

Create a user in managed mode

Use this procedure to create VMware vCenter role with required permissions to run PowerFlex Manager in managed mode. You can create a role using the default permissions or selecting granularly defined permissions.

Steps

1. To do default:
   a. Log in to VMware vSphere client, select **Administration** > **Users and Groups**.
   b. Click **Add User** to create a user account and enter the username and password.
   c. In root view of the VMware vSphere client, click **Administration** and select **Roles**.
   d. Click **Hosts and Clusters** and right-click the VMware vCenter.
      i. Choose **Add permission** and select the user account previously created.
      ii. Select the name of the role you created and check **Propagate to children**.
   e. Log in to PowerFlex Manager.
   f. Click **Settings** and create a new credential of type **vCenter**.

      NOTE: Ensure the username and password coincide with the vSphere credentials created earlier.

   g. Create a user credential for the vCenter server that matches the account created in vCenter earlier.
   h. Add the vCenter server object to the inventory using those credentials from PowerFlex Manager. For more information on the PowerFlex Manager credential creation see the *Dell PowerFlex Manager User Guide*.

2. To do more granular permissions:
   a. Log in to VMware vSphere client, select **Administration** > **Users and Groups**.
   b. Click **Add User** to create a user account and enter the username and password.
   c. In root view of the VMware vSphere client, click **Administration** and select **Roles**.
   d. Create a new role and select the following permissions:

Table 39. Role permissions

Category	Permissions to select
Alarms	• Acknowledge alarm • Create alarm • Disable alarm action • Modify alarm • Remove alarm • Set alarm status
Permissions	• Modify permission • Modify privilege • Modify role • Reassign role permissions
AutoDeploy	• Host • AssociateMachine • Image Profile • Create • Edit • Rule • Create • Delete • Edit • RuleSet • Activate • Edit
Certificates	Manage certificiates
Certificate Management	• Create/Delete (Admins priv) • Create/Delete (below Admins priv)
CNS	Searchable
Content Library	• Add library item • Create a subscription for a published library • Create local library • Create subscribed library • Delete library item • Delete local library • Delete subscribed library • Delete subscription of a published library • Download files • Evict library item • Evict subscribed library • Import storage • Probe subscription information • Publish a library item to its subscribers • Publish a library to its subscribers • Read storage • Sync library item • Sync subscribed item • Type introspection

Table 39. Role permissions (continued)

Category	Permissions to select
	• Update configuration settings • Update files • Update library • Update library item • Update local library • Update subscribed library • Update subscription of a published library • View configuration settings
Content	• Add disk • Clone • Decrypt • Direct Access • Encrypt • Encrypt new • Manage KMS • Manage encryption policies • Manage keys • Migrate • Recrypt • Register VM • Register host
dVPort group	• Create • Delete • IPFIX operation • Modify • Policy operation • Scope operatin
Distributed switch	• Create • Delete • Host operation • IPFIX operation • Modify • Move • Network I/O control operation • Policy operation • Port configuration operation • Port setting operation • VSPAN operation
Datacenter	• Create datacenter • Move datacenter • Network protocol profile configuration • Query IP pool allocation • Reconfigure datacenter • Release IP allocation • Remove datacenter • Rename datacenter
Datastore	• Allocate space • Browse datastore • Configure datastore • Low level file operations • Move datastore

Table 39. Role permissions (continued)

Category	Permissions to select
	• Remove datastore • Remove file • Rename datastore • Update virtual machine files • Update virtual machine metadata
ESX Agent Manager	• Config • Modify • View
Extension	• Register extension • Unregister extension • Update extension
External stats provider	• Register • Unregister • Update
Folder	• Create folder • Delete folder • Move folder • Rename folder
Global	• Act as vCenter Server • Cancel task • Capacity planning • Diagnostics • Disable methods • Enable methods • Global tag • Health • Licenses • Log event • Manage custom attributes • Proxy • Script action • Service managers • Set custom attribute • Settings • System tag
Health update provider	• Register • Unregister • Update
Host	• CIM • CIM interaction • Configuration • Advanced settings • Authentication Store • Change PciPassthru settings • Change SNMP settings • Change data and time settings • Change settings • Connection • Firmware • Hyperthreading

Table 39. Role permissions (continued)

Category	Permissions to select
	• Image configuration • Maintenance • Memory configuration • NVDIMM • Network Configuration • Power • Quarantine • Query patch • Security profile and firewall • Storage partition configuration • System Management • System resources • Virtual machine autostart configuration • Inventory • Add host to cluster • Add standalone host • Create cluster • Modify cluster • Move cluster or standalone host • Move host • Remove cluster • Remove host • Rename cluster • Local operations • Add host to vCenter • Create virtual machine • Delete virtual machine • Manage user groups • Reconfigure virtual machine • vSphere Replication • Manage replication
vSphere Tagging	• Assign or Unassign vSphere tag • Create vSphere tag • Create vSphere tag category • Delete vSphere tag • Delete vSphere tag category • Edit vSphere tag • Edit vSphere tag category • Modify UsedBy Field for Category • Modify UsedBy Field for Tag
Network	• Assign network • Configure • Move network • Remove
Performance	Modify intervals
Host Profile	• Clear • Create • Delete • Edit • Export • View

Table 39. Role permissions (continued)

Category	Permissions to select
Resource	• Apply recommendation • Assign vApp to resource pool • Assign virtual machine to resource pool • Create resource pool • Migrate powered off virtual machine • Migrate powered on virtual machine • Modify resource pool • Move resource pool • Query vMotion • Remove resource pool • Rename resource pool
Scheduled task	• Create tasks • Modify task • Remove task • Run task
Sessions	• Impersonate user • Message • Validate session • View and stop sessions
Datastore cluster	Configure a datastore cluster
Profile-driven storage	• Profile-driven storage update • Profile-driven storage view
Storage views	• Configure service • View
Tasks	• Create task • Update task
Transfer service	• Manage • Monitor
vApp	• Add virtual machine • Assign resource pool • Assign vApp • Clone • Create • Delete • Export • Import • Move • Power off • Power on • Rename • Suspend • Unregister • View OVF environment • vApp appliacation configuration • vApp instance configuration • vApp managedBy configuration • vApp resource configuration
VMware vSphere Update Manager	• Configure • Configure service

Table 39. Role permissions (continued)

Category	Permissions to select
	• Manage baseline • Attach baseline • Manage baseline • Manage Patches and Upgrades • Remediate to Apply Patches, Extensions. and Upgrades • Scan for Applicable Patches, Extensions, and Upgrades • Stage Patches and Extensions • View Compliance Status • Upload file • Upload upgrade images and offline bundles
Virtual machine	• Change Configuration • Acquire disk lease • Add existing disk • Add new disk • Add or remove device • Advanced configuration • Change CPU count • Change Memory • Change Settings • Change Swapfile placement • Change resource • Configure Host USB device • Configure Raw device • Configure managedBy • Display connection settings • Extend virtual disk • Modify device settings • Query fault tolerance compatibility • Query unowned files • Reload from path • Remove disk • Rename • Reset guest information • Set annotation • Toggle disk change tracking • Toggle fork parent • Upgrade virtual machine compatibility • Edit Inventory • Create from existing • Create new • Move • Register • Remove • Unregister • Guest operations • Guest operation alias modification • Guest operation alias query • Guest operation modifications • Guest operation program execution • Guest operation queries • Interaction • Answer question • Backup operation on virtual machine

Table 39. Role permissions (continued)

Category	Permissions to select
	• Configure CD media • Configure floppy media • Connect devices • Console interaction • Create screenshot • Defragment all disks • Drag and drop • Guest operating system management by VIX API • Inject USB HID scan codes • Install VMware Tools • Pause or Unpause • Perform wipe or shrink operations • Power off • Power on • Record session on virtual machine • Replay session on virtual machine • Reset • Resume Fault Tolerance • Suspend • Suspend Fault Tolerance • Test failover • Test restart Secondary VM • Turn off Fault Tolerance • Turn on Fault Tolerance • Provisioning • Allow disk access • Allow file access • Allow read-only disk access • Allow virtual machine download • Allow virtual machine files upload • Clone template • Clone virtual machine • Create template from virtual machine • Customize guest • Deploy template • Mark as template • Mark as virtual machine • Modify customization specification • Promote disks • Read customization specifications • Service configuration • Allow notifications • Allow polling of global event notifications • Manage service configurations • Modify service configuration • Query service configurations • Read service configuration • Snapshot management • Create snapshot • Remove snapshot • Revert to snapshot • vSphere Replication • Configure replication

Table 39. Role permissions (continued)

Category	Permissions to select
	• Manage replication • Monitor replication
vSAN	• Cluster • ShallowRekey
vService	• Create dependency • Destroy dependency • Reconfigure dependency configuration • Update dependency

e. Assign a name to the new role.
f. Click **Hosts and Clusters** and right-click the VMware vCenter.
   i. Choose **Add permission** and select the user account previously created.
   ii. Select the name of the role you created and check **Propagate to children**.
g. Log in to PowerFlex Manager.
h. Click **Settings** and create a new credential of type **vCenter**.

   NOTE: Ensure the username and password coincide with the vSphere credentials created earlier.

i. Create a user credential for the vCenter server that matches the account created in vCenter earlier.
j. Add the vCenter server object to the inventory using those credentials from PowerFlex Manager. For more information on the PowerFlex Manager credential creation see the *Dell PowerFlex Manager User Guide*.

21

System logs and audit logs

PowerFlex Manager logs

Generating a troubleshooting bundle

You can generate a troubleshooting bundle which is a compressed file that contains logging information for PowerFlex Manager managed components.

About this task

If necessary, download the bundle and send it to Dell Technologies Support for issue debugging.

The troubleshooting bundle includes the following logs:

- ASM deployer
- iDRAC life cycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods
- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes
- PowerFlex Block logs
- PowerFlex data server and device gateway target logs

NOTE: You can also generate troubleshooting bundles from the **Resource Groups** page.

Steps

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Generate Troubleshooting Bundle**.
3. Select **Include PowerFlex Block Core Dump logs** to include core dump logs.
4. Select a **PowerFlex Log Level**:
   - Default Node Logs
   - Default Node Logs plus additional MDM information
   - Latest Logs only (Most recent copy of all logs)
5. Select a **Node Selection**:
   - Logs from all nodes
   - Select Specific Nodes

   If you select the **Select Specific Nodes** option and select the number of nodes for which you want to generate the log, the **View/Select Nodes** button is displayed. Click the button to view the list of nodes in the **Node Selection** window. Select the required nodes from the **Available Nodes** list and click **>>** to view them in the **Selected Nodes** list. Click **Save** to return to the **Generate Troubleshooting bundle** page.

   For the **Select Specific Nodes** option **Generate** is enabled only if a node is selected in the **Node Selection** window.
6. Choose a **Delivery Option**:

Option	Description
Send to Configured Dell Technologies connectivity services	If you are using Dell Technologies connectivity services, **Send to Configured Dell Technologies connectivity services** is selected by default. Leave this default setting. If Dell Technologies connectivity services is not configured, this option is disabled.
Download to Share	Provide a path using the following format with credentials: `CIFS:\\1.1.1.1\uploadDirectory`. Enter a username and password, and click **Test Connection** to verify the connection to the CIFS share before generating the bundle.
Download Locally	Download the troubleshooting bundle to a local file: `\\IP Address\Any folder`

7. Click **Generate**.

Next steps

Sometimes, the troubleshooting bundle does not include log information for all the nodes. The log collection may appear to succeed, but the log for one or more of the nodes may be missing. You may see an error message in the scaleio.trace.log file that says `Could not run get_info script`. If you see this message, you must generate the troubleshooting bundle again to include information for all the logs.

VMware vCenter logs

Collect VMware vCenter log files to a single location.

Steps

1. In the VMware vSphere Client menu, click **Administration** > **Deployment** > **System Configuration**.
2. Select a vCenter Server node and click **Export Support Bundle**.
3. Select the support bundle type.
4. Click **Export**.

CloudLink logs

Collect diagnostic log files.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Diagnostics**.
3. Click **View Logs**.
4. In the **Select Logs to View** dialog box, select a log file, and enter the required values for which you want to create a diagnostic file.
5. Click **Show Logs**.
6. Click **Actions** > **Generate Diagnostics**.

Network logs

Gather logs from the network switch

Generate logs to troubleshoot your network switch.

Steps

1. Open an SSH session with the switch using PuTTY or a similar SSH client.
2. Log in with admin or other credentials with privileges and type **`show tech-support`**.

3. Enable session logging. If using PuTTY, right-click the menu bar and go to **Change settings** > **Sessions** > **Logging**.
4. Select **All session output**.
5. Type a log file name and click **Apply**.
6. In the switch CLI, type the following:

Table 40. Gather logs by switch type

Switch type	Code required
Dell PowerSwitch	`show tech-support` `show process cpu`

Cisco Nexus	`show tech-support details | no-more`
	`show tech-support vpc | no-more`
	`show process cpu history| no more`

Gathering logs from the Cisco Nexus network for troubleshooting

It is important to export the technical support file from your Cisco Nexus switch before attempting to troubleshoot the switch or contact Dell Technologies Support.

About this task

Use the `show tech-support` command to obtain the technical support file from your Cisco Nexus switch, which contains log output and a snapshot of the switch at the time of failure.

Steps

1. Open an SSH session with the Cisco Nexus switch using PuTTY or a similar SSH client.
2. Log in with admin or other credentials with privileges to run `show tech-support`.
3. Right-click the menu bar and select **Change Settings...** and enter the following information:
   - **Session logging:** select **All session output**
   - **Log file name:** `putty.log`
4. Click **Apply**.
5. In the Cisco NX-OS CLI, type the following:

```
show tech-support details | no-more
show tech-support vpc | no-more
show process cpu history | no-more
```

Gathering logs from the Dell network for troubleshooting

It is important to export the technical support file from your Dell PowerSwitch switch before attempting to troubleshoot the switch or contact Dell Technologies Support.

About this task

Use the `show tech-support` command to obtain the technical support file from your Dell PowerSwitch switch, which contains log output and a snapshot of the switch at the time of failure.

Steps

In the Dell OS CLI, type the following:

```
show tech-support
show processes node-id node-id-number [pid process-id]
```

Configuring syslogs and audit logs

This section covers the procedures to configure system logs/syslogs and audit logs.

The audit log is supported for components like iDRAC, storage virtual machines, PowerFlex system, and CloudLink. Audit log configuration must be created at the source except in PowerFlex deployments.

NOTE: If iDRAC sources are not configured, no iDRAC events are generated. After iDRAC sources are added, the iDRAC traps should appear in the syslog output.

The logs can be transmitted through either of the following modes to the remote server:

- Directly
- Using PowerFlex Manager

For the audit log transmitted using PowerFlex Manager, Facility 13/Audit is used.

The log retention for all components is available locally for 15 days.

Table 41. Components and the source configurations

Components	Source configuration
iDRAC	Manual
Cisco switches/Dell switches	Manual
VMware vCenter/ESXi	Manual
CloudLink	Manual
PowerFlex	Manual
Storage-only nodes	Manual

Consider the following while configuring the syslogs and audit logs:

- TLS is supported if the remote syslog server is configured directly.
- TLS is not supported if PowerFlex Manager is used to configure the remote syslog server.

  Ensure the PowerFlex Manager user interface access and credentials are available to configure the remote syslog server.

To configure an external source and a destination, see Configure an external source and Configure a destination.

Configure syslogs using VMware ESXi

Use this procedure to configure syslogs using VMware ESXi. All system logs are forwarded to a centralized location.

Steps

1. Log in to the VMware vSphere client.
2. Go to **Manage** > **System** > **Advanced Settings**.
3. To export logs to a central logging server, search for `Syslog.global.LogHost`.
4. Enter the protocol (UDP), IP address or FQDN, and the port number.
   For example, **`udp://192.168.1.1:514`** (or) **`tcp://192.168.1.1:514`**.
5. Click **Save**.
6. To ensure syslog is started and ports are added, go to **Networking** > **Firewall rules** and search for **`Syslog`**.

Configure and forward the syslogs using VMware vCenter

Use this procedure to configure and forward the syslogs using VMware vCenter.

Steps

1. Log in to the vCenter server management interface as root: **<https://vcenterIP:5480>**.
2. In the vCenter server management interface, select **Syslog**.
3. From the **Forwarding Configuration** pane, click **Configure** if you have not configured any remote syslog hosts.
   Click **Edit** if you already have configured hosts.
4. In the **Create Forwarding Configuration** pane, enter the server IP address of the destination host. The maximum number of supported destination hosts is three.
5. From the **Protocol** drop-down menu, select the protocol to use.
6. In the **Port** text box, enter the port number to use for communication with the destination host.
7. In the **Create Forwarding Configuration** pane, click **Add** to enter another remote syslog server and click **Save**.
8. Verify that the remote syslog server is receiving messages.
9. In the **Forwarding Configuration** section, click **Send Test Message**.
10. Verify on the remote syslog server that the test message was received.
    The new configuration settings are displayed in the **Forwarding Configuration** section.

Configure syslog and audit logs in CloudLink

CloudLink supports four multiple syslog configurations from CloudLink 8.1. The audit log and system log are retrieved locally and available for 15 days.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog** > **Add Syslog Server**. The **Add** window appears.
   a. Provide the syslog IP address, port number, and select the facility and protocol. Click **Add**.

      NOTE: For audit logs, choose the facility as audit if the logs are transmitted using PowerFlex Manager.

   b. If the service status of the syslog server is not active, click **Resume**. In the **Confirm Syslog Mode Change** window, click **Resume**.
      The service status is Active.

      NOTE: The **Change Syslog Format** and **Postpone** options can only be done globally.

3. To retrieve the logs locally:
   a. To enable SSH from CloudLink, click **Server** > **Network** > **Change SSH**.
   b. Select the **Enabled** option and click **Change**.
   c. SSH to the CloudLink IP address. The logs are in the `/var/log/CLC_audit.log` file.

Configure audit logs in PowerFlex storage-only nodes

Use this procedure to configure and maintain audit logs of a user in a separate file `commands.log` in PowerFlex storage-only nodes and transmit them to the remote servers.

Steps

1. Log in to each PowerFlex storage-only node using SSH.
2. Run the following commands to record all the commands executed by the users:
   a. Edit the system-wide BASH runtime configuration file:
      #sudo -e /etc/bash.bashrc
   b. Append the following to the end of the file:

```
export PROMPT_COMMAND='RETRN_VAL=$?;logger -p security.debug "$(whoami) [$$]: $
(history 1 | sed "s/^[ ]*[0-9]\+[ ]*//" ) [$RETRN_VAL]"'
```

c. Press **Esc**+**:wq** to save the file.

d. To setup the audit logging for `local6` with a new file as `commands.log`, run the following command to edit the remote syslog BASH runtime configuration file:

```
# sudo -e /etc/rsyslog.d/bash.conf
```

e. Enter the following:

```
security.* /var/log/commands
```

f. Press **Esc**+**:wq** to save the file.

3. To forward the audit log file `/var/log/audit/audit.log` to the remote syslog server, edit **/etc/audit/plugins.d/syslog.conf** and do the following:

a. Set **active=yes**

b. Append **security** in args:

For example:

```
#vi /etc/audit/plugins.d/syslog.conf
active = yes  # changes made from No to Yes
direction = out
path = /sbin/audisp-syslog
type = always
args = LOG_INFO LOG_security  # append security to write the audit logs to remote
server
format = string
```

4. To transmit the logs to the remote server:

a. Edit the `rsyslog.conf` file:

```
# vi /etc/rsyslog.conf
```

b. Add the following lines to route the logs to the remote server:

```
$ModLoad imfile
$InputFileName /var/log/audit/audit.log
$InputFileTag tag_audit_log:
$InputFileStateFile audit_log
$InputFileSeverity info
$InputRunFileMonitor
local6.* @<remoteserverIP>:514  #Remote server IP address to be updated
```

In PowerFlex Manager, both TCP and UDP connections are supported and port number 514 is used for the syslog transfer.

5. Restart the remote syslog service:

```
systemctl restart rsyslog.service
```

6. To manage the log file `/etc/logrotate.conf`:

a. To edit the file, type **# vi /etc/logrotate.conf**.

b. Update the following lines:

```
## AUDIT LOG###
/var/log/commands {
compress
dateext
maxage 14
rotate 2
missingok
notifempty
size +4096k
}
```

c. Type **:wq** to save the file.

Configure iDRAC for audit logging

Use this procedure to manage the alerts, configure bulk alerts, granular alerts, and multiple syslog destinations in iDRAC.

Prerequisites

Before you configure iDRAC for audit logging, see the following for iDRAC related information:

- iDRAC **Help** in the iDRAC user interface
- Dell EMC iDRAC Service Module Configuration Guide
- Managing Logs

About this task

There are 30 audit alerts that can be enabled globally or per alert. Severities can also be configured globally or per alert: `critical`, `warning`, and `informational`. Audit logs cannot be sent to a separate destination than the syslog. Separation of iDRAC audit logs can be done by parsing syslog at the syslog destination. For example:

```
Tue Jan 31 14:36:17 2023;10.234.93.12; <174>2023-01-31T22:36:42.113015-06:00 idrac-
C0763W2 Severity: Informational, Category: Audit, MessageID: RAC1195, Message: User root
via IP 10.234.94.12 requested state / configuration change to Alert Configuration using
GUI.
```

For managing logs with RACADM, see Integrated Dell Remote Access Controller 9 (iDRAC9) Version 3.00.00.00 User's Guide.

Steps

1. Log in to the iDRAC using the root credentials.
2. To enable or disable alerts, click **Configuration** > **System Settings** > **Alert Configuration** > **Alerts**.
3. To bulk configure alerts by category, severity, and destination type, click **Configuration** > **System Settings** > **Alert Configuration** > **Quick Alert Configuration**.
4. To configure the granular alerts, click **Configuration** > **System Settings** > **Alert Configuration** > **Alerts** > **Alert Configuration**.
5. To configure multiple syslog destinations, click **Configuration** > **System Settings** > **Alert Configuration** > **Remote Syslog Settings**.

Configuring syslogs for Dell PowerSwitch and Cisco Nexus switches

For the Dell PowerSwitch and Cisco Nexus switches, syslog can be transmitted through TLS and audit logs. Ensure TACACS (terminal access controller access-controller system) is used for the audit logs.

Table 42. Switches and their product documentation

Product	Link to the product documentation
Dell PowerSwitch switches	Dell SmartFabric OS10 User Guide Release 10.5.4
Cisco Nexus switches	Cisco Nexus 9000 Series NX-OS System Management Configuration Guide, Release 10.3(x)
TACACS logs	Cisco Identity Services Engine Administrator Guide, Release 3.2

PowerFlex audit log

This section describes how to enable audit logging in PowerFlex Manager.

It is recommended to be using PowerFlex Manager 4.6 before using the new audit log configuration. It will continue to work on older versions of PowerFlex Manager.

Using the audit policy and set up (instead of events to syslog, and syslog to syslog) narrows down the events and messages to "audit" (Facility 13).

Audit logs filter the Facility 13 messages. Adding syslog and events to rsyslog is allowed.

Ingress and events propagation to syslog

Use this procedure to ingress and events propagation to syslog.

Steps

1. Go to `/opt/platform-provisioner/support_scripts` and run the `./install_audit.sh` script once from any of the MVM nodes. Provide the PowerFlex user credentials at the credential prompt.

```
# Enter the node
cd /opt/platform-provisioner/support_scripts; ./install_audit.sh
# Installing observability, If there is issue break and repeat
# PFXUSER must be defined to the powerflex admin user, something like:
# PFXUSER="admin" PFXPASSWORD="Scaleio123!" ./install_audit.sh
# Enter your PFX USER : Enter PFX PASSWORD :
```

   This may take 20 minutes, do not close this terminal until this script completes successfully.
2. User credentials must be supplied in the environment or as a variable in the command line above. If the variable is missing, it prompts for it to be entered.
3. If the process failed or was interrupted, there might be a job running in the platform manager:

```
"{"title": "An execution is already in progress", "description": "Please refer
bedrock log for more details"}++ wait_for_done"
...
No ID - check error
```

   The process can be run again when this job completes.

   The script checks for having 10 observability services as a check that it successfully completed the install observability part. Then it completes the platform installer configuration change.

Define the audit policy notification policy

Use this procedure to define a notification policy to forward events in the PowerFlex system to the rsyslog-forwarder (also known as the syslog-listener). Then, the rsyslog-forwarder forwards the events to the external destinations that are defined in the policy.

Steps

1. Create a source:
   a. Click **Add a source** and from the **Source Type** drop-down menu, select **auditlog**.
   b. Add the name and description.
2. Create a destination:
   a. Go to **Settings** > **Events and Alerts** > **Notification Policies**.

      You can also use the following REST API: `dispatch-destinations~1/post`
   b. Enter the destination name and description.
   c. From the **Destination Type** menu, select **Audit Log**.
   d. Click **Next** and enter the IP address, port, and protocol (TCP) of the target SIEM. Ensure that the SIEM IP address, port, and protocol are reachable.
3. Create a new policy:
   a. Go to **Settings** > **Events and Alerts** > **Notification Policies**.

      You can also use the following REST API: `dispatch-policies~1/post`
   b. Click **Create New Policy**.
   c. Enter a name and a description for the notification policy. For the policy name, you can enter: **`Powerflex audit events to external Syslog`**

d. Set the **Source Type** to **Auditlog**.
e. Set the **Resource Domain** to **All** (all resource domains must be selected).
f. For **Severity**, select all the check boxes (all severity levels must be selected).
g. Select one or more of the destinations you created above.

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Migrating to NVMe over TCP on VMware ESXi

Use the following guidelines when migrating a VMFS datastore and clusters from SDC to NVMe over TCP.

PowerFlex supports three options for migrating from SDC to NVMe over TCP on VMware ESXi:

Online migration using Storage vMotion (VMFS only)

This method is a standard way of using Storage vMotion to move storage and it also supports switching protocols by migrating to a new datastore. This is the method that is described in this guide.

Offline conversion (VMFS only)

Offline conversion is an option for converting an existing VMFS datastore from SCSI (SDC) to NVMe over TCP without having to copy all the data over the network. This option is covered in the Dell Knowledge Base article SCSI to NVMe VMware VMFS Datastore Offline Migration Steps.

Conversion for clustered applications using RDMs

This option converts existing SDC- and RDM-based application clusters to shared VMDKs on NVMe over TCP datastores. This option is covered in the Dell Knowledge Base article PowerFlex: SDC to NVMe/TCP conversion for clustered applications utilizing RDMs on vSphere.

NOTE:

- Dell Technologies recommends using VMware Storage vMotion to migrate data from a volume that is presented by SDC to one presented by NVMe over TCP.
- Linux environments are not discussed in this guide.

Requirements

See the following VMware vSphere 7.0 product documentation link for information about Storage vMotion requirements and limitations: Storage vMotion Requirements and Limitations.

See the following VMware vSphere 8.0 product documentation link for information about requirements and limitations of VMware NVMe Storage: Requirements for VMware NVMe Storage.

See the following VMware KB article for more details about storage migration (Storage vMotion), with the virtual machine powered on: Migrating virtual machines with Raw Device Mappings (RDMs).

Ensure that you satisfy these requirements before you begin the migration:

- The VMware ESXi version is 8.x or later.
- The PowerFlex version is 5.0 or later.

Workflow

The following steps summarize the workflow that you must follow to migrate from SDC to NVMe over TCP using Storage vMotion:

1. Create and map a new volume of equal or greater size than the current VMFS datastore using NVMe over TCP to the same host.
2. Scan for the newly mapped volume.
3. Create a datastore on the NVMe over TCP volume.
4. Perform the standard data migration using Storage vMotion by using a nondisruptive process.

Prepare the VMware ESXi node for mapping NVMe over TCP volumes

Before migrating the data, you must prepare the VMware ESXi node for mapping NVMe over TCP volumes.

Ensure that you satisfy these requirements before you begin:

- You must have an NVMe over TCP target supported PowerFlex system (version 4.5.x or later).
- Deployed VMware ESXi compute-only nodes.
- The host must be at VMware ESXi version 8.x or later.

Enable the NVMe over TCP VMkernel ports

Use this procedure to enable the NVMe over TCP VMkernel ports.

Steps

1. Log in to the VMware vSphere Client.
2. Click **Home/Inventory** and select the host.
3. Select **Configure** > **VMkernel adapters**.
4. Edit **PowerFlex-Data 1**.
5. Select the **NVMe over TCP** check box and click **OK**.
6. Repeat these steps for the remaining PowerFlex data networks.
7. Repeat the steps for the remaining hosts in the cluster.

Add NVMe over TCP software storage adapter

Use this procedure to add an NVMe over TCP software storage adapter.

Steps

1. Log in to the VMware vSphere Client.
2. Click **Home** > **Inventory** > **Hosts and Clusters**.
3. Select the cluster of PowerFlex compute-only nodes that you want to configure.
4. From the right pane, click **Configure** > **Storage Adapters**.
5. From the right pane, click **Add Software Adapter**.
6. Click **Add NVMe over TCP adapter**.
7. Select the first VMNIC and click **OK**.
8. Click **Add NVMe over TCP adapter**.
9. Select the second VMNIC and click **OK**.

Copy the host NQN

Use this procedure to copy the host NQN to the copy buffer. The host NQN details are required when you add the host to PowerFlex.

Steps

1. Log in to VMware vSphere Client.
2. Select the first host.
3. From the right pane, click **Configure** > **Storage Adapters**.
4. Select the first VMware NVMe over TCP storage adapter. For example, **vmhba6x**.
5. From the pane, select **Controllers/Add Controller**.

   The host NQN is listed at the top of the form.
6. Click **COPY** and place the host NQN in the copy buffer.
7. Click **CANCEL**.
8. Repeat the steps for all the hosts.

Add a host to PowerFlex

Use this procedure to add a host to PowerFlex.

Steps

1. Log in to PowerFlex Manager.
2. Click **Block** > **Hosts**.
3. Click **Add Host**.
4. Enter the hostname and paste the host NQN from the copy buffer.
5. Enter the **Number of Paths Per Volume**. The default number of paths is four per volume, and the maximum number of paths is eight per volume.
6. Enter the **Number of System Ports per Protection Domain**. The default number of system ports is ten per protection domain, and the maximum number of system ports is 16 per protection domain.
7. Click **Add**.

Create a volume

Use this procedure to create a volume.

Steps

1. From PowerFlex Manager, click **Block** > **Volumes**.
2. Click **Create Volume**.
3. Enter the number of volumes and the name of the volumes.
4. Select **Thick** or **Thin**. Thin is the default.
5. Enter the required volume size in GB.
6. Select the NVMe storage pool and click **Create**.

Map a volume to the host

Use this procedure to map a volume to the host.

Steps

1. From PowerFlex Manager, click **Block** > **Volumes**.
2. Select the volume check box and click **Provision** > **Map**.

3. Select the protocol **NVMe**.
4. Select the check box for the host to which you are mapping the volume.
5. Click **Map**.

Discover and connect the NVMe over TCP Target

Use the procedure to discover and connect the NVMe over TCP Target PowerFlex system. Use the `esxcli` command to perform the operation.

Steps

1. Log in to the ESXi server using ssh.
2. Run the discovery query on each adapter:

```
#esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
#esxcli nvme fabrics discover -a vmhba6y -i 192.168.x.y -p 8009
```

   In the first example above, 6x is the first NVMe over TCP software adapter and 6y is the second NVMe over TCP software adapter.

   In the second example above, 192.168.x.x is first data IP Address and 192.168.x.y is second data IP Address, depends on which VMNIC is enabled for which software NVMe over TCP adapter.

3. Connect to the PowerFlex system by appending "-c" to the discovery query command.

```
#esxcli nvme fabrics discover -a vmhba64 -i 192.168.x.x -p 8009 -c
#esxcli nvme fabrics discover -a vmhba65 -i 192.168.x.y -p 8009 -c
```

4. Get the controller list by verifying the connected controllers.

```
#esxcli nvme controller list
```

Perform a rescan of the storage

Use this procedure to perform a rescan of the storage.

Steps

1. In the vSphere Client object navigator, browse to a host, a cluster, a data center, or a folder that contains hosts.
2. From the right-click menu, select **Storage** > **Rescan Storage**.
3. Specify the extent of the rescan.

Option	Description
Scan for New Storage Devices	Rescan all adapters to discover new storage devices. If new devices are discovered, they appear in the device list.
Scan for New VMFS Volumes	Rescan all storage devices to discover new datastores that have been added since the last scan. Any new datastores appear in the datastore list.

Create a VMFS datastore on the NVMe over TCP volume

Use this procedure to create a VMFS datastore on the NVMe over TCP volume.

Steps

1. Log in to VMware vCenter.
2. Select **Home/Inventory** and select the storage icon.
3. Right-click the cluster of PowerFlex compute-only nodes and select **Storage/New Datastore**.

4. Select VMFS and click **NEXT**.
5. Enter the name of the datastore and select a host from the list.
6. You should see a new disk called NVMe TCP Disk (`eui.######`).
7. Select the disk and click **NEXT**.
8. Select VMFS 6 and click **NEXT**.
9. Leave the default configuration unless you have been otherwise instructed and click **NEXT**.
10. Review your selections and click **FINISH**.

Results

You are ready to migrate the data with Storage vMotion.

Migrate the data with Storage vMotion

After you prepared the node, you can migrate the data with Storage vMotion.

Follow the standard VMware vSphere 7.0 procedure to migrate the virtual machines from an SDC-based datastore to an NVMe over TCP-based datastore: Migrate a Virtual Machine to New Storage

Migrate VMFS datastores from SDC to NVMe-oF

You can migrate VMFS6 datastores from SCSI-controlled storage data clients (SDCs) to Non-Volatile Memory Express over Fabrics (NVMe-oF) by using an offline migration method.

CAUTION: These instructions apply only to VMFS6 datastores.

Offline VMFS datastore migration from a SCSI-controlled SDC to NVMe-oF does not involve data movement. The device previously presented to a VMware ESXi host or cluster as an SDC device is unpresented and presented again as an NVMe-oF device. The VMFS datastore is then signatured again and made available to hosts, and retains all VM contents.

Migration workflow

The migration process happens in three phases: pre-migration, migration, and post-migration. The following table provides a migration workflow checklist.

Table 43. Migration workflow checklist

Pre-migration	Migration	Post-migration
Pre-migration includes preparatory checks and validation. 1. Check feature impact 2. Verify VMFS locking mode is `ATS` and not `ATS+SDC (SCSI)` 3. Check for RDM usage on VMs 4. Review claim rules and settings 5. Verify device and path limits 6. Check for unsupported features 7. Assess feature impact post-migration	Migration is the actual migration process. 1. Power off and unregister all VMs 2. Unmount VMFS volume from all hosts 3. Check VMFS metadata consistency 4. Detach SDC volume from ESXi hosts 5. Stop presenting SDC volume from array 6. Present device as NVMe-oF to one ESXi host 7. Rescan adapters on the host 8. Check VMFS metadata consistency after conversion 9. Re-signature the VMFS volume 10. (Optional) Rename the datastore 11. Check VMFS metadata consistency after re-signature	Post-migration includes using vSphere Client or another interface to verify that the VMs are successfully running.

Table 43. Migration workflow checklist

Pre-migration	Migration	Post-migration
	12. Present device as NVMe-oF to all ESXi hosts 13. Register and power on all VMs	

Additional resources and support

Using vSphere On-disk Metadata Analyzer (VOMA): Checking Metadata Consistency with vSphere On-disk Metadata Analyzer.

Contact VMware Support if you encounter issues during migration. Consult your storage array vendor's documentation for array-specific procedures.

Offline migration feature impact

Offline migration impacts the features listed in the following tables.

CAUTION: Review the following tables to understand which features are unavailable and which perform differently after migration.

Unsupported features

The following features are not available after offline migration.

Table 44. Unsupported features

Feature	Description	Notes
Guest clustering	Clustered virtual machine disk (VMDK) feature supporting high availability solutions such as Windows Server Failover Cluster (WSFC)	A VMFS datastore with clustered VMDK enabled cannot be migrated
Site Recovery Manager (SRM)	Array-based replication with SRM is not supported by NVMe-oF	Migrating datastores involved with SRM array replication renders the solution useless

Performance impact on supported features

The following features are available but perform differently after offline migration.

Table 45. Performance impact on supported features

Feature	Nature of impact	Action required
Hardware Accelerated Move (XCOPY)	No equivalent command to XCOPY. VMware software data mover is used instead. This may decrease performance of operations like cloning or SvMotion.	None
WRITE SAME command with the UNMAP bit	If NVMe-oF device does not support NVMe-oF equivalent of write zeroes or unmap, there may be performance impact.	None

Complete pre-migration checks

You must complete checks to understand environment characteristics and features in use. These checks determine if offline migration is possible and help you understand the post-migration impact.

Prerequisites

Before you begin, ensure the following:

- The ESXI hosts must be configured for NVMe-oF.
- The Storage Data Target (SDT) is configured in PowerFlex.

CAUTION:

- **All VMs in the VMFS datastore must be powered off prior to starting the migration.**
- **The volume must support ATS mode. Only attempt migration if the VMFS6 datastore uses ATS-only locking mode and does not use SDC SCSI-2 reservations.**

Steps

1. Check the VMFS volume locking mode. Run the following command to check the locking mode:

```
esxcli storage vmfs lockmode list -l <volume name/label>
```

Example output for a volume that supports offline migration:

```
Volume Name UUID                                  Type   Locking Mode ATS Compatible ATS
Upgrade Modes ATS Incompatibility Reason
----------- ------------------------------------ ------ ------------ --------------
----------------- -------------------------
testVol1    5d1c5b0f-xxxxxxxx-xxxx-246e9xxxxdb0 VMFS-6 ATS          true           No
upgrade needed
```

CAUTION:

- **Offline migration is supported only if the locking mode is `ATS`. The `ATS+SDC (SCSI)` mode is not supported.**
- **Local drives on the targeted server may respond with `Device does not support ATS`. You can still perform offline migration if this response is returned.**

Example output for a volume that does NOT support offline migration:

```
Volume Name UUID                                  Type       Locking Mode       ATS
Compatible    ATS Upgrade Modes    ATS Incompatibility Reason
----------- ------------------------------------ ------ ------------ --------------
----------------- -------------------------
testVol2    63510e51-xxxxxxxx-xxxx-246e9xxxxde6   VMFS-6    ATS+SDC (SCSI)     false
   None              Device does not support ATS
```

2. Check for Raw Device Mapping (RDM) usage. Use the Dell Virtual Storage Integrator (VSI) plug-in to view disk types for each VM. The plug-in lists VMs by Runtime Name and notes if the VM has an RDM.

   NOTE: If a VM in the selected datastore has an RDM, the VM cannot be migrated to NVMe-oF.

   Do ONE of the following if you find an RDM:

   - Remove the RDM from the VM.
   - Convert the RDM.
   - Move the VM to another datastore before migration.
3. Check the claim rules and settings. Custom claim rules on the SDC device that exist before migration are not applied when the device is presented through NVMe-oF.
   - NVMe-oF devices do not present separate vendor and model fields through inquiry.
   - Fields are combined, which requiring a new claim rule if needed.
   - Claim rules based on device identifiers, such as World Wide Name (WWN), will fail because SDC and NVMe-oF identifiers are distinct.
   - By default, VMware claims newly presented NVMe-oF devices with the default pathing in the High-Performance Plug-in (HPP).
4. Check the device and path limits.

   NVMe-oF supports fewer devices and paths per ESXi host than SDC.

Protocol	Devices per Host	Paths per Host
SDC	1024	4096
NVMe-oF	256	2048

If SDC devices exceed NVMe-oF limits, do the following:

- Employ more ESXi hosts.
- Use VMware Storage vMotion to consolidate datastores before or after conversion.

5. Check for unsupported features. Some VMware features are not supported when you use NVMe-oF. Before migration, check feature support in your VMware ESXi version's release notes on the Broadcom TechDocs site.
6. Check for feature impact post-migration. The following features may have performance or functional changes when using NVMe-oF compared to SDC:

Feature	Nature of Impact	Action to be taken
WRITE SAME command with the UNMAP bit	If an NVMe-oF device does not support the NVMe-oF equivalent of write zeroes or unmap, there could be a performance impact.	None

Power off VMs and unmount volumes

Power off and unregister all VMs hosted on the datastore that you are migrating. Then, unmount the VMFS volumes from all hosts.

Prerequisites

Ensure that you have a list of the VMs in the datastore and the folder where the `VMX` file resides so you can register the VMs again after migration.

About this task

CAUTION: Do not delete the VMs, only unregister them.

Steps

1. To power off and unregister all VMs and templates, log in to vSphere Client.
2. Right-click the VM.
3. Select **Power** > **Power Off**.
   If VMware Tools is installed, Dell Technologies recommends using **Shut Down Guest OS**.
4. Click **Yes** to confirm.
5. Right-click the VM again.
6. Click **Remove from Inventory**.
7. Click **Yes** to confirm.

Unmount the VMFS volume from all ESXi hosts to ensure the volume is not in use during the consistency check and during migration. Do the following in vSphere Client:

8. Go to **Storage**.
9. Select the datastore.
10. Right-click the datastore.
11. Click **Unmount Datastore**.
12. On the **Unmount Datastore** page, select the hosts.
13. Click **OK**.

Check VMFS volume metadata consistency

Before you initiate migration, check the consistency of VMFS on-disk metadata to ensure no inconsistencies exist.

Prerequisites

The VMs are powered off and unregistered, and the VMFS volumes are unmounted.

About this task

Check the consistency of VMFS on-disk metadata and correct any inconsistencies.

Steps

1. Run VMware On-Disk Metadata Analyzer (VOMA) in check mode:

```
voma -m vmfs -f check -d /vmfs/devices/disks/<DEVICE>:<PARTITION> -s <OUTPUT_FILE>
```

In the previous command:

- *<DEVICE>* is the SDC device hosting the VMFS6 volume being migrated.
- *<PARTITION>* is the partition number on which the VMFS volume is formatted.
- *<OUTPUT_FILE>* is the absolute path of the file to save command output. This can be in /tmp or any VMFS volume other than the one being migrated.

For example:

```
voma -m vmfs -f check -d /vmfs/devices/disks/naa.60000970000120200302533030313031:1
-s /tmp/voma.out
```

Example output:

```
voma -m vmfs -f check -d /vmfs/devices/disks/eui.c70fd6f05a6ab90fc01b0dc500000005:1
-s tmp/voma.out
```

```
Running VMFS Checker version 2.1 in check mode
Initializing LVM metadata, Basic Checks will be done
Checking for filesystem activity
Performing filesystem liveness check..|Scanning for VMFS-6 host activity (4096 bytes/HB,
1024         SDC (SCSI) 2 reservation successful
             \
Phase 1: Checking VMFS header and resource files
   Detected VMFS-6 file system (labeled:'SDS2-storage') with UUID:69eafa45-51188a9f-
a0ac-58a2e139218c, Version 6:82
Phase 2: Checking VMFS heartbeat region
Phase 3: Checking all file descriptors.
Phase 4: Checking pathname and connectivity.
Phase 5: Checking resource reference counts.
Total Errors Found:           0
```

NOTE: If the command returns the error `VOMA failed to check device: Device or Resource busy`, the VMFS is not properly unmounted.

2. Analyze the output file for metadata inconsistencies. If you find inconsistencies, address them by running VOMA in advanced fix mode before continuing.

```
voma -m vmfs -f fix -d /vmfs/devices/disks/<DEVICE>:<PARTITION>
```

For example:

```
voma -m vmfs -f fix -d naa.60000970000120200302533030313031:1
```

```
Running VMFS Checker version 2.1 in fix mode
Initializing LVM metadata, Basic Checks will be done
Checking for filesystem activity
         SDC (SCSI) 2 reservation successful                        st activity (4096
bytes/HB, 1024 HBs).
Phase 1: Checking VMFS header and resource files
   Detected VMFS-6 file system (labeled:'SRM_UPGRADE_1') with UUID:6418928f-d0fb0a78-
fa29-34800d0ed39c, Version 6:82
Phase 2: Checking VMFS heartbeat region
Phase 3: Checking all file descriptors.
Phase 4: Checking pathname and connectivity.
Phase 5: Checking resource reference counts.
Total Errors Found:           0
Total Errors Fixed:           0
```

Collect the VMFS metadata dump. Save a metadata dump for reference if any metadata inconsistencies are seen in subsequent steps.

For more details on using VOMA in check mode, advanced fix mode, or dump mode, see Checking Metadata Consistency with vSphere On-disk Metadata Analyzer.

Detach and stop presenting the SDC

Detach the SDC device from ESXi hosts, and stop presenting the SDC device from the array.

Prerequisites

You checked the VMFS volume metadata consistency.

Steps

1. Log in to vSphere Client.
2. Select an ESXi host.
3. Under **Configure**, select **Storage** > **Storage Devices**.
4. Right-click the Network Address Authority (NAA) ID of the device.
5. Click **Detach**.
6. In the **Confirm Detach Device** window:
   a. Select **Detach the selected device from multiple hosts**.
   b. Check the listed hosts.
   c. Click **Yes**.

Use the following steps to stop presenting the SDC device that is specific to your storage array. Consult your array-specific documentation for the procedure. The following steps are for PowerFlex Manager:

7. Log in to PowerFlex Manager.
8. On the dashboard, select **Block** > **Volumes**.
9. Select the volume by checking the box.
10. Select **Provision** > **Unmap**.
11. In the **Unmap** pane:
    a. Check the hosts you want to unmap.
    b. Click **Unmap**.

Present and scan the device

Present the device as NVMe-oF to the ESXi host, and initiate device rescan on the host.

Prerequisites

You detached and stopped presenting the SDC.

About this task

The steps to present the device again using NVMe-oF are storage array-specific. Consult your array-specific documentation for the procedure. The following steps are for PowerFlex Manager:

Steps

1. Log in to PowerFlex Manager.
2. On the dashboard, select **Block** > **Volumes**.
3. Select the volume by checking the box.
4. Select **Provision** > **Map**.
5. In the **Map Volumes to Hosts** window:
   a. Click the **NVMe-oF** option.

b. Check the host or hosts that you want to map.
c. Click **Map**.

Once the device is presented using NVMe-oF, discovery is typically immediate. If the device does not appear, scan the adapters using vSphere UI or CLI.

6. Log in to vSphere Client.
7. Right-click the cluster.
8. Select **Storage** > **Rescan Storage**.
9. In the **Rescan Storage** window:
   a. Check both options: **Scan for new Storage Devices** and **Scan for new VMFS Volumes**.
   b. Click **OK**.
10. Optional: You can also run this command in the vSphere CLI:

```
esxcli storage core adapter rescan -a
```

Check VMFS volume metadata consistency after conversion

After converting a device to NVMe-oF, check the VMFS metadata consistency again.

Prerequisites

You presented the device as NVMe-oF to the ESXi host.

About this task

On the ESXi host with access to the device, run VOMA in check mode again to validate that VMFS on-disk metadata is still consistent.

CAUTION: VOMA uses the SDC (SCSI-2) reserve command to lock the device. NVMe-oF devices do not support an equivalent of SDC (SCSI-2) reservation. Because of this limitation, you must pass the `-N` option to VOMA when the backend device is NVMe-oF.

Steps

1. Run the following command:

```
voma -m vmfs -f check -N -d /vmfs/devices/disks/<DEVICE>:<PARTITION> -s <OUTPUT_FILE>
```

The following warning is displayed:

```
########################################################################
# Warning !!!                                                          #
#                                                                      #
# You are about to execute VOMA without device reservation.            #
# Any access to this device from other hosts when VOMA is running      #
# can cause severe data corruption                                     #
#                                                                      #
# This mode is supported only under VMware support supervision.        #
########################################################################
VMware ESXi Question:
Do you want to continue (Y/N)?
0) _Yes
1) _No

Select a number from 0-1:
```

CAUTION: This warning notifies you that it is your responsibility to prevent the volume from being mounted or accessed concurrently from other hosts while VOMA is running. If you correctly followed the previous steps and the device is mapped and discovered on only one ESXi host, it is safe to proceed.

Enter **0** at the prompt to continue with VOMA check mode.

Example command and output:

```
voma -m vmfs -f check -N -d /vmfs/devices/disks/eui.c01b0dc5000000056
```

```
Running VMFS Checker version 2.1 in check mode
Initializing LVM metadata, Basic Checks will be done
Checking for filesystem activity
         Reservation Support is not present for NVMEOF devices st activity (4096
bytes/HB, 1024
Performing filesystem liveness check..|
###########################################################################
#   Warning !!!                                                           #
#                                                                         #
#   You are about to execute VOMA without device reservation.             #
#   Any access to this device from other hosts when VOMA is running       #
#   can cause severe data corruption                                      #
#                                                                         #
#   This mode is supported only under VMware support supervision.         #
###########################################################################
Do you want to continue (Y/N)?
0) _Yes
1) _No
Select a number from 0-1: 0
Phase 1: Checking VMFS header and resource files
   Detected VMFS-6 file system (labeled:'SDS2-storage') with UUID:69eafa45-51188a9f-
a0ac-58a2
Phase 2: Checking VMFS heartbeat region
Phase 3: Checking all file descriptors.
Phase 4: Checking pathname and connectivity.
Phase 5: Checking resource reference counts.
Total Errors Found:           0
```

2. If you find errors, investigate before proceeding.

Update VMFS volume signature

After a device is presented as NVMe-oF, you must update the signature on the datastore. The current signature is based partly on the World Wide Name (WWN) of the device when presented using SDC. Because the NVMe-oF device ID is different, you must generate a new signature.

Prerequisites

You checked the VMFS volume metadata consistency again.

Steps

1. Log in to vSphere Client.
2. Go to **Storage** and select the cluster.
3. Right-click the cluster.
4. Click **Storage** > **New Datastore**.
5. On the **New Datastore** page:
   a. Select the **VMFS datastore** type.
   b. Click **Next**.
6. Assign a new name to the datastore.
7. Select the host.
8. Select the **NVMe-oF TCP** disk.
   The **Snapshot Volume** column confirms that ESXi has successfully recognized a previous VMFS SCSI signature on this specific disk.
9. Click **Next**.
10. On the **Mount Option** page:
    a. Select **Assign a new signature**.
    b. Click **Next**.
11. Click **Finish**.

Alternately, you can also run these commands in the vSphere CLI:

12. Dell Technologies recommends that you scan the file system again by running the following command:

```
esxcli storage filesystem rescan
```

13. Get a list of VMFS snapshot devices:

```
esxcli storage vmfs snapshot list
```

Check that the newly presented NVMe-oF device is present. Other snapshots unrelated to this process may also appear.

14. Update the VMFS volume signature:

```
esxcli storage vmfs snapshot resignature --volume-label=<label>|--volume-uuid=<id>
```

15. Optional: Rename the VMFS datastore.

When a VMFS volume is re-signatured, the VMFS volume label is prefixed with `snap` followed by an alphanumeric string.

For example, if the original datastore was `sds-storage`, it may now be named `snap-5c42a2bc-sds-storage`.

To revert to the original name:

a. Right-click the newly added datastore. For example, `snap-5c42a2bc-sds-storage`.
b. Click **Rename**.
c. Rename the datastore back to the original name by removing the `snap-` prefix.

Check the VMFS volume metadata consistency after applying the new signature.

16. Validate that the on-disk VMFS metadata is consistent after re-signature. Run the following VOMA command to check the VMFS volume:

```
voma -m vmfs -f check -N -d /vmfs/devices/disks/<DEVICE>:<PARTITION> -s <OUTPUT_FILE>
```

In the previous command:

- *<DEVICE>* is the SDC device hosting the VMFS6 volume being migrated.
- *<PARTITION>* is the partition number on which the VMFS volume is formatted.
- *<OUTPUT_FILE>* is the absolute path of the file to save command output. This can be in /tmp or any VMFS volume other than the one being migrated.

CAUTION: Verify if VOMA reports any inconsistencies. Proceed only if VOMA does not report any errors.

17. If there were no issues in previous steps, present the device using NVMe-oF to all ESXi hosts in the cluster.

NOTE: NVMe-oF devices are recognized immediately. If not visible, rescan adapters through vSphere UI or CLI.

18. Required: For verification purposes, ensure that the VMFS6 volume is mounted and accessible on all hosts.

Register and power on all VMs and templates

Register all VMs hosted on the datastore and power them on. Verify that VMs power on successfully and that you can access the virtual machine disks (VMDKs).

Prerequisites

You updated the VMFS volume signature.

About this task

Register and power on all VMs and templates.

Steps

1. Log in to vSphere Client.
2. Go to **Storage**.

3. Right-click the new datastore.
4. Select **Register VM**.
5. Select the VMs to register.
6. Click **OK**.
7. On the **Register Virtual Machine** page:
   a. Select the datacenter.
   b. Click **Next**.
8. Select the cluster to place the VM.
9. Click **Next**.
10. Click **Finish**.

To power on VMs:

11. Go to **Hosts and Clusters**.
12. Right-click the VM.
13. Click **Power On**.

> **NOTE:** When powering on VMs from the vCenter UI, you may see a dialog box asking if the VM was copied or moved. Select **I Copied it** in the dialog box.

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Performing a single component upgrade

The single component upgrade allows for upgrading the software and firmware on individual components, such as applying security patches, without performing a full RCM/IC upgrade. It is recommended to use this option for component-specific upgrades, such as patch updates, and to ensure compliance with the standard RCM/IC version.

Single component upgrades are not engineering-validated upgrades. You should understand the implication and risks. Dell Technologies recommends aligning with RCM/IC as the official-validated catalog. If you consider deviating to align with the security or functionality requirements, consider testing the compatibility.

A single component upgrade involves the following steps:

1. Upgrading the firmware and software components.
2. Extracting the compliance report.
3. Modifying and publishing the existing template.

CAUTION: Since Dell Technologies only validates the released RCM/IC code catalogs, you must be aware of the implications and risks that are involved in a single component upgrade. If any issues are identified related to the RCM/IC deviation, you may be asked to modify the code or firmware.

If an upgrade file, for example iDRAC FW 7.xx.xx.xx, exceeds 200 MB, use the SMB or CIFS share option. Browser uploads are not supported for upgrade files larger than 200 MB.

The use of single component upgrade enables the PowerFlex platform to remain in compliance, using the deviated code/ firmware, which means future upgrades will not be blocked due to non-compliance.

The single component upgrade option enables the PowerFlex platform to remain in compliance. The future upgrades are not blocked due to non-compliance.

The following limitations are associated with the single component upgrade:

- VMware ESXi version changes are restricted. Upgrading from VMware ESXi 8.x to 9.x or downgrading is not supported.
- Updates to embedded operating systems, such as SLES or CentOS are not supported.

Upgrade the firmware components

To upgrade firmware on a PowerFlex node, authenticate as an administrator, choose a bundle, modify the component, and apply the updates.

Prerequisites

You must verify that you are using administrator credentials.

About this task

An upgrade of the following firmware components is supported:

- Drive firmware
- iDRAC Lifecycle Controller (LCC)
- BIOS firmware
- Network Interface Card (NIC) firmware
- BOSS card firmware
- Switch firmware

The upgrade of the PowerFlex core is not supported.

Steps

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. From the **Compliance Versions** page, select the current RCM/IC bundle and click **View Bundles**.

3. From the list of bundles in the **Firmware Bundles** tab, verify the bundles that you want to modify.
   To modify components on the PowerFlex R660 node, select **System Bundle (Windows x64) PER660 v850** or select the appropriate components such as switches, CloudLink, or operating system bundles.
4. Select the appropriate bundle to upgrade and click **View**.

   NOTE: Only one component can be modified at a time.

5. Select the specific component to upgrade and click **Modify**.
6. On the **Update Bundle Component** page, perform the following:
   a. Select either **Download from local network path** or **Upload from local**.
   b. Provide the **New Bundle Version** (if it is not picked automatically) to match the existing bundle version.
   c. In the **Type "MODIFY COMPLIANCE FILE" to confirm** field, enter `Modify Compliance File` and click **Proceed**.
7. Repeat the steps through 5 and 7 for the remaining components.
8. Click **Settings** > **Repositories** > **Compliance Versions** and verify that a wrench icon and `Modified` is appearing for the modified RCM/IC.
9. To update the resources, perform the following:
   a. On the menu bar, click **Resources** and select the PowerFlex nodes to update and click **Run Inventory**.
   b. Click **Lifecycle** > **Resource Groups**.
   c. Select the resource group to be modified and click **View Details**.
   d. Click **View Compliance Report**.
   e. Select the modified PowerFlex node or other components and click **Update Resources**.
   f. From the **Apply Resource Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now**.
   g. Select the appropriate maintenance mode and click **Apply**.
10. To monitor and confirm that the component is updated with the new version, perform the following:
    a. Click **Lifecycle** > **Resource Groups** to verify that the required component is upgraded.
    b. Select the resource group that you already modified and click **View Details**.
    c. On the **Resource Group Information** page, verify that the resource group compliance status is `Compliant`.

    If an upgrade failure occurs, contact Dell Technologies Support for assistance.

Upgrade the software components

Use the following procedures to upgrade the software components using PowerFlex Manager. These procedures are applicable only when you consider deviating from the standard upgrade procedure.

NOTE: Ensure you upload identical versions of the VMware ESXi patch bundle and the ESXi ISO file to PowerFlex Manager.

A software component upgrade involves operating system updates such as the following:

- Applying ESXi patches
- Upgrading the CloudLink virtual machine

The following limitations are associated with the single component upgrade:

- ESXi version changes are restricted; upgrading from ESXi 8.x to 9.x or downgrading is not supported.
- Updates to embedded operating systems, such as SLES or CentOS are not supported.

Upgrade VMware ESXi patch and ISO images

You can upgrade the VMware ESXi patch and ISO images using PowerFlex Manager.

Steps

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. Verify that the required RCM/IC is uploaded and is available.
3. Select the RCM/IC bundle to be modified and click **View Bundles**.
4. From the **Software Bundles** tab, select the **ESXI Software Bundle** checkbox, clear the checkbox for other software bundles, and click **View**.

5. Select **Upload Software Depot** and select either of the following options to upload the ESXi path depot file:
   - **Upload from local**: Select this option when the depot file is smaller than 500 MB.
   - **Download from local network path**: Select this option when the depot file is larger than 500 MB. Enter the CIFS file path and user credentials to access the file on the local network drive.
6. For the **Upload from local** option, complete the following:
   a. To provide a local file, click **Choose File**.
   b. To confirm that you want to modify the existing file, enter **`Modify Compliance File`** and click **Proceed**.
   c. Wait for the compliance file to synchronize and verify that the status of the compliance file is `Available`.
   d. Verify that a wrench icon and `Modified` displays for the modified RCM/IC.
7. For the **Download from local network path** option, complete the following:
   a. Enter the `CIFS` file or the `HTTPS` address to access the file.
   b. Enter the username and password.
   c. Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file or the file that is at the HTTPS address.
   d. To confirm that you want to modify the existing file, enter **`Modify Compliance File`** and click **Proceed**.
   e. Wait for the compliance file to synchronize and verify that the status of the compliance file is `Available`.
   f. Verify that a wrench icon and `Modified` displays for the modified RCM/IC.
8. **NOTE:** You must complete steps 5 through 7 before you start step 8.

   To upload the `ESXI.depot.zip` and `ESXi.iso` files, complete the following:
   a. From **Repositories** > **Compliance Versions**, select the RCM/IC bundle, and click **Add**.
   The **Add Compliance File** dialog box displays.
   b. Enter the file path, username, and password for both the installation ISO and the depot ZIP file. Ensure the files match with the existing VMware ESXi patch and ISO files.
   c. Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file or the file that is at the HTTPS address.
   d. Click **Save** and wait for the synchronization and download to complete.
   e. On the **Compliance Versions** page, verify that the files are added and the RCM/IC is in `Available` state.

Update resource inventory and resource groups

Once you upgrade the VMware ESXi patch and ISO images, use this procedure to update the resource inventory and resource groups.

Steps

1. On the menu bar, click **Lifecycle** > **Resources**.
2. Select the PowerFlex node to update, and click **Run Inventory**.
3. Click **Lifecycle** > **Resource Groups** and select the service to upgrade.
4. Click **Non-Compliant** in the **Resource Group Information** tab.
5. From the **Software Components** tab, select all the listed nodes, and click **Update Resources**.
6. Select the appropriate **Maintenance Mode** and click **Apply**.
7. To monitor the job and verify that the required component is upgraded, perform the following:
   a. Click **Lifecycle** > **Resource Groups**.
   b. Select the resource group that you modified and click **View Details**.
   c. On the **Resource Group Information** tab, verify that the resource group compliance status is `Compliant`.

   If the upgrade fails, contact Dell Technologies Support for assistance.

Upgrade CloudLink Center

Using the single component option, you can upgrade the CloudLink Center version.

Prerequisites

CloudLink ISO is required to start the CloudLink version upgrade using the single component upgrade option.

You must verify that you are using the administrator credentials.

Steps

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Version**.
2. Select the RCM/IC bundle, and click **View Bundles**.
3. From the **Firmware Bundles** tab, select the **System Bundle CloudLink** and click **View**.
4. Select the component **clc** that you want to upgrade and click **Modify**.
5. From the **Update Bundle Component** page, select **Download from local network path** option and upload the CloudLink ISO file.
6. Enter the **New Bundle Version** if it is not picked automatically, to match with the existing bundle version.
7. Type **`Modify Compliance File`** in the field and click **Proceed**.
8. Click **Repositories** > **Compliance Versions** and verify that the modified RCM/IC appears with a wrench icon and `Modified`.
9. To update the resources, perform the following:
   a. From **Lifecycle** > **Resources**, select both the CloudLink resources and click **Run Inventory**. Wait for two minutes to complete the inventory and refresh your browser.
   b. Select the CloudLink Center to upgrade and click **Non-Compliant** and click **Update Resource**.
   c. From the **Apply Resource Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now** and click **Apply Updates**. Wait till the deployment status becomes `Available`.
   d. On the **Resources** page, click **CloudLink Center** > **View Details** to verify the upgraded CloudLink Center version. Wait for three to four minutes. Both CloudLink Centers are updated with the new version.
10. To manually upgrade the SecureVM agent on all the PowerFlex nodes from CloudLink Center, perform the following:
    a. In your web browser, type the URL for CloudLink Center in the following format: **`https://clc_address`**
    b. Log in with the **`secadmin`** credentials.
    c. Click **Agents** > **Machines** and select the node.
    d. Click **Actions** > **Upgrade**. The upgrade option is disabled if an upgrade is not required for the SecureVM agent.
    e. Repeat step 10a through 10d for every PowerFlex node that requires a CloudLink agent upgrade.
    f. Click **Agents** > **Machines**, select the node, and verify the SecureVM agent version.
    g. After the agents are updated on the nodes, from CloudLink Center, click **Agents** > **Agent Download** to download the `64-bit SUSE RPM` package or `64-bit CentOS RPM` package depending on the operating system running on the nodes and modify the SecureVM agent component in the RCM/IC bundle.
11. To modify the SecureVM agent component in the RCM/IC bundle, perform the following:
    a. Verify that you are using the administrator credentials.
    b. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**, select the current RCM/IC bundle and click **View Bundles**.
    c. Click **Software Bundles** > **CloudLink SecureVM** > **View**. You will notice that there are two CloudLink SecureVM bundles with the same name. One bundle is for SUSE and the other bundle is for CentOS. Select the bundle and click **View** to differentiate the bundle. Select the bundle based on the operating system installed on the PowerFlex nodes.
    d. Click **Modify**.
    e. From the **Update Bundle Component** page, select the **Download from local network path** or **Upload from local** and select the downloaded agent.
    f. Enter the **New Bundle Version** to match the existing bundle version if it is not picked automatically.
    g. Type **`Modify Compliance File`** in the field and click **Proceed**.
12. To update the resource group details, perform the following:
    a. On the menu bar, click **Lifecycle** > **Resource Groups**.
    b. Select the resource group to modify and click **View Details**.
    c. Click **More Actions** > **Update Resource Group Details**.
    d. On the **Resource Group Information** page, verify that the resource group compliance status is `Compliant`.

If the upgrade fails, contact Dell Technologies Support for assistance.

Next steps

Exit the PowerFlex nodes from service mode.

Export the compliance report

You can export a compliance report using PowerFlex Manager. The compliance report identifies the list of modified components and their versions. It also compares component versions during upgrades from a modified RCM/IC to a standard RCM/IC.

Prerequisites

Before exporting the compliance report, you must ensure that the default RCM/IC version is selected. To verify that the default version is selected, go to **Settings** > **Repositories** > **Compliance Versions**.

About this task

When transitioning from a modified RCM to a desired RCM, ensure the following:

- The component versions in the modified RCM are equal to or lower than the component versions in the desired RCM.
- If any component in the modified RCM has a higher version, use the **Extract Compliance** tool to identify the components.
- Use the single component upgrade to add those components with a higher version to the desired RCM to ensure version alignment and compatibility,

Steps

1. On the menu bar, click **Lifecycle** > **Resources** > **Export Report**.
2. Select **Export Compliance PDF Report** or **Export Compliance CSV Report**.
3. Identify the modified components and verify the number of components.

Modify and publish the template

After completing a single component upgrade, you can modify and publish the template. This process involves updating the existing template by incorporating a modified ESXi ISO image. The updated template is then published and used for future system expansions.

About this task

Ensure the modified RCM/IC is the same version as the **Target Version** under the **Existing Resource Group** page. While adding a resource group on an existing PowerFlex system, ensure that you select the latest uploaded operating system images.

Steps

1. On the menu bar, click **Templates** > **My Templates**.
2. Select the template that you want to modify and click **Modify**.
3. Select the node and click **Modify**.
   The node details display.
4. Click **Continue**.
   The **OS Settings** display.
5. Under **OS Settings** > **OS Image**, ensure that the newly uploaded image is selected from the dropdown.
6. Click **Save** > **Publish Template**.
   A confirmation message displays asking whether you are sure that you want to publish the template, click **Yes**.

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Host setup and configuration

Use the following procedures to manually install a storage data client on various operating systems.

Installing the SDC manually

If a host of the SDC type was not installed on front-end servers during initial installation activities, follow the appropriate instructions for your front-end servers' operating system.

See the *Dell PowerFlex and PowerFlex Manager Support Matrix* to ensure that the intended SDC operating system of the host is compatible for use with your PowerFlex system.

You can use any of the following methods to install the SDC manually:

- VMware ESXi
- Linux
- Windows

Install the SDC on a VMware ESXi 8.x server

Install the SDC on a VMware ESXi 8.x server using command-line esxcli commands.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this task, to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host.
- The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- A GUID string for the SDC. These strings can be generated by tools that are freely available online. The GUID must conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

About this task

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

NOTE: This procedure requires two servers reboots.

Steps

1. On the VMware ESXi-based server on which you are installing the SDC, set the acceptance level:

   ```
   esxcli software acceptance set --level PartnerSupported
   ```

2. Install the SDC:
   a. Change the directory where you have copied SDC.ZIP file.
   b. Unzip the SDC.ZIP file, run the following command: **`unzip <filename.zip>`**.
   c. Install the SDC package, run the following command: **`esxcli software vib install -v <path_to_vib>`**.

Where **<path_to_vib>** is the path to SDC.VIB file which is unzipped in the above command.

3. Get permission to reboot the VMware ESXi-based server, and perform a graceful reboot cycle to load the SDC driver on the server.

   In a first-time install, the SDC will not automatically boot up at this point. Further configuration and an additional reboot are required, as described in this procedure.

4. Set the IP address of the MDM:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
   IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
   ```

   Where

   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses, or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a "," symbol.
     - Separate multiple MDM clusters with the + symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one:

       ```
       10.20.30.40,50.60.70.80+11.22.33.44
       ```

   - *<XXXXXX>* is the user-generated GUID string. Example GUID: 12345678-90AB-CDEF-1234-567890ABCDEF
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.

   CAUTION: A few applications require an advanced settings in ESXi for specific vital product data (VPD) pages. For more information about changing `blkScsiUnitSerialLenByStandard` parameter on an ESXi host for PowerFlex volumes, see Procedure to change parameter blkScsiUnitSerialLenByStandard on ESXi for PowerFlex volumes. Perform this procedure only if your applications require this setting. This is not a common setting.

5. Back up the parameters:

   ```
   /bin/auto-backup.sh
   ```

6. Perform another graceful reboot cycle.
7. Ensure that the driver is loaded:

   ```
   vmkload_mod -l | grep scini
   ```

   Output similar to the following indicates that the driver is loaded:

   ```
   scini 2 808
   ```

   The SDC component is installed on the VMware ESXi-based server.

Results

The SDC is installed on the VMware ESXi-based server and is connected to PowerFlex.

Install the SDC on a VMware VCF/ESXi 9.x based server

Use this procedure to install the SDC on a VMware VCF/ESXi 9.x based server.

About this task

Install the SDC with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more SDCs to an existing system, and for adding SDCs to a two-layer system during initial deployment activities.

NOTE: This procedure requires multiple servers reboots.

Prerequisites

Ensure that you have the following:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.

  NOTE: Obtain and include the IP addresses of as many potential manager MDMs as possible when performing this procedure to make it easier to switch MDMs in the future. You can add a total of eight IP addresses.

- Login credentials for the intended SDC host.
- The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- A GUID string for the SDC. These strings can be generated by tools that are available online, for example GUID generator or UUID generator. The GUID must conform to OSF DCE 1.1. The expected format is xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx where each x can be a digit (0–9) or a letter (a–f).

Steps

1. On the VMware ESXi-based server on which you are installing the SDC, set the acceptance level, type the following: **`esxcli software acceptance set --level PartnerSupported`**
2. Put the VMware ESXi node into maintenance mode: **`esxcli system maintenanceMode set --enable true`**
3. Install the SDC agent package: **`esxcli software component apply -d /<PATH>/scaleio-daemon-esx_5.x.x.xxx-x.zip`**

   Where **`<PATH>`** is the directory containing the SDC agent package (.zip file).

4. Install the SDC driver package: **`esxcli software component apply -d /<PATH>/scaleio-sdc-esx9x_5.x.x.xxx-xOEM.xxx.x.xxxxxxxx.zip`**.
5. Get permission to reboot the VMware ESXi-based server, and perform a graceful reboot cycle to load the SDC driver: **`reboot`**
6. Ensure that the driver is loaded: **`vmkload_mod -l | grep scini`**.

   Output similar to the following indicates that the driver is loaded:

   ```
   scini 2 808
   ```

7. Set the module parameters:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=<XXXXXX>
   IoctlMdmIPStr=<LIST_VIP_MDM_IPS> IoctlMdmPasswordStr=<ip>-<password>"
   ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the virtual IP addresses, or the MDM IP address. Do not include tiebreaker addresses here.
     - Separate the IP addresses of the same MDM cluster with a `,` symbol.
     - Separate multiple MDM clusters with the + symbol.

       In the following example, there are two MDM clusters. The first has two IP addresses and the second has only one, use the following format for the `IoctlMdmIPStr` parameter: `10.20.30.40,50.60.70.80+11.22.33.44`
   - <XXXXXX> is the user-generated GUID string. Example GUID: `12345678-90AB-CDEF-1234-567890ABCDEF`
   - Optional: `IoctlMdmPasswordStr` is used for SDC authentication to the MDM for access to the PowerFlex system.

   For example:

   ```
   esxcli system module parameters set -m scini -p "IoctlIniGuidStr=a9745569-
   eed1-43fe-82fd-b28199dd7ffb  IoctlMdmIPStr=192.168.211.93,172.168.211.108"
   ```

8. Reboot the VMware ESXi host to set the parameters.
9. Exit the VMware ESXi host from maintenance mode: **`esxcli system maintenanceMode set --enable false`**
10. Back up the parameters: **`/bin/auto-backup.sh`**.
11. Log in to the primary MDM or PowerFlex Manager to verify that the SDC is connected, and proceed with volume mapping.

Install the SDC on a Linux-based server

Install the storage data client on a Linux-based server with the appropriate parameters to connect it to an existing PowerFlex system. This procedure is relevant both for adding more storage data clients to an existing system, and for adding storage data clients to a two-layer system when deploying a new system.

Prerequisites

Ensure that you have:

- The virtual IP address or MDM IP address of the existing system. If an MDM virtual IP address is not in use, obtain the IP addresses of all the MDM managers.
- Login credentials for the intended SDC node.
- The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the Dell Technologies support site).
- Secure boot is not supported; ensure that secure boot is disabled in the BIOS.

About this task

This section describes how to manually install an external SDC on a Linux-based server using the command line.

The steps in the procedure apply to Citrix Hypervisor(XenServer)/Red Hat Enterprise Linux/Oracle Linux operating systems.

Steps

1. Install the GPG key on every server on which SDC is installed. From the PowerFlex installation folder, run the following command on every server:

   ```
   rpm --import RPM-GPG-KEY-ScaleIO
   ```

2. From the command line, install the SDC:
   - Red Hat Enterprise Linux/Oracle Linux/SLES

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> rpm -i <SDC_PATH>.rpm
     ```

   - Ubuntu (Extract these files before use, as described above.)

     ```
     MDM_IP=<LIST_VIP_MDM_IPS> dpkg -i <SDC_PATH>.deb
     ```

   Where
   - *<LIST_VIP_MDM_IPS>* is a comma-separated list of all the MDM IP addresses or the virtual IP address of the MDM. Do not include tiebreaker addresses here.
   - *<SDC_PATH>* is the path where the SDC installation package is located
3. Verify that the driver is loaded using one of the following commands:
   - SLES 15:

     ```
     systemctl status scini
     ```

   - All other Linux:

     ```
     service scini status
     ```

Results

The SDC is installed on the Linux-based server, and is connected to PowerFlex.

Install SDC on Windows-based PowerFlex compute-only node

Use this procedure to install SDC on the Windows-based PowerFlex compute-only node.

Steps

1. Download PowerFlex LIA and the SDC plugin.

2. Double-click the PowerFlex LIA setup file.
3. Accept the terms and license agreement and click **Install**.
4. Click **Finish**.
5. Do the following for Windows-based SDC node before deploying the SDC component using the command prompt with administrator privileges:
   a. Type **bcdedit /debug ON**
   b. Type **bcdedit /**set testsigning ON
   c. Type **reboot windows**
6. If you know the MDMs VIPs before installing the SDC component, run the command:

```
msiexec /i HYPERLINK "http://
vm-sio-fs.xiolab.lab.emc.com/sw_dev/Artifacts/Build-All-WIN16/5.0.0.12015/release/EMC-
ScaleIO-sdc-5.0-0.12015.msi"EMC-ScaleIO-sdc-5.x-x.xxxxx.msi msi
MDM_IP=<LIST_VIP_MDM_IPS>
```

   For example:

```
msiexec /i HYPERLINK "http://
vm-sio-fs.xiolab.lab.emc.com/sw_dev/Artifacts/Build-All-WIN16/5.0.0.12015/release/EMC-
ScaleIO-sdc-5.0-0.12015.msi"EMC-ScaleIO-sdc-5.x-x.xxxxx.msi msi
MDM_IP=192.168.151.185,192.168.152.185
```

7. Accept the terms and license agreement and click **Install**.
8. Click **Finish**.
9. On confirmation from the customer to reboot the Windows server, perform a reboot to load the SDC driver on the server.
10. If the MDMs VIPs are not provided before installing the SDC component:
    a. Double-click the PowerFlex SDC setup.
    b. Accept the terms in the license agreement and click **Install**.
    c. Click **Finish**.
    d. Type **c:\Program Files\EMC\scaleio\sdc\bin>drv_cfg.exe --add_mdm --ip <VIPs_MDMs>** to configure the PowerFlex node in the PowerFlex software.

Mapping volumes to Windows-based PowerFlex compute-only nodes

Use this procedure to map volumes to Windows-based PowerFlex compute-only nodes.

Steps

1. Log in to PowerFlex Manager.
2. Verify that the Windows-based PowerFlex compute-only nodes are listed under **Lifecycle** > **Storage nodes**.
3. Click **Block** > **Volumes**.
4. Select the **Volume** that you want to map.
5. Click **Provision** > **MAP**.
6. Select the newly added Windows host and click **MAP**.
7. Log in to a Windows-based PowerFlex compute-only node and select **Disk Management**.
8. Rescan the disk and bring it online.
9. Click **Initialize disk**.
10. Create **New Simple Volume** and click **Next**.
11. Select **Default** and click **Next**.
12. Assign the drive letter.
13. Select **Default** and click **Next**.
14. Click **Finish**.

Enabling or disabling SDC authentication

PowerFlex allows authentication and authorization be enabled for all SDCs connected to a cluster. Once authentication and authorization are enabled, older SDC clients and SDCs without a configured password will be disconnected.

The SDC procedures are not applicable for the PowerFlex management cluster.

NOTE: If SDC authentication is enabled in a production environment, data unavailability may occur if clients are not properly configured.

Log in to PowerFlex using scli

The PowerFlex MDM cluster will use mTLS authentication now instead of legacy TLS authentication with username and password.

About this task

Mutual Transport Layer Security (mTLS) is a method for mutual authentication. mTLS ensures that the parties at each end of a network connection are who they claim to be by verifying that they both have the correct private key.

Steps

1. Generate the certificate:
   a. To copy the management certificate to the root location, type:
      `cp /opt/emc/scaleio/mdm/cfg/mgmt_ca.pem /`
   b. To generate the certificate, type **`scli --generate_login_certificate --management_system_ip <MNO_IP> --username <USER> --password <PASS> --p12_path <P12_PATH> --p12_password <P12_PASS> --insecure`**

      where:

      management_system_ip is Ingress IP address and username

      username is keycloak username (Ingress UI username)

      password is keycloak user password (Ingress UI password). If not provided in the command line then CLI will prompt for it.

      `--p12_path <P12_PATH>` is optional. If not provided then file will be created users home directory

      p1_password is the password for p12 bundle. Same password needs to be provided for generation of the certificate and for login operation
2. To add the certificate, type **`cd /opt/emc/scaleio/mdm/cfg; scli --add_certificate --certificate_file mgmt_ca.pem`**
3. To log in using the certificate, type **`scli --login --p12_path <P12_PATH> --p12_password <P12_PASS>`**

Prepare for storage data clients authentication

Prepare the storage data clients for authentication.

Prerequisites

Ensure that you have the following information:

- Primary and secondary MDM IP address
- PowerFlex cluster credentials

Steps

1. Log in to the primary MDM: **`scli --login --username admin --management_system_ip <management_system_ip>`**
2. Authenticate with the PowerFlex cluster using the credentials provided.
3. Type **`scli --query_all_sdc`** and record all the connected SDCs (any of the identifier - NAME, GUID, ID, or IP address):

4. For each SDC in your list, use the identifier recorded to generate and record a CHAP password. Type **scli --generate_sdc_password --sdc_id <id> or --sdc_ip <ip> or --sdc_name <name> or --sdc_guid <guid> --reason "CHAP setup"**.

   This password is specific to that SDC and cannot be reused for subsequent SDC entries.

   For example:

   scli --generate_sdc_password --sdc_IP 172.16.151.36 --reason "CHAP setup"

   Sample output:

```
# scli --generate_sdc_password --sdc_ip 172.16.151.36 --reason "CHAP setup"
Successfully generated SDC with IP 172.16.151.36 password:
AQAAAAAAAAAAAAA8UKVYp0LHCDFD59BrnEXNPVKSlGfLrwAk
```

Configure storage data client to use authentication

Perform this procedure to configure the storage data clients for authentication.

About this task

For each storage data client, populate the generated CHAP password. On a VMware ESXi host, this requires setting a new `scini` parameter through the esxcli tool. Use the procedure to perform this configuration change. For Windows and Linux SDC hosts, the included `drv_cfg` utility is used to update the driver and configuration file in real time.

NOTE: Reboot the VMware ESXi hosts for the new parameter to take effect.

Prerequisites

- Generate the pre-shared passwords for all the storage data clients to be configured.
- Ensure that you have the following information:
  - Primary and secondary MDM IP addresses or names
  - Credentials to access all VMware ESXi hosts running storage data clients

Steps

1. Using SSH log in to the VMware ESXi host using the provided credentials.
2. Type **esxcli system module parameters list -m scini | grep Ioctl** to list the hosts current `scini` parameters:

```
IoctlIniGuidStr                      string  d30ff770-b64c-40b5-a341-58d18927e523
                    Ini Guid, for example: 12345678-90AB-CDEF-1234-567890ABCDEF
IoctlMdmIPStr                        string
192.168.151.20,192.168.152.20,192.168.153.20,192.168.154.20  Mdms IPs, IPs for MDM in
same cluster should be comma separated. To configure more than one cluster use '+'
to separate between IPs.For Example: 10.20.30.40,50.60.70.80+11.22.33.44. Max 1024
characters
IoctlMdmPasswordStr                  string
                                         Mdms passwords. Each value
is <ip>-<password>, Multiple passwords separated by ';' signFor example: 10.20.30.40-
AQAAAAAAAACS1pIywyOoC5t;11.22.33.44-tppW0eap4cSjsKIcMax 1024 characters
```

   NOTE: The third parameter `IoctlMdmPasswordStr` is empty.

3. Using ESXCLI, configure the driver with the existing and new parameters. To specify multiple IP addresses, use a semicolon (;) between the entries, as shown in the following example. Additional data IP addresses, data3, and data4 can be used, if required.

```
esxcli system module parameters set -m
scini -p "IoctlIniGuidStr=10cb8ba6-5107-47bc-8373-5bb1dbe6efa3
IoctlMdmIPStr=192.168.151.20,192.168.152.20 IoctlMdmPasswordStr=192.168.151.20-
AQAAAAAAAAA8UKVYp0LHCFD59BrnExNPvKSlGfLrwAk;192.168.152.20-
```

```
AQAAAAAAAAA8UKVYp0LHCFD59BrnExNPvKSlGfLrwAk bBlkDevIsPdlActive=1
blkDevPdlTimeoutMillis=60000"
```

NOTE: There are spaces between `Ioctl` parameter fields and the opening quotes. The example is entered on a single line.

4. Manually migrate the workloads to the other hosts if required, and place the VMware ESXi host in maintenance mode.
5. Reboot the VMware ESXi host.
6. Once the host has completed rebooting, remove it from maintenance mode.
7. Repeat this procedure for each VMware ESXi SDC host.

Examples - Windows and Linux SDC nodes

Windows and Linux hosts have access to the `drv_cfg` utility, which allows driver modification and configuration in real time.

The **`--file`** option allows for persistent configuration to be written to the driver's configuration file (so that the SDC remains configured after a reload or reboot).

NOTE: Only one IP address is needed for the command to identify the MDM to modify.

Windows (from within a PowerShell prompt):

```
C:\Program Files\EMC\scaleio\sdc\bin\drv_cfg --set_mdm_password --ip <MDM IP> --port
6611 --password <secret>
```

Linux:

```
/opt/emc/scaleio/sdc/bin/drv_cfg --set_mdm_password --ip <MDM IP> --port 6611 --password
<secret> --file /etc/emc/scaleio/drv_cfg.txt
```

Iterate through the relevant SDCs, using the command examples along with the recorded information.

Enable storage data client authentication

Perform this procedure to enable storage data client authentication.

Prerequisites

- Make sure that all storage data clients are running PowerFlex, and are configured with their appropriate CHAP password. Any older or unconfigured storage data client will be disconnected from the system when authentication is turned on.
- Ensure that you have the following information:
  - Primary MDM IP address
  - Credentials to access the PowerFlex cluster

Steps

1. SSH into the primary MDM.
2. Type **`scli --login --username admin --management_system_ip <management_system_ip>`** to log in to the PowerFlex cluster using the provided credentials.
3. Type **`scli --set_sdc_authentication --enable`** to enable storage data client authentication feature.
4. Type **`scli --check_sdc_authentication_status`** to verify that the storage data client authentication and authorization is on, and that the storage data clients are connected with passwords.

   Sample output:

   ```
   # scli --check_sdc_authentication_status
   SDC authentication and authorization is enabled.
   Found 4 SDCs.
   The number of SDCs with generated password: 4
   The number of SDCs with updated password set: 4
   ```

5. If the number of storage data clients does not match or any storage data clients are disconnected, storage data clients, list any or all of the disconnected storage data clients and then disable the storage data client authentication by typing the following commands:

```
scli --query_all_sdc | grep "State: Disconnected"

scli --set_sdc_authentication --disable
```

6. Recheck the disconnected storage data clients to make sure that they have the proper configuration applied. If necessary, regenerate their shared password and reconfigure the storage data client. If you are unable to resolve the storage data client disconnection, leave the feature disabled and contact Dell support as needed.

Disable SDC authentication

Use this procedure to disable SDC authentication.

Prerequisites

Ensure all SDCs are configured with their appropriate CHAP secret. Any older or unconfigured SDC will be disconnected from the system when authentication is turned on.

Ensure you have the following information:

- primary MDM IP address
- Credentials to access the PowerFlex cluster

Steps

1. SSH to the primary MDM address.
2. Log in to the PowerFlex cluster using the provided credentials.
3. Disable the SDC authentication, type: **`scli --set_sdc_authentication --disable`**
   Once disabled, SDCs will reconnect automatically unless otherwise configured.

Results

Once disabled, the SDCs reconnect automatically unless otherwise configured.

Expand an existing PowerFlex cluster with SDC authentication enabled

Once a PowerFlex cluster has SDC authentication that is enabled, new SDCs must have the configuration step that is performed after the client is installed. This procedure is not applicable for the PowerFlex management cluster (ESXi). For Windows PowerFlex compute-only nodes, only firmware upgrades are supported.

Prerequisites

Ensure you have the following information:

- Primary MDM IP address
- Credentials for the PowerFlex cluster
- The IP address of the new cluster members

Ensure you have added the SDC authentication enabled on the PowerFlex cluster.

Steps

1. Install and add the SDCs as per normal procedures (whether using PowerFlex Manager or manual expansion process).

   NOTE: New SDCs will show as **Disconnected** at this point, as they cannot authenticate to the system.

2. SSH to the primary MDM.
3. Log in to the PowerFlex cluster using the scli tool.
4. For each of your newly added SDCs, generate and record a new CHAP secret, type: **`scli --generate_sdc_password --sdc_IP <IP of SDC> --reason "CHAP setup - expansion."`**
5. SSH and log in to the SDC host.
6. If the new SDC is a VMware ESXi host, follow the rest of this procedure.

7. Type **-m scini | grep Ioctl** and **esxcli system module parameters list -m scini** to list the current scini parameters of the host.
8. Using esxcli, type **esxcli system module parameters set -m scini -p** to configure the driver with the existing and new parameters.
   For example, **esxcli system module parameters set -m scini -p "IoctlIniGuidStr=*09bde878-281a-4c6d-ae4f-d6ddad3c1a8f* IoctlMdmIPStr=*10.234.134.194,192.168.152.199,192.168*"**.
9. At this stage, the SDC's configuration is ready to be applied. On VMware ESXi nodes a reboot is necessary for this to happen.
10. Place the VMware ESXi host in maintenance mode. No workloads should be running on the node, as we have not yet configured the SDC.
11. Reboot the VMware ESXi host.
12. Once the host has completed rebooting, remove it from maintenance mode.
13. Take the storage nodes out of maintenance mode (if present).
14. Repeat steps 5 through 15 for all VMware ESXi SDC hosts.

Add a Windows or Linux authenticated SDC

Use the **drv_cfg** utility on a Windows or Linux machine to modify both a running and persistent configuration. Use the following examples to perform the task on a Windows or Linux based PowerFlex node.

About this task

For Windows PowerFlex compute-only nodes, only firmware upgrades are supported.

Prerequisites

Only one IP address is required for the command to identify the MDM to modify.

Steps

1. Press **Windows +R**.
2. To open the command line interface, type `cmd`.
3. For Windows, type **drv_cfg --set_mdm_password --ip <MDM IP>** in the drv_cfg utility. For example:

   drv_cfg --set_mdm_password --ip <MDM IP> --port 6611 --password <secret>

4. For Linux, type **/opt/emc/scaleio/sdc/bin/drv_cfg --set_mdm_password --ip <MDM IP>**. For example:

   /opt/emc/scaleio/sdc/bin/drv_cfg --set_mdm_password --ip <MDM IP> --port 6611 --password <secret> --file /etc/emc/scaleio/drv_cfg.txt

5. Repeat until all new SDCs are connected.

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Configuring Dell PowerSwitch switches using SONiC networking

Use this section to configure the SONiC network on the Dell PowerSwitch switches.

NOTE: SONiC networking only supports partial network automation (PNA) and the configurations in this section are for reference only.

NOTE: To review the network architecture designs, see the *PowerFlex Appliance Network Planning Guide*.

Network configuration workflow

The following is the workflow for configuration requirements and reference examples for customer switch configuration prior to deployment:

1. Initialize the switch (if needed).
2. Upgrade the firmware (if needed).
3. Configure the basic requirements.
4. Configure the VLANs.
5. Configure the access/aggregation switches.
6. Configure the first installer node access ports.
7. Configure customer uplinks.

NOTE: Steps 1 through 6 are completed by the factory and step 7 is completed by the Dell Networking team.

Information required from the EPS form

Ensure you have the following information from the EPS form:

- The customer uplink number, type, and speed
- VLAN names and numbers
- Layer-2 and layer-3 information
- Interface types (access, port-channel, Link Aggregation Control Protocol)
- Mclag

Supported switches

The following Dell PowerSwitch switches support Dell Enterprise SONiC.

Table 46. Supported switches

Type of switch	Supported switches
Management	Dell PowerSwitch S3248T-ON
Access	Dell PowerSwitch S5448F-ON
Aggregation	Dell PowerSwitch Z9432F-ON
Management aggregation switch	Dell PowerSwitch S5448F-ON

Configuring Dell PowerSwitch management network

Use this section to configure a Dell PowerSwitch management network using SONiC.

Configure the Dell PowerSwitches management switches

Use this procedure to configure the Dell PowerSwitches management switches.

About this task

Ensure to be in the configuration mode to perform the commands in the following procedures.

1. Enter the SONiC management framework CLI, after `admin@sonic:~$` type the following: **sonic-cli**
2. Enter the configuration mode in the CLI, type the following: **configure terminal**.
3. Fully exit the configuration mode, type the following: **end**.

Steps

1. Change the interface naming mode to standard extended, after `sonic(config)#` type the following: **Interface-naming standard extended**

   NOTE: Write memory followed by system reboot is required.

2. Specify a hostname, after `sonic(config)#` type the following: **hostname <name>**.

   NOTE: Users running **sonic-cli** are suggested to restart your session.

3. Set a global username and password and an enable mode password, type the following: **username <admin> password <admin> role <sysadmin>**
4. Configure the global switch settings:

```
ip vrf mgmt
ip route vrf mgmt 0.0.0.0/0 <next-hop gateway>
spanning-tree mode rapid-pvst
spanning-tree edge-port bpdufilter default
login exec-timeout 600
login lockout period 30
login lockout max-retries 5
login lockout console-exempt
mac address-table aging-time 600
```

5. Configure the IP address for the management port:

```
interface mgmt 0
description "oob mgmt port"
mtu 1500
autoneg on
speed <>
ip address <ip address>/<mask>
end
```

6. Enable SSH, complete the following:
   a. Regenerate keys for the SSH server in EXEC mode, enter the following: **crypto ssh-key {rsa {2048}**

      Example output:

```
Processing SSH Key Gen request ...
Generated 2048-bit rsa key!!
```

   b. Display the SSH public keys in EXEC mode, type the following: **show crypto ssh-key rsa**
   c. Set the SSH login attempts, enter the following: **password-attributes max-retry 5 lockout-period 30**

7. Enable message of the day (MOTD):

```
banner motd %
*********************************************
*  WARNING: THIS IS A PRIVATE COMPUTER SYSTEM.
This computer system including all related equipment, network devices,
are provided only for authorized use.
All computer systems may be monitored for all lawful purposes, including
those activities that are authorized for management of the system.
All information including personal information, stored or sent over this
system may be monitored.
Uses of this system, authorized or unauthorized, constitutes consent to
monitoring of this system.
Unauthorized use may subject you to criminal prosecution.

WARNING: Unauthorized access to this system is forbidden and will be
prosecuted by law.
By accessing this system, you agree that your actions may be monitored.           *
*********************************************
%
```

8. Configure SNMP, type the following:

Table 47. SNMP types

SNMP type	Code

SNMP V2c	`snmp-server community <test>`
	` snmp-server contact < >`
	`snmp-server location < >`
	`snmp-server enable trap`
	`snmp-server host <ip address> community dell`
	`snmp-server host <ip address> community dell informs timeout 200 retries 10`

SNMPv3	` snmp-server view view-name {oid-tree [included | excluded]}`
	
	`snmp-server group group-name {any | v2c | v3 {auth | noauth | priv}} [read view-name] [write view-name] [notify view-name]`
	
	`snmp-server user user-name [group group-name] [encrypted] [auth {md5 | sha | noauth} [auth-password password]] [priv {DES | AES-128} [priv-password password]`
	
	`snmp-server contact < >`
	`snmp-server location < >`
	`snmp-server engine engine-id`
	`snmp-server enable trap`
	`snmp-server host <ip address> community <community name>`
	`snmp-server host <ip address> community <community name> informs timeout <value> retries <value>`

9. Enable LLDP, type the following: **`lldp enable`**
10. Configure the NTP and DNS, type the following:

```
ntp server <NTP1> minpoll 6 maxpoll 10 prefer true
ntp server <NTP2> minpoll 6 maxpoll 10
ntp server <NTP3> minpoll 6 maxpoll 10
ntp source-interface Management0
!
ip name-server <DNS1> vrf mgmt
```

```
ip name-server <DNS2> vrf mgmt
ip name-server source-interface management0
```

11. Configure the VLANs, type the following:

```
Interface vlan <vlan number>
description <”vlan-name”>
ip address <ip address>/<mask>
no shutdown
```

12. Configure the interfaces, type the following:

```
Sonic#Config t
Sonic(config)# interface ethernet <1/1/X>
description Dell-Flex-Engineering-Service-Port-DO-NOT-DISABLE
no shutdown
switchport access vlan <vlan id>
spanning-tree port type edge
```

13. Configure the iDRAC/management port for the switches or nodes, type the following:

```
Sonic(config)#interface ethernet <1/1/X>
description iDRAC/mgmt port for switches/nodes
no shutdown
switchport access vlan <vlan id>
spanning-tree port type edge
```

NOTE: To remove the created VLAN, type the following: **`Sonic(config)# no interface vlan <VLAN ID>`**

14. Configure the OOB management network port for the PowerFlex management nodes, type the following command:

```
sonic(config)#interface Ethernet <x/y>
description <controller node> switchport mode trunk
switchport trunk allowed vlan <vlan id>
spanning-tree port type edge trunk
spanning-tree bpduguard enable
spanning-tree guard root
mtu 9216
speed <speed>
no shutdown
```

NOTE: The default MTU for SONiC networking is 9100. The default setting is maintained during deployment, and the customer can modify as needed.

15. Configure uplinks to management aggregation, type the following:

```
sonic(config)#interface breakout port 1/X mode 1x10G
sonic(config)#interface breakout port 1/Y mode 1x10G
!
sonic(config)#interface Eth1/x/1
description mgmt-agg uplink A
switchport access Vlan <>
switchport trunk allowed Vlan <>
spanning-tree guard loop
no shutdown
!
sonic(config)#interface Eth1/Y/1
description mgmt-agg uplink B
switchport access Vlan <>
switchport trunk allowed Vlan <>
spanning-tree link-type point-to-point
spanning-tree guard loop
no shutdown
```

Configure the Dell PowerSwitch management aggregation switches

Use this procedure to configure the Dell PowerSwitch management aggregation switches.

About this task

Ensure to be in the configuration mode to perform the commands in the following procedures.

1. Enter the SONiC management framework CLI, after `admin@sonic:~$` type the following: **`sonic-cli`**
2. Enter the configuration mode in the CLI, type the following: **`configure terminal`**
3. Fully exit the configuration mode, type the following: **`end`**

Steps

1. Change the interface naming mode to standard extended, after `sonic(config)#` type the following: **`Interface-naming standard extended`**

   NOTE: Write memory followed by system reboot is required.

2. Specify a hostname, after `sonic(config)#` type the following: **`hostname <name>`**

   NOTE: Users running **`sonic-cli`** are suggested to restart your session.

3. Set a global username and password and an enable mode password, type the following: **`username <admin> password <admin> role <sysadmin>`**
4. Configure the global switch settings:

```
Interface vlan <vlan list>
ip vrf mgmt
ip route vrf mgmt 0.0.0.0/0 <next-hop gateway>
spanning-tree mode rapid-pvst
spanning-tree edge-port bpdufilter default
switching-mode cut-through
aaa authentication login default local
aaa authentication login console local
login exec-timeout 600
login lockout period 30
login lockout max-retries 5
login lockout console-exempt
mac address-table aging-time 600
```

5. Configure the IP address for the management port:

```
interface mgmt 0
description "oob mgmt port"
mtu 1500
autoneg on
speed 1000
ip address <ip address>/<mask>
end
```

6. Set DNS server, type the following: **`ip name-server <ip address> vrf mgmt`**
7. Enable SSH, complete the following:
   a. Regenerate keys for the SSH server in EXEC mode, enter the following: **`crypto ssh-key {rsa {2048}`**

      Example output:

      ```
      Processing SSH Key Gen request ...
      Generated 2048-bit rsa key!!
      ```

   b. Display the SSH public keys in EXEC mode, type the following: **`show crypto ssh-key rsa`**
   c. Set the SSH login attempts, enter the following: **`password-attributes max-retry 5 lockout-period 30`**

8. Enable message of the day (MOTD):

```
banner motd %
**********************************************
*  WARNING: THIS IS A PRIVATE COMPUTER SYSTEM.
This computer system including all related equipment, network devices,
are provided only for authorized use.
All computer systems may be monitored for all lawful purposes, including
those activities that are authorized for management of the system.
All information including personal information, stored or sent over this
system may be monitored.
Uses of this system, authorized or unauthorized, constitutes consent to
monitoring of this system.
Unauthorized use may subject you to criminal prosecution.

WARNING: Unauthorized access to this system is forbidden and will be
prosecuted by law.
By accessing this system, you agree that your actions may be monitored.           *
**********************************************
%
```

9. Configure SNMP, type the following:

Table 48. SNMP configurations

SNMP type	Code

SNMP V2c	`snmp-server community <test>`
	`snmp-server contact < >`
	`snmp-server location < >`
	`snmp-server enable trap`
	`snmp-server host <ip address> community dell`
	`snmp-server host <ip address> community dell informs timeout 200 retries 10`

SNMPv3	`snmp-server view view-name {oid-tree [included | excluded]}`
	
	`snmp-server group group-name {any | v2c | v3 {auth | noauth | priv}} [read view-name] [write view-name] [notify view-name]`
	
	`snmp-server user user-name [group group-name] [encrypted] [auth {md5 | sha | noauth} [auth-password password]] [priv {DES | AES-128} [priv-password password]`
	
	`snmp-server contact < >`
	`snmp-server location < >`
	`snmp-server engine engine-id`
	`snmp-server enable trap`
	`snmp-server host <ip address> community <community name>`
	`snmp-server host <ip address> community <community name> informs timeout <value> retries <value>`

10. Enable LLDP, type the following: **`lldp enable`**
11. Configure the VLANs, type the following:

```
interface vlan <vlan number>
description <”vlan-name”>
no shutdown
```

NOTE:

- To remove the created VLAN, after `sonic(config)#` type the following command: **`no interface vlan <VLAN ID>`**.
- To describe the created VLAN, after `Dell(config VLAN ID)#` enter the interface mode and type the following command: **`description <enter the description>`**.

12. Configure the OSPF management aggregation peer-link, type the following:

```
interface Vlan<>
description "OSPF Agg-Peer"
mtu 9216
mclag-separate-ip
ip address <ip address/mask>
ip ospf area <area-id>
ip ospf network point-to-point

router ospf
ospf router-id <router-id>
passive-interface default
redistribute connected
area <area-id>
no passive-interface Eth <Uplink to customer>
no passive-interface Eth <Uplink to customer>
no passive-interface <vlan>
```

13. Configure the peer-link port channel, peer-link to the management aggregation switch and MCLAG domain:

MCLAG in Dell SonicOS refers to multi-chassis link aggregation group. It is a networking feature that allows you to aggregate multiple physical links across two separate devices (chassis) into a single logical link for redundancy and load balancing.

NOTE: Ensure the following:

- Domain ID: The mclag id number recommended for an access switch is 50 and aggregation switch is 60, but any number from 1 to 256 can be used.
- Peer link port channel IDs and domain IDs must be identical
- System MAC must be identical on both switches
- Layer 3 point-to-point VLAN: SVI (VLAN interface) is used as the MCLAG keepalive link between two switches in an MCLAG pair. Use a **`/30`** subnet for point-to-point

For switch-A and switch-B, complete the following steps:

a. Configure the layer-3 point-to-point VLAN, type the following:

```
interface VLAN <use mclag domainID>
descripton "peerlink keepalive "
mclag-separate-ip
ip address <ip address>/<mask>
exit
```

b. Configure peer-link port channel, type the following:

```
interface PortChannel <use mclag domain ID>
description "MCLAG peer link”
switchport trunk allowed Vlan <defined vlans>
fast_rate
mtu 9216
no shutdown
```

c. Configure peer-link to the management aggregation switch, type the following:

```
interface Eth1/x
description "Peer link to management aggregation2"
mtu 9216
speed 400000
fec RS

unreliable-los auto
```

```
channel-group <>
no shutdown

interface Eth1/y
description "Peer link to management aggregation 2"
mtu 9216
speed 400000
fec RS
unreliable-los auto
channel-group <>
no shutdown
```

d. Configure the MCLAG domain, type the following:

```
mclag domain id <domain id>
source-ip <peerlink keepalive IP of source switch >
peer-ip <peerlink keepalive IP of peer switch>
peer-link PortChannel <use mclag domain ID>
mclag-system-mac 00:00:00:00:00:01
keepalive-interval 1
session-timeout 30
delay-restore 300
fast-convergence
backup-keepalive source-ip <mgnt agg switch1-mgmt0>
backup-keepalive peer-ip <mgnt agg switch2-mgmt0>
backup-keepalive interval 1
backup-keepalive session-vrf mgmt
```

14. Configure the management aggregation downlinks, type the following:

```
interface Ethernetx/x
description "mgmt-switch downlink"
speed 10000
unreliable-los auto
switchport trunk allowed vlan <>
spanning-tree guard loop
no shutdown
```

Configuring Dell PowerSwitch access-aggregation network

Use this section to configure a Dell PowerSwitch access-aggregation network.

NOTE: Access/aggregation switches are configured in the factory for deployment and Dell Networking services work with the customer for final configurations, hence the following configuration is only for reference purpose.

Configure the Dell PowerSwitch access switches

Use this procedure to configure two Dell PowerSwitch access switches for deployment and to configure the PowerFlex management node. Perform the steps in this procedure on both switches.

About this task

Ensure to be in the configuration mode to perform the commands in the following procedures.

1. Enter the SONiC management framework CLI, after `admin@sonic:~$` type the following: **`sonic-cli`**
2. Enter the configuration mode in the CLI, type the following: **`configureterminal`**.
3. Fully exit the configuration mode, type the following: **`end`**.

Steps

1. Change the interface naming mode to standard extended, after `sonic(config)#` type the following: **`Interface-naming standard extended`**

NOTE: Write memory followed by system reboot is required.

2. Specify a hostname, after `sonic(config)#` type the following: **hostname <name>**.

   NOTE: Users running **sonic-cli** are suggested to restart your session.

3. Set a global username and password and an enable mode password, type the following: **username <admin> password <admin> role <sysadmin>**
4. Configure the global switch settings:

```
Interface vlan <vlan list>
ip vrf mgmt
ip route vrf mgmt 0.0.0.0/0 <next-hop gateway>
spanning-tree mode rapid-pvst
spanning-tree edge-port bpdufilter default
switching-mode cut-through
aaa authentication login default local
aaa authentication login console local
login exec-timeout 600
login lockout period 30
login lockout max-retries 5
login lockout console-exempt
mac address-table aging-time 600
```

5. Configure the IP address for the management port:

```
interface mgmt 0
description "oob mgmt port"
mtu 1500
autoneg on
speed <>
ip address <ip address>/<mask>
end
```

6. Set DNS server, type the following: **ip name-server <ip address> vrf mgmt**
7. Enable SSH, complete the following:
   a. Regenerate keys for the SSH server in EXEC mode, enter the following: **crypto ssh-key {rsa {2048}**

      Example output:

```
Processing SSH Key Gen request ...
Generated 2048-bit rsa key!!
```

   b. Display the SSH public keys in EXEC mode, type the following: **show crypto ssh-key rsa**
   c. Set the SSH login attempts, enter the following: **password-attributes max-retry 5 lockout-period 30**
8. Enable message of the day (MOTD):

```
banner motd %
**********************************************
*  WARNING: THIS IS A PRIVATE COMPUTER SYSTEM.
This computer system including all related equipment, network devices,
are provided only for authorized use.
All computer systems may be monitored for all lawful purposes, including
those activities that are authorized for management of the system.
All information including personal information, stored or sent over this
system may be monitored.
Uses of this system, authorized or unauthorized, constitutes consent to
monitoring of this system.
Unauthorized use may subject you to criminal prosecution.

WARNING: Unauthorized access to this system is forbidden and will be
prosecuted by law.
By accessing this system, you agree that your actions may be monitored.           *
**********************************************
%
```

9. Configure SNMP, type the following:

Table 49. SNMP types

SNMP type	Code

SNMP V2c	`snmp-server community <test>`
	` snmp-server contact < >`
	`snmp-server location < >`
	`snmp-server enable trap`
	`snmp-server host <ip address> community dell`
	`snmp-server host <ip address> community dell informs timeout 200 retries 10`

SNMPv3	` snmp-server view view-name {oid-tree [included | excluded]}`
	
	`snmp-server group group-name {any | v2c | v3 {auth | noauth | priv}} [read view-name] [write view-name] [notify view-name]`
	
	`snmp-server user user-name [group group-name] [encrypted] [auth {md5 | sha | noauth} [auth-password password]] [priv {DES | AES-128} [priv-password password]`
	
	`snmp-server contact < >`
	`snmp-server location < >`
	`snmp-server engine engine-id`
	`snmp-server enable trap`
	`snmp-server host <ip address> community <community name>`
	`snmp-server host <ip address> community <community name> informs timeout <value> retries <value>`

10. Enable LLDP, type the following: **`lldp enable`**
11. Configure the VLANs, type the following:

```
interface vlan <vlan number>
description <”vlan-name”>
no shutdown
```

NOTE:

- Remove the created VLAN, after `sonic(config)#` type the following command: **`no interface vlan <VLAN ID>`**.
- Describe the created VLAN, after `Dell(config VLAN ID)#` enter the interface mode and type the following command: **`description <enter the description>`**.

12. Configure the peer link port channel, peerlink to the access switch and MCLAG domain:

MCLAG in Dell SonicOS refers to multi-chassis link aggregation group. It is a networking feature that allows you to aggregate multiple physical links across two separate devices (chassis) into a single logical link for redundancy and load balancing.

NOTE: Ensure the following:

- Domain ID: The mclag id number recommended for an access switch is 50 and aggregation switch is 60, but any number from 1 to 256 can be used.
- Peer link port channel IDs and domain IDs must be identical
- System MAC must be identical on both switches

- Layer 3 point-to-point VLAN: SVI (VLAN interface) is used as the MCLAG keepalive link between two switches in an MCLAG pair. Use a **/30** subnet for point-to-point

a. Configure the layer-3 point-to-point VLAN, type the following:

```
interface VLAN <use mclag domainID>
descripton "peerlink keepalive "
mclag-separate-ip
ip address <ip address>/<mask>
exit
```

b. Configure peer-link port channel, type the following:

```
interface PortChannel <use mclag domain ID>
description "MCLAG peer link”
switchport trunk allowed Vlan <defined vlans>
fast_rate
mtu 9216
no shutdown
```

c. Configure peerlink to the access switch, type the following:

```
interface Eth1/x
description "Peer link1 to access2"
mtu 9216
speed 400000
fec RS
unreliable-los auto
channel-group <mclag domain ID>
no shutdown

interface Eth1/y
description "Peer link2 to access2"
mtu 9216
speed 400000
fec RS
unreliable-los auto
channel-group <mclag domain ID>
no shutdown
```

d. Configure the MCLAG domain, type the following:

```
mclag domain id <domain id>
source-ip < peerlink keepalive IP of source switch >    (Layer 3 point-to-point
vlan ip of source switch )
peer-ip < peerlink keepalive IP of peer switch>  (Layer 3 point-to-point vlan ip of
peer switch )
peer-link PortChannel <mclag domain id>
mclag-system-mac 00:00:00:00:00:01   (should be same on other switch)
keepalive-interval 1
session-timeout 30
delay-restore 300
fast-convergence
backup-keepalive source-ip <access switch1mgmt0>
backup-keepalive peer-ip <access switch2-mgmt0>
backup-keepalive interval 1
backup-keepalive session-vrf mgmt
```

13. Configure the port-channels for uplink to the aggregation switch, type the following:

```
interface PortChannel <>
description "To agg switches"
switchport trunk allowed Vlan <defined vlans>
fast_rate
lacp individual
mtu 9216
```

```
no shutdown
mclag <mclag domain id>
```

14. Configure uplink member port, type the following:

```
interface Eth <>
description "Link to Agg A"
mtu 9216
speed 400000
unreliable-los auto
channel-group <>
no shutdown

interface Eth <>
description "Link to Agg B"
mtu 9216
speed 400000
unreliable-los auto
channel-group <>
no shutdown
```

15. Configure the PowerFlex node facing port-channel interfaces on the access switch, type the following:

NOTE: If you are using a 25 G PowerFlex management node, run the following command to split 100 G to 25 G for:

```
interface breakout port 1/5 mode 1x25g
interface breakout port 1/6 mode 1x25g
```

This procedure is applicable for the PowerFlex management controller, storage-only, and compute-only nodes.

Repeat this procedure based on the configuration data.

NOTE: For partial node automation, switch interface configurations are completed manually.

Table 50. Configuration type

Trunk	LACP

`interface <interface number>`
`mtu 9216`
`speed 25000`
`unreliable-los auto`
`switchport trunk allowed Vlan <vlan-list>`
`spanning-tree bpduguard`
`spanning-tree port type edge`
`spanning-tree guard root`
`no shutdown`
	`interface PortChannel <>`
	`description "bond0 to node"`
	`switchport trunk allowed Vlan < >`
	`mtu 9216`
	`mclag <mclag domain id>`
	`fast_rate`
	`lacp individual`
	`spanning-tree bpduguard`
	`spanning-tree port type edge`
	`spanning-tree guard root`
	`exit`
	
	`interface PortChannel <>`
	`description "bond1 to node"`
	`switchport trunk allowed Vlan < >`
	`mtu 9216`
	`mclag <mclag domain id>`
	`fast_rate`
	`lacp individual`
	`spanning-tree bpduguard`
	`spanning-tree port type edge`
	`spanning-tree guard root`
	`exit`
	
	`interface Eth <>`
	`description "connected to bond0:node"`
	`mtu 9216`
	`speed 25000`
	`unreliable-los auto`
	`channel-group <>`
	`no shutdown`

Table 50. Configuration type

Trunk	LACP

	`interface Eth <>`
	`description "connected to bond1:node "`
	`mtu 9216`
	`speed 25000`
	`unreliable-los auto`
	`channel-group <>`
	`no shutdown`

Configure the Dell PowerSwitch aggregation switches

Use this procedure to configure two Dell PowerSwitch aggregation switches for deployment. Perform the steps in this procedure on both switches.

About this task

Ensure to be in the configuration mode to perform the commands in the following procedures.

1. Enter the SONiC management framework CLI, after `admin@sonic:~$` type the following: **`sonic-cli`**
2. Enter the configuration mode in the CLI, type the following: **`configureterminal`**.
3. Fully exit the configuration mode, type the following: **`end`**.

Steps

1. Change the interface naming mode to standard extended, after `sonic(config)#` type the following: **`Interface-naming standard extended`**

   NOTE: Write memory followed by system reboot is required.

2. Specify a hostname, after `sonic(config)#` type the following: **`hostname <name>`**.

   NOTE: Users running **`sonic-cli`** are suggested to restart your session.

3. Set a global username and password and an enable mode password, type the following: **`username <admin> password <admin> role <sysadmin>`**
4. Configure the global switch settings:

```
Interface vlan <vlan list>
ip vrf mgmt
ip route vrf mgmt 0.0.0.0/0 <next-hop gateway>
spanning-tree mode rapid-pvst
spanning-tree edge-port bpdufilter default
switching-mode cut-through
aaa authentication login default local
aaa authentication login console local
login exec-timeout 600
login lockout period 30
login lockout max-retries 5
login lockout console-exempt
mac address-table aging-time 600
```

5. Configure the IP address for the management port:

```
interface mgmt 0
description "oob mgmt port"
mtu 1500
autoneg on
speed 1000
ip address <ip address>/<mask>
end
```

6. Set DNS server, type the following: **ip name-server <ip address> vrf mgmt**
7. Enable SSH, complete the following:
   a. Regenerate keys for the SSH server in EXEC mode, enter the following: **crypto ssh-key {rsa {2048}**

      Example output:

      ```
      Processing SSH Key Gen request ...
      Generated 2048-bit rsa key!!
      ```

   b. Display the SSH public keys in EXEC mode, type the following: **show crypto ssh-key rsa**
   c. Set the SSH login attempts, enter the following: **password-attributes max-retry 5 lockout-period 30**
8. Enable message of the day (MOTD):

```
banner motd %
**********************************************
*  WARNING: THIS IS A PRIVATE COMPUTER SYSTEM.
This computer system including all related equipment, network devices,
are provided only for authorized use.
All computer systems may be monitored for all lawful purposes, including
those activities that are authorized for management of the system.
All information including personal information, stored or sent over this
system may be monitored.
Uses of this system, authorized or unauthorized, constitutes consent to
monitoring of this system.
Unauthorized use may subject you to criminal prosecution.

WARNING: Unauthorized access to this system is forbidden and will be
prosecuted by law.
By accessing this system, you agree that your actions may be monitored.          *
**********************************************
%
```

9. Configure SNMP, type the following:

Table 51. SNMP configurations

SNMP type	Code

SNMP V2c	`snmp-server community <test>`
	`snmp-server contact < >`
	`snmp-server location < >`
	`snmp-server enable trap`
	`snmp-server host <ip address> community dell`
	`snmp-server host <ip address> community dell informs timeout 200 retries 10`

SNMPv3	`snmp-server view view-name {oid-tree [included | excluded]}`
	
	`snmp-server group group-name {any | v2c | v3 {auth | noauth | priv}} [read view-name] [write view-name] [notify view-name]`
	
	`snmp-server user user-name [group group-name] [encrypted] [auth {md5 | sha | noauth} [auth-password password]] [priv {DES | AES-128} [priv-password password]`
	
	`snmp-server contact < >`
	`snmp-server location < >`
	`snmp-server engine engine-id`
	`snmp-server enable trap`
	`snmp-server host <ip address> community <community name>`
	`snmp-server host <ip address> community`

Table 51. SNMP configurations (continued)

SNMP type	Code
	`<community name> informs timeout <value> retries <value>`

10. Enable LLDP, type the following: **`lldp enable`**
11. Configure the VLANs, type the following:

```
interface vlan <vlan number>
description <”vlan-name”>
no shutdown
```

NOTE:

- To remove the created VLAN, after `Sonic(config)#` type the following command: **`no interface vlan <VLAN ID>`**.
- To describe the created VLAN, after `Dell(config VLAN ID)#` enter the interface mode and type the following command: **`description <enter the description>`**.

12. Configure the OSPF aggregation peer-link, type the following:

```
interface Vlan<>
description "OSPF Agg-Peer"
mtu 9216
mclag-separate-ip
ip address <ip address/mask>
ip ospf area <area-id>
ip ospf network point-to-point

router ospf
ospf router-id <router-id>
passive-interface default
redistribute connected
area <area-id>
no passive-interface Eth <Uplink to customer>
no passive-interface Eth <Uplink to customer>
no passive-interface <vlan>
```

13. Configure the peer-link port channel, peer-link to the aggregation switch and MCLAG domain:

    MCLAG in Dell SonicOS refers to multi-chassis link aggregation group. It is a networking feature that allows you to aggregate multiple physical links across two separate devices (chassis) into a single logical link for redundancy and load balancing.

    NOTE: Ensure the following:

    - Domain ID: The mclag id number recommended for an access switch is 50 and aggregation switch is 60, but any number from 1 to 256 can be used.
    - Peer link port channel IDs and domain IDs must be identical
    - System MAC must be identical on both switches
    - Layer 3 point-to-point VLAN: SVI (VLAN interface) is used as the MCLAG keepalive link between two switches in an MCLAG pair. Use a **`/30`** subnet for point-to-point

    For switch-A and switch-B, complete the following steps:

    a. Configure the layer-3 point-to-point VLAN, type the following:

```
interface VLAN <use mclag domainID>
descripton "peerlink keepalive "
mclag-separate-ip
ip address <ip address>/<mask>
exit
```

b. Configure peer-link port channel, type the following:

```
interface PortChannel <use mclag domain ID>
description "MCLAG peer link”
switchport trunk allowed Vlan <defined vlans>
fast_rate
mtu 9216
no shutdown
```

c. Configure peer-link to the aggregation switch, type the following:

```
interface Eth1/x
description "Peer link to aggregation2"
mtu 9216
speed 400000
fec RS

unreliable-los auto
channel-group <mclag domain id>
no shutdown

interface Eth1/y
description "Peer link to aggregation 2"
mtu 9216
speed 400000
fec RS
unreliable-los auto
channel-group <mclag domain id>
no shutdown
```

d. Configure the MCLAG domain, type the following:

```
mclag domain id <domain id>
source-ip <peerlink keepalive IP of source switch >
peer-ip <peerlink keepalive IP of peer switch>
peer-link PortChannel <use mclag domain ID>
mclag-system-mac 00:00:00:00:00:01
keepalive-interval 1
session-timeout 30
delay-restore 300
fast-convergence
backup-keepalive source-ip <agg switch1-mgmt0>
backup-keepalive peer-ip <agg switch2-mgmt0>
backup-keepalive interval 1
backup-keepalive session-vrf mgmt
```

14. Configure the port-channels for downlink to the access switch, type the following:

```
interface port-channel <port channel number>
description <downlink to access switch>
switchport trunk allowed vlan <allowed vlan list
fast_rate
mtu 9216
mclag <mclag domain id>
```

15. Configure downlink member port, type the following:

```
interface Ethernet<1/x>
description <Connection to access switch>
speed <speed>
mtu 9216
unreliable-los auto
channel-group <>
no shutdown

interface Eth1/y
description <”Connection to access switch”>
mtu 9216
speed 100000
```

```
unreliable-los auto
channel-group <>
no shutdown
```

16. Configure the aggregation switch SVI, type the following:

```
interface vlan <vlan-no>
description <vlan-name>
no shutdown
ip anycast-address <ip address/mask>
mtu 9216
neigh-suppress
```

NOTE: The Dell Networking services team are responsible for customer uplink configuration.

Verify the network configuration

Use this procedure to verify configuration before adding a VLAN to the network.

Steps

1. Verify that the multi-chassis link aggregation group (MCLAG) is configured between the access switches, type: **`show mclag brief`**.
2. Verify the port channel between servers, hosts, and access switches are configured, type: **`show port-channel summary`**.
3. Verify the port channel between aggregation switches and access switches are configured, type: **`show port-channel summary`**.
4. Verify that the VLAN is created and enabled on the port channel between the aggregation switches and the access switches, type: **`show vlan`**.
5. Verify that SVI is configured on the aggregation switches, type: **`show ip interfaces`**.
6. Verify that you have the VLAN IP address, mask, and gateway to verify the VLAN configuration, type: **`show running-configuration vlan <>`**.
7. Display hardware identity information stored in the system EEPROM, such as the service tag, base MAC address, serial number, product ID, and ONIE (Open Network Install Environment) details, type the following: **`show platform syseeprom`**.
8. Display the operational status of all interfaces on the device, type the following: **`show interface status`**.
9. View the network topology information and verifying physical connectivity between devices, type the following: **`show lldp table`**.

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Troubleshooting

Use this section for troubleshooting.

System validation and troubleshooting

Use this section for reviewing and validating a PowerFlex management cluster (KVM), including QCOW2 discovery, virtual machine/host mapping, console access, network validation, and cluster health.

Console access overview

PowerFlex management cluster (KVM) deployments support two console access methods. Both access a virtual machine, and each serves a different purpose.

Serial console access (default)

The `virsh` console connects directly to a virtual machines serial console through **`libvirt`**.

It does not require SSH keys or a graphical environment, and works even if the virtual machine has no network configured.

Use for:

- Deploying or configuring CloudLink virtual machines
- A virtual machine has no graphical console or network
- Early boot access is required
- Troubleshooting low level virtual machine issues

This command opens a text-based serial console session directly to the virtual machine:

```
sudo virsh console cloudlink-1
```

Example output:

```
Connected to domain 'cloudlink-1'
Escape character is ^]

cloudlink login: admin
Password:
cloudlink-1:~ #
```

Graphical console access

The virt-viewer provides a full GUI console (VNC/SPICE), similar to VMware or Hyper-V consoles.

It connects via SSH to the KVM host, so SSH key authentication is required.

Use for:

- A GUI (installer screens, GUI-based tools)
- A visual bootloader view
- Working from a remote workstation

Generate SSH key (required for virt viewer):

```
ssh-keygen -N ""
```

This command creates an SSH key pair used to authenticate to the KVM hypervisor.

Copy key to KVM host:

```
ssh-copy-id delladmin@pfm-node-1
```

This command allows passwordless SSH authentication required for virt viewer.

Launch the viewer:

```
Launch virt viewer
virt-viewer --connect qemu+ssh://delladmin@pfm-node-1/system cloudlink-1
```

This command opens a GUI console to the VM through **`libvirt`** over SSH.

QCOW2 discovery and validation

The following commands are used on a PowerFlex management cluster (KVM) to inspect, verify, and troubleshoot QCOW2 virtual disk files.

Table 52. QCOW2 file management and validation commands

Task	Command	Description

List all QCOW files	`find /var/lib/libvirt/images -name "*.qcow2" -ls`	Shows all QCOW2 images stored on the PowerFlex management cluster (KVM). Useful for confirming CloudLink, MVM, and PowerFlex management platform VM disk files.
		
		Example output:
		`12582913 /var/lib/libvirt/images/cloudlink/cloudlink-1.qcow2`
		`12582914 /var/lib/libvirt/images/cloudlink/cloudlink-2.qcow2`
		`12582915 /var/lib/libvirt/images/mvm/mvm-1.qcow2`

Inspect QCOW image metadata	`qemu-img info /var/lib/libvirt/images/cloudlink/cloudlink-1.qcow2`	Displays QCOW2 file format, size, and internal structure.
		
		Example output:
		`image: cloudlink-1.qcow2`
		`file format: qcow2`
		`virtual size: 64G`
		`disk size: 8.2G`

Validate QCOW integrity	`qemu-img check /var/lib/libvirt/images/cloudlink/cloudlink-1.qcow2`	Checks QCOW2 for corruption or metadata errors. Example output: `No errors were found on the image.`

Table 52. QCOW2 file management and validation commands (continued)

Task	Command	Description

Check available storage	`df -h /var/lib/libvirt/images`	Ensures enough free space exists for QCOW growth and virtual machine operation.
		
		Example output:
		
		`Filesystem Size Used Avail Use% Mounted on`
		`/dev/sda3 500G 172G 329G 35% /var/lib/libvirt/images`

VM-QCOW-host mapping

The following `virsh` commands are used to list all the virtual machines, view virtual machine configurations and identify associated QCOW backing files.

Table 53. `virsh` commands for VM-QCOW-host mapping

Task	Command	Description

List all virtual machines	`sudo virsh list --all`	Lists running and stopped VMs on the PowerFlex management cluster (KVM).
		
		Example:
		
		`3 cloudlink-1 running`
		`4 cloudlink-2 running`
		`7 mvm-1 running`
		`- pfmp-installer shut off`

Inspect virtual machine configuration	`sudo virsh dominfo cloudlink-1`	Shows CPU, memory, and virtual machine state information.
Identify QCOW backing file	`sudo virsh dumpxml cloudlink-1 | grep -A4 "<disk"`	Extracts the QCOW2 file path used by the virtual machine.

Hypervisor validation (PowerFlex management cluster (KVM))

The following commands verify that `libvirt` is running and confirm that the host has sufficient CPU and memory for virtual machine operations.

Table 54. PowerFlex management cluster (KVM) service and resource validation

Task	Command	Description
Verify `libvirt` is running	`sudo systemctl status libvirtd`	Ensures the virtualization layer is active.
Check CPU and memory availability	`lscpu` `free -h`	Confirms the host has sufficient resources for virtual machines.

Virtual machine internal state validation

The following table outlines key network diagnostic commands used within a virtual machine to verify interface configuration, routing, DNS resolution, and time synchronization.

Table 55. Virtual machine internal state validation commands

Component	Command	Description
Interfaces	`ip address`	Lists NICs and IP addresses in the virtual machine.
Routing	`ip route`	Shows default gateway and routing configuration.
DNS	`cat /etc/resolv.conf`	Confirms correct DNS servers for CloudLink cluster joining.
NTP	`chronyc tracking`	Validates time synchronization (required for CloudLink cluster).

PowerFlex management cluster (KVM) host networking

The following table lists the commands that validate key networking components on a PowerFlex management cluster (KVM), ensuring correct interface, VLAN, bridge, routing, and MTU configurations.

Table 56. PowerFlex management cluster (KVM) host networking components and commands

Component	Command	Description
Physical NICs	`ip link show`	Displays interface status and MTU values.
VLAN Tagging	`ip -d link show p1p1`	Shows VLAN ID and tagging configuration.
Linux Bridges	`brctl show`	Lists virtual switches and attached virtual machine NICs.
Routing	`ip route`	Shows hypervisor routing table.
Jumbo MTU Test	`ping -M do -s 8972 <peer-ip>`	Validates MTU 9216 for PowerFlex data networks.

Unexpected power loss

This section is to assist you in responding to unexpected power loss.

The following sections are general guidelines and has not been tested in every scenario.

If at any point there is a failure, contact Dell Technologies Support.

PowerFlex management cluster (ESXi)

Checklist for the PowerFlex management cluster (ESXi).

Management

- PDUs powered on
- Power on the switches
  - On all switches, confirm **Green** status lights
  - For out-of-band management switches, log in and confirm:

        - Validate the configuration, type **`show running-configuration`**
        - Validate uplinks, type **`show interface brief`**
        - Validate iDRAC connectivity, type **`show interface brief`**
        - Validate swtich hardware, type **`show hardware (cisco only)`**
    - For non-management switches, log in and confirm:
        - Validate the configuration, type **`show running-configuration`**
        - Validate uplinks, type **`show interface brief`**
        - Validate port channels, type **`show port-channel summary`**
        - Validate switch hardware, type **`show hardware (cisco only)`**
- Power on the management nodes
    - Confirm the status LED indicator light is off on each management node. The light on the front of the node should not be blinking amber.
    - Log in to each iDRAC to confirm no hardware alarms or errors and confirm the OS booted successfully.
    - Log in to each VMware ESXi host using the web client and ensure all storage virtual machines are booted.
        - Navigate to **Virtual Machines** and confirm the storage virtual machine is powered on. If not, power it on.
        - Allow PowerFlex storage to boot before you continue.
    - Ensure all management virtual machines are started, and if not, power them on:
        - CloudLink
        - Secure connect gateway
        - Jump server
        - VMware vCenter
        - Management virtual machines
        - Customer workloads
    - After 15 minutes, log in to PowerFlex Manager.
- Verify the following in PowerFlex Manager:
    - Check alerts: Go to **Monitoring** > **Alerts**.
    - Check management diagnostic status (for PowerFlex Manager 4.8 and above), and ensure all statuses are green.
        - Go to **Monitoring** > **Management Diagnostics**.
        - Run all management diagnostic tests.
    - SDC
        - Go to **Block** > **Hosts**.
        - Ensure the connected state is **Connected**.
    - Verify health of resources: Go to the **Resources** tab and confirm the **Health Status**.

Production

- Verify power on the storage-only nodes, using the PowerFlex Manager storage only resource group to identify nodes. Ensure the resource group is healthy.
- Verify power on compute-only nodes, using the PowerFlex Manager storage only resource group to identify nodes. Ensure the resource group is healthy.
- If applicable, verify CloudLink status using PowerFlex Manager. Go to **Resource Group** > **Overview**.

Single ESXi PowerFlex management node

Management

- PDUs powered on
- Power on the switches
    - On all switches, confirm **Green** status lights
    - For out-of-band management switches, log in and confirm:
        - Validate the configuration, type **`show running-configuration`**
        - Validate uplinks, type **`show interface brief`**
        - Validate iDRAC connectivity, type **`show interface brief`**

        - Validate swtich hardware, type **`show hardware (cisco only)`**
    - For non-management switches, log in and confirm:
        - Validate the configuration, type **`show running-configuration`**
        - Validate uplinks, type **`show interface brief`**
        - Validate port channels, type **`show port-channel summary`**
        - Validate switch hardware, type **`show hardware (cisco only)`**
- Power on the management nodes
    - Confirm the status LED indicator light is off on each management node. The light on the front of the node should not be blinking amber. See for more information.
    - Log in to each iDRAC to confirm no hardware alarms or errors and confirm the OS booted successfully.
    - Log in to each VMware ESXi host using the web client and ensure all storage virtual machines are booted.
        - Navigate to **Virtual Machines** and confirm the storage virtual machine is powered on. If not, power it on.
        - Allow PowerFlex storage to boot before you continue.
    - Ensure all management virtual machines are started, and if not, power them on:
        - CloudLink
        - Secure connect gateway
        - Jump server
        - VMware vCenter
        - Management virtual machines
        - Customer workloads
    - After 15 minutes, log in to PowerFlex Manager.
- Verify the following in PowerFlex Manager:
    - Check alerts: Go to **Monitoring** > **Alerts**.
    - Check management diagnostic status (for PowerFlex Manager 4.8 and above), and ensure all statuses are green.
        - Go to **Monitoring** > **Management Diagnostics**.
        - Run all management diagnostic tests.
    - SDC
        - Go to **Block** > **Hosts**.
        - Ensure the connected state is **Connected**.
    - Verify health of resources: Go to the **Resources** tab and confirm the **Health Status**.

Production

- Verify power on the storage-only nodes, using the PowerFlex Manager storage only resource group to identify nodes. Ensure the resource group is healthy.
- Verify power on compute-only nodes, using the PowerFlex Manager storage only resource group to identify nodes. Ensure the resource group is healthy.
- If applicable, verify CloudLink status using PowerFlex Manager. Go to **Resource Group** > **Overview**.

PowerFlex management cluster (KVM)

The PowerFlex management cluster (KVM) feature is considered as technology preview. Contact PowerFlex product management for details on this option.

Management

- Power on the switches
    - On all switches, confirm **Green** status lights
    - For out-of-band management switches, log in and confirm:
        - Validate the configuration, type **`show running-configuration`**
        - Validate uplinks, type **`show interface brief`**
        - Validate iDRAC connectivity, type **`show interface brief`**
        - Validate swtich hardware, type **`show hardware (cisco only)`**
    - For non-management switches, log in and confirm:
        - Validate the configuration, type **`show running-configuration`**

        - Validate uplinks, type **`show interface brief`**
        - Validate port channels, type **`show port-channel summary`**
        - Validate switch hardware, type **`show hardware (cisco only)`**
- Power on the management nodes
    - Confirm the status LED indicator light is off on each management node. The light on the front of the node should not be blinking amber.
    - Log in to each iDRAC to confirm no hardware alarms or errors and confirm the OS booted successfully.
    - From the management PowerFlex Manager, SSH to any management node and type **`sudo crm status`** to verify that all nodes are online and the resources are running.
- Verify the following in the management PowerFlex Manager:
    - Check alerts: Go to **Monitoring** > **Alerts**.
    - Check management diagnostic status (for PowerFlex 4.8 and above), and ensure all statuses are green.
        - Go to **Monitoring** > **Management Diagnostics**.
        - Run all management diagnostic tests.
    - SDC
        - Go to **Block** > **Hosts**.
        - Ensure the connected state is **Connected**.
    - Verify health of resources: Go to the Resources tab and confirm the **Health Status**.

Production

- Log in to each production PowerFlex Manager.
- Check alerts: Go to **Monitoring** > **Alerts**.
- Check management diagnostic status (for PowerFlex Manager 4.8 and above), and ensure all statuses are green.
    - Go to **Monitoring** > **Management Diagnostics**.
    - Run all management diagnostic tests.
- SDC
    - Go to **Block** > **Hosts**.
    - Ensure the connected state is **Connected**.
- Verify health of resources: Go to the **Resources** tab and confirm the **Health Status**.
- Verify power on the storage-only nodes, using the PowerFlex Manager storage only resource group to identify nodes. Ensure the resource group is healthy.
- Verify power on compute-only nodes, using the PowerFlex Manager storage only resource group to identify nodes. Ensure the resource group is healthy.